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Machine Automation Basic Training

TIA Portal Part 1 : SIMATIC S7-1200

www.siemens.com/s7-1200

S7-1200 : Processor

CPU 1211C



50 KB, DI 6x24 V DC, DQ 4x24 V DC
or 4xRLY, AI 2x10 bit 0–10 V DC,
expandable to 3 CM

DC/DC/DC	6ES7 211-1AE40-0XB0
AC/DC/RLY	6ES7 211-1BE40-0XB0
DC/DC/RLY	6ES7 211-1HE40-0XB0

CPU 1212C



75 KB, DI 8x24 V DC, DQ 6x24 V DC
or 6xRLY, AI 2x10 bit 0–10 V DC,
expandable to 3 CM + 2 SM

DC/DC/DC	6ES7 212-1AE40-0XB0
AC/DC/RLY	6ES7 212-1BE40-0XB0*
DC/DC/RLY	6ES7 212-1HE40-0XB0

CPU 1214C



100 KB, DI 14x24 V DC, DQ 10x24 V DC
or 10xRLY, AI 2x10 bit 0–10 V DC,
expandable to 3 CM + 8 SM

DC/DC/DC	6ES7 214-1AG40-0XB0
AC/DC/RLY	6ES7 214-1BG40-0XB0*
DC/DC/RLY	6ES7 214-1HG40-0XB0

CPU 1215C



125 KB, DI 14x24 V DC, DQ 10x24 V DC
or 10xRLY, AI 2x10 bit 0–10 V DC,
AQ 2x10 bit, 0 to 20 mA,
expandable to 3 CM + 8 SM

DC/DC/DC	6ES7 215-1AG40-0XB0
AC/DC/RLY	6ES7 215-1BG40-0XB0
DC/DC/RLY	6ES7 215-1HG40-0XB0

CPU 1217C



150 KB, DI 10x24 V DC, 4x1.5 V differential,
DQ 6x24 V DC, 4x1.5 V differential,
AI 2x10 bit 0–10 V DC, AQ 2x10 bit 0–20 mA,
line driver IO for (1 MHz ±1.5 V),
expandable to 3 CM + 8 SM

DC/DC/DC	6ES7 217-1AG40-0XB0
----------	---------------------

Fail-safe modules

CPU 1212FC



100 KB, DI 8x24 V DC, DQ 6x24 V DC
or 6xRLY, AI 2x10 bit 0–10 V DC,
expandable to 3 CM + 2 SM

DC/DC/DC	6ES7 212-1AF40-0XB0
DC/DC/RLY	6ES7 212-1HF40-0XB0

CPU 1214FC



125 KB, DI 14x24 V DC, DQ 10x24 V DC
or 10xRLY, AI 2x10 bit 0–10 V DC

DC/DC/DC	6ES7 214-1AF40-0XB0
DC/DC/RLY	6ES7 214-1HF40-0XB0

CPU 1215FC



150 KB, DI 14x24 V DC, DQ 10x24 V DC or
10xRLY, AI 2x10 bit 0–10 V DC, AI 2x10 bit,
0 to 20 mA

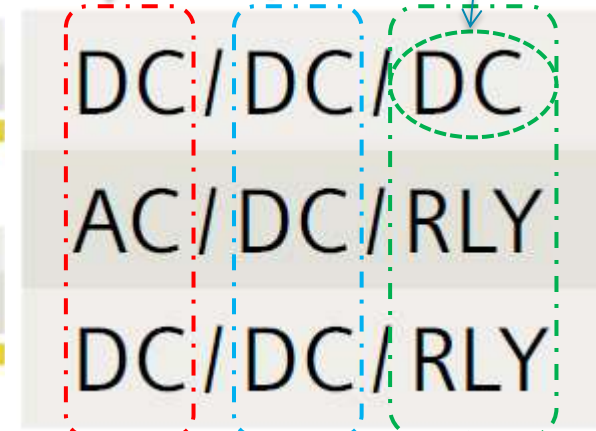
DC/DC/DC	6ES7 215-1AF40-0XB0
DC/DC/RLY	6ES7 215-1HF40-0XB0

*also available as RAIL-variant

ชนิด Power supply

ชนิด DI

ใช้ PTO
เลือก DC



ชนิด DO

S7-1200 : Board

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Communications board

Article No.

CB 1241 RS485

6ES7 241-1CH30-1XB0

> Download product
overview

Signal boards

Article No.

SB 1221 DC* 200 kHz

DI 4x5 V DC*

6ES7 221-3AD30-0XB0

DI 4x24 V DC*

6ES7 221-3BD30-0XB0

SB 1222 DC 200 kHz

DQ 4x5 V DC 0.1 A

6ES7 222-1AD30-0XB0

DQ 4x24 V DC 0.1 A

6ES7 222-1BD30-0XB0

SB 1223 DC/DC

DI 2x24 V DC/DQ 2x24 V DC 0.5 A

6ES7 223-0BD30-0XB0

SB 1223 DC*/DC 200 kHz

DI 2x5 V DC*/DQ 2x5 V DC 0.1 A

6ES7 223-3AD30-0XB0

DI 2x24 V DC*/DQ 2x24 V DC 0.1 A

6ES7 223-3BD30-0XB0



SB 1232 AQ

AQ 1 x 12 bit ± 10 V DC or 0–20 mA

6ES7 232-4HA30-0XB0

SB 1231 AI

AI 1 x 12 bit ± 10 V DC, ± 5 V DC,
 ± 2.5 V DC or 0–20 mA

6ES7 231-4HA30-0XB0

SB 1231 RTD

AI 1 x RTD x 16 bit, type: Platinum (Pt)

6ES7 231-5PA30-0XB0

SB 1231 TC

AI 1 x TC x 16 bit, types: J, K
voltage range: ± 80 mV

6ES7 231-5QA30-0XB0

*Sourcing input

S7-1200 : Communication (CM)



CM 1241 RS232

6ES7 241-1AH32-0XB0



CM 1241 RS422/485

6ES7 241-1CH32-0XB0



CM 1243-2 AS-i master

3RK7 243-2AA30-0XB0

DCM 1271 AS-i data decoupling

3RK7 271-1AA30-0AA0



CM 1242-5 PROFIBUS DP slave

6GK7 242-5DX30-0XE0



CM 1243-5 PROFIBUS DP master

6GK7 243-5DX30-0XE0



CP 1242-7 GPRS

6GK7 242-7KX31-0XE0



CP 1243-7 LTE

6GK7 243-7KX30-0XE0



CP 1243-1 Security

6GK7 243-1BX30-0XE0



CP 1243-1 DNP3 protocol

6GK7 243-1JX30-0XE0



CP 1243-1

IEC 60870-5-104 protocol

6GK7 243-1PX30-0XE0



CP 1243-1 PCC

(Plant Cloud Connect)

6GK7 243-1HX30-0XE0



CP 1243-8 IRC ST7 protocol

6GK7 243-8RX30-0XE0

> Download product
overview

S7-1200 : Signal module (SM)

Signal modules – digital

Article No.

SM 1221 DC

DI 8x24 V DC

6ES7 221-1BF32-0XB0

DI 16x24 V DC

6ES7 221-1BH32-0XB0

SM 1222 DC

DQ 8x24 V DC 0.5 A

6ES7 222-1BF32-0XB0

DQ 16x24 V DC 0.5 A

6ES7 222-1BH32-0XB0

SM 1222 RLY

DQ 8xRLY 30 V DC/250 V AC 2 A

6ES7 222-1HF32-0XB0

DQ 16xRLY 30 V DC/250 V AC 2 A

6ES7 222-1HH32-0XB0

DQ 8xRLY switchover
30 V DC/250 V AC 2 A

6ES7 222-1XF32-0XB0

SM 1223 DC/DC

DI 8x24 V DC, DQ 8x24 V DC 0.5 A

6ES7 223-1BH32-0XB0

DI 16x24 V DC, DQ 16x24 V DC 0.5 A

6ES7 223-1BL32-0XB0

SM 1223 DC/RLY

DI 8x24 V DC,
DQ 8xRLY 30 V DC/250 V AC 2 A

6ES7 223-1PH32-0XB0

DI 16x24 V DC,
DQ 16xRLY 30 V DC/250 V AC 2 A

6ES7 223-1PL32-0XB0

SM 1223 AC/RLY

DI 8x120/250 V AC,

DQ 8xRLY 30 V DC/250 V AC 2 A

6ES7 223-1QH32-0XB0

Signal modules – analog

Article No.

SM 1231 AI

AI 4x13 bit ± 10 V DC, ± 5 V DC,
 ± 2.5 V DC or 4–20 mA

6ES7 231-4HD32-0XB0

AI 8x13 bit ± 10 V DC, ± 5 V DC,
 ± 2.5 V DC or 4–20 mA

6ES7 231-4HF32-0XB0

AI 4x16 bit ± 10 V DC, ± 5 V DC,
 ± 2.5 V DC, ± 1.25 V DC or 4–20 mA

6ES7 231-5ND32-0XB0

SM 1231 RTD

AI 4xRTD x 16 bit

6ES7 231-5PD32-0XB0

AI 8xRTD x 16 bit

6ES7 231-5PF32-0XB0

SM 1231 TC

AI 4xTC x 16 bit

6ES7 231-5QD32-0XB0

AI 8xTC x 16 bit

6ES7 231-5QF32-0XB0

SM 1232 AQ

AQ 2x14 bit ± 10 V DC or 4–20 mA

6ES7 232-4HB32-0XB0

AQ 4x14 bit ± 10 V DC or 4–20 mA

6ES7 232-4HD32-0XB0

SM 1234 AI/AQ

AI 4x13 bit ± 10 V DC, ± 5 V DC,
 ± 2.5 V DC or 4–20 mA,
AQ 2x14 bit ± 10 V DC or 4–20 mA

6ES7 234-4HE32-0XB0

SM 1238 Energy Meter 480VAC

Energy measuring module for data
acquisition in 1- and 3-phase
networks (TN, TT) up to 480 V AC;
Current range: 1 A, 5 A

6ES7 238-5XA32-0XB0

[Download product
overview](#)

S7-1200 : Engineering framework

Automation of engineering tasks and flexible extended functions



STEP7 WinCC Startdrive

Integrated machine and plant diagnostics



STEP7 WinCC Safety

Safe monitoring of kinematic motions in space



STEP7 Safety

Cloud-based management of engineering software



STEP7 WinCC Startdrive

Long-term archiving of production data



WinCC

Acceptance and documentation of safety functions in the SINAMICS drive system



Startdrive Safety

Web-based dashboards for transparent production data



WinCC

Creation, use, and transfer of recipes



WinCC

From integrated energy data acquisition to proactive load management



STEP7 WinCC

Centralized user management in TIA Portal



STEP7 WinCC Startdrive

Increased SCADA system availability thanks to redundant architectures



WinCC

Use model-based Matlab closed-loop controller and machine simulations in the controller



STEP7

SIMATIC STEP 7 (TIA Portal) V16TRIAL Download

Entry type: Download, Entry ID:
109772803, Entry date: 12/03/2019



➤ Software in TIA Portal

The complete package for your automation solution optimizes your engineering processes



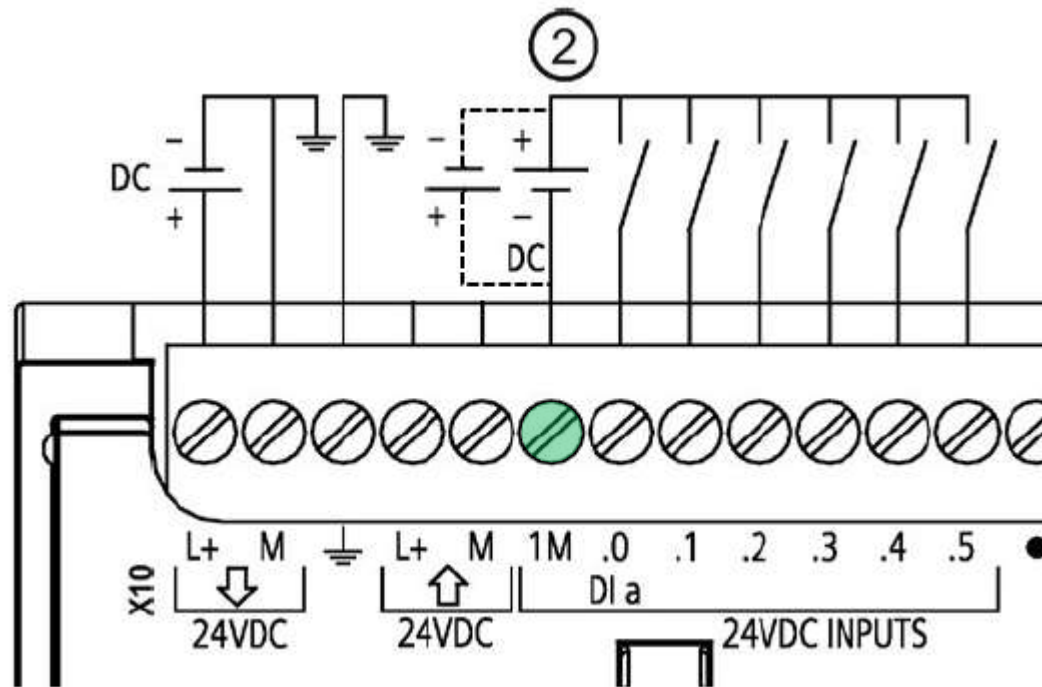
Firmware Update 4.4

Extended communication functions of the SIMATIC S7-1200 enable an improved, comprehensive data transfer!

- **OPC UA Data Access** as a server enables standardized horizontal and vertical communication as well as compliance with industry-specific standards. The Siemens OPC UA Modelling Editor (SiOME) enables easy adaptation to industry-specific Companion Specifications.
- **Cloud connectivity** enables data storage and analysis, e.g. in the Siemens MindSphere, thus supporting plant process efficiency and predictive maintenance.
- **Secure e-mail transmission** – optionally with attachment - enables the transfer of even sensitive machine data.



Common can be any + or -



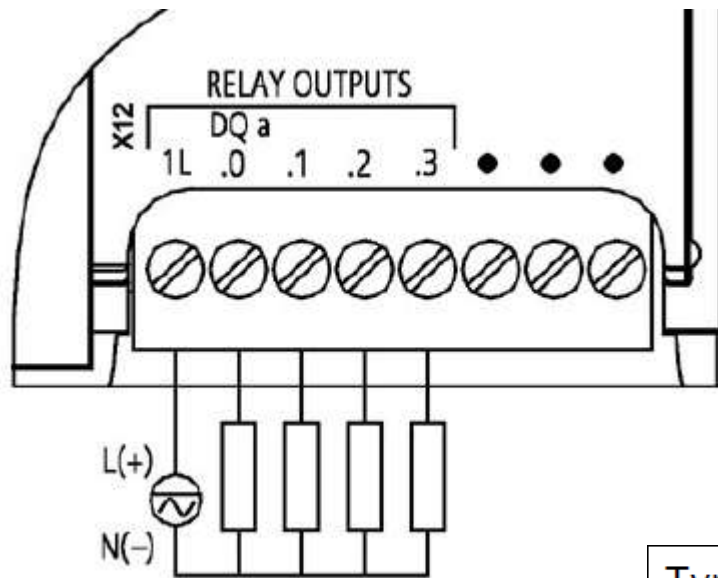
- ② For sinking inputs, connect "-" to "M" (shown).
For sourcing inputs, connect "+" to "M".

Digital Output (Relay)

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Type	Relay, mechanical
Voltage range	5 to 30 V DC or 5 to 250 V AC
Logic 1 signal at max. current	--
Logic 0 signal with 10 K Ω load	--
Current (max.)	2.0 A

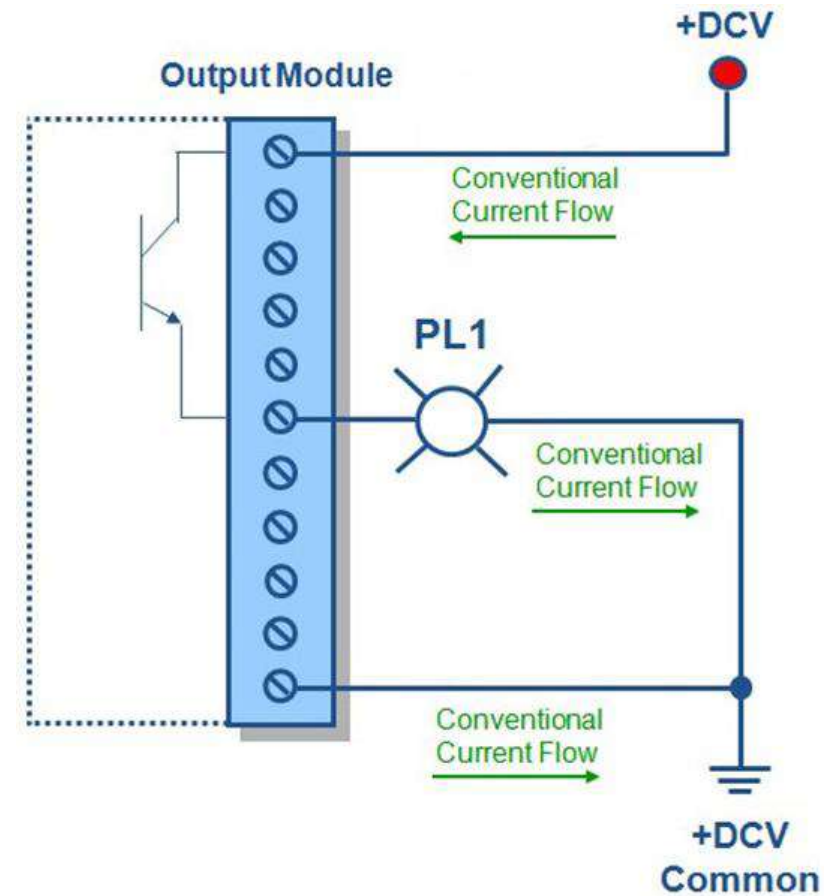
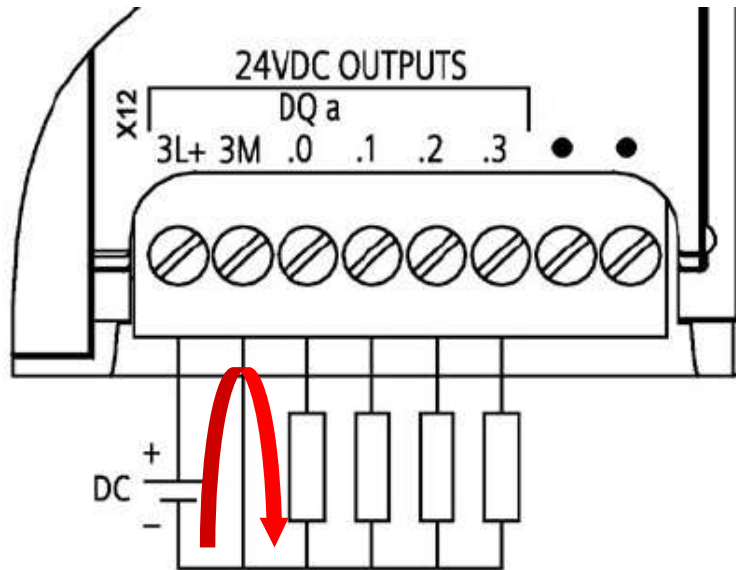
Digital Output (Transistor)

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Only source output for S7-1200 !!!

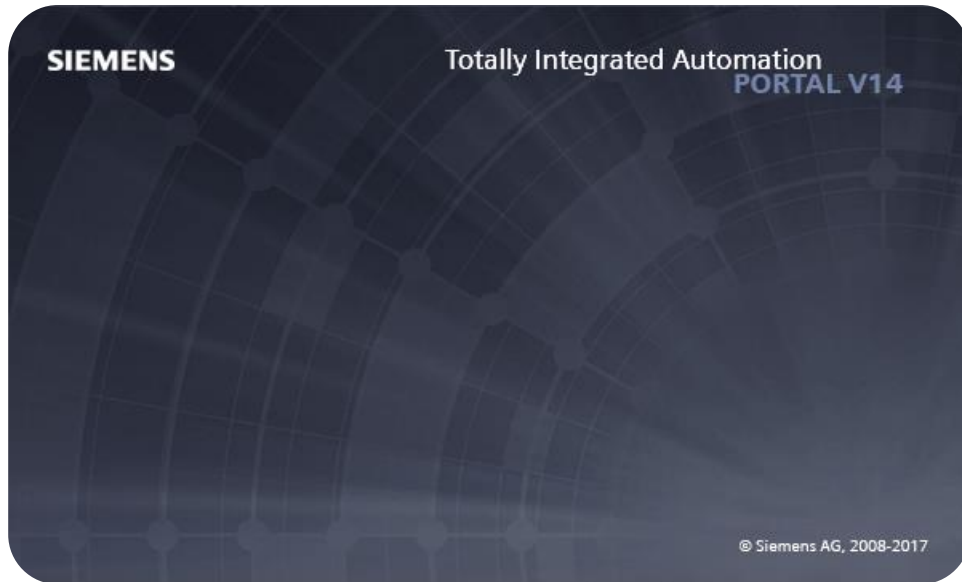


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Programming Procedure

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Basic programming 1

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Basic programming 2

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PLC Concept & Programming Concept

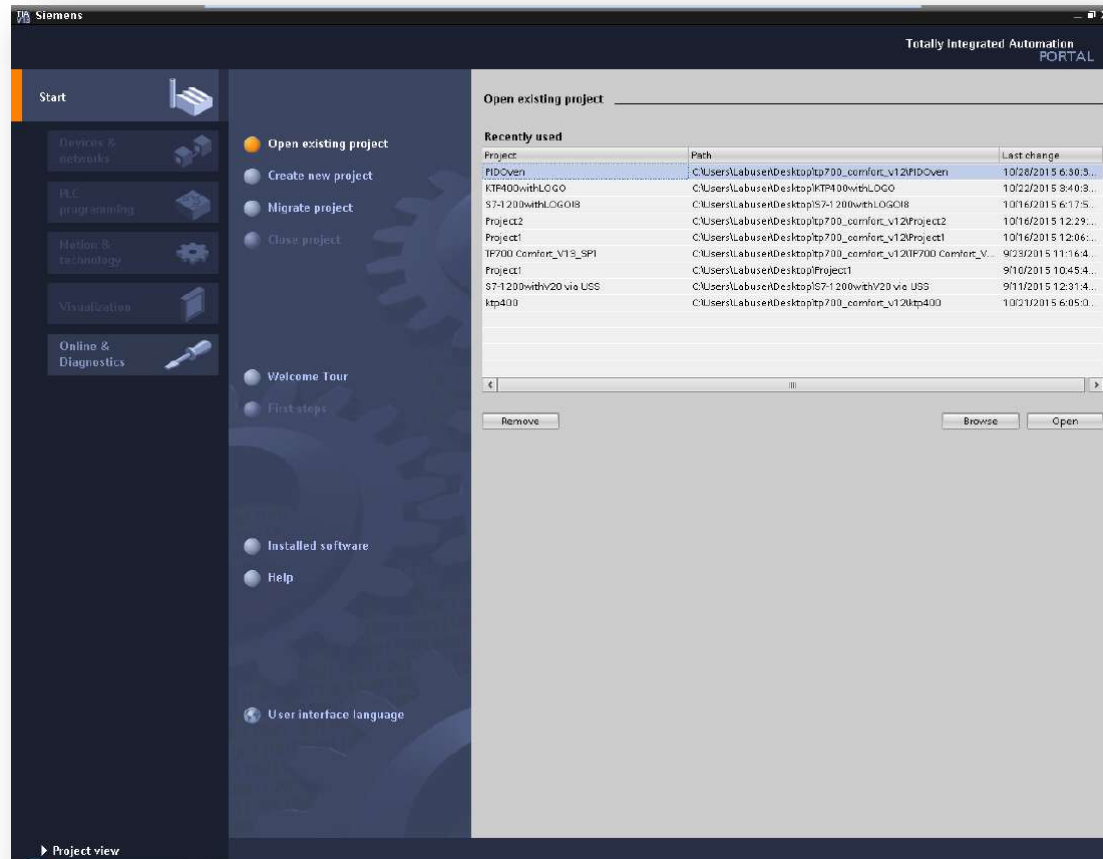
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Technical note

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Appendix

1. Introduction to SIMATIC Step 7



1

System requirements

(ความต้องการของระบบ)

2

Software Environment

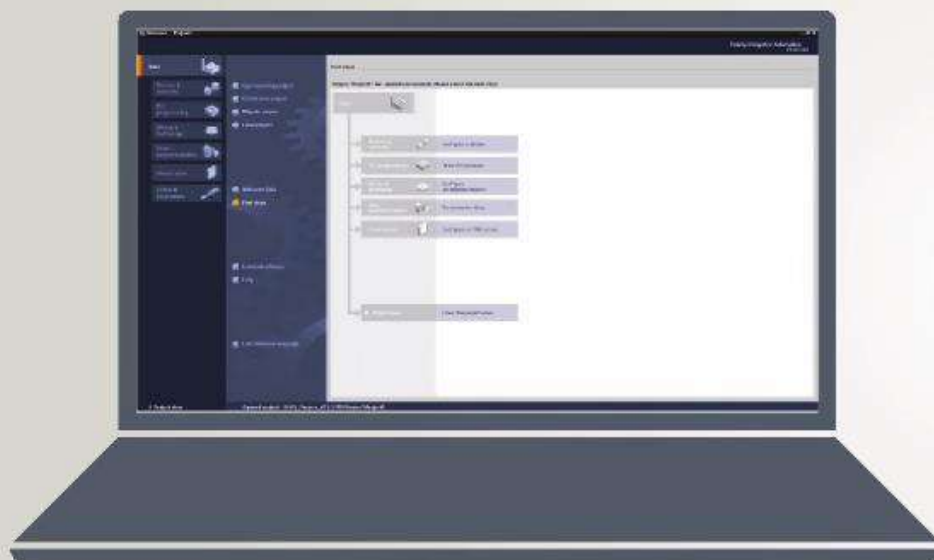
(ส่วนประกอบของโปรแกรม)

1.1 System requirements "SIMATIC STEP 7 Basic"

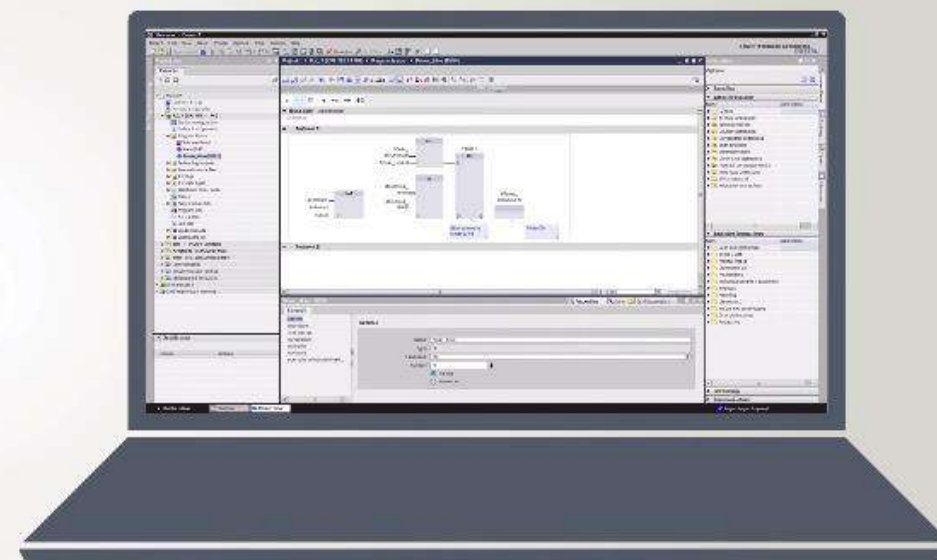
Hardware/software	Requirements
Processor type	Intel® Core™ i5-3320M 3.3 GHz or better
RAM	8 GB
Available hard disk space	2 GB on system drive C:\
Operating systems	You can use STEP 7 with the following operating systems (64-bit only): • Microsoft Windows 7 Home Premium SP1 or higher (STEP 7 Basic only, not supported for STEP 7 Professional) • Microsoft Windows 7 or higher (Professional SP1, Enterprise SP1, Ultimate SP1) • Microsoft Windows 8.1 (STEP 7 Basic only, not supported for STEP 7 Professional) • Microsoft Windows 8.1 (Professional, Enterprise) • Microsoft Server 2008 R2 StdE SP1 (STEP 7 Professional only) • Microsoft Server 2012 R2 StdE
Graphics card	32 MB RAM 24-bit color depth
Screen resolution	1920 x 1080 (recommended)
Network	10 Mbit/s Ethernet or faster, for communication between STEP 7 and the CPU
Optical drive	DVD-ROM

1.2 Software Environment : Portal View vs Project View

Totally Integrated Automation Portal

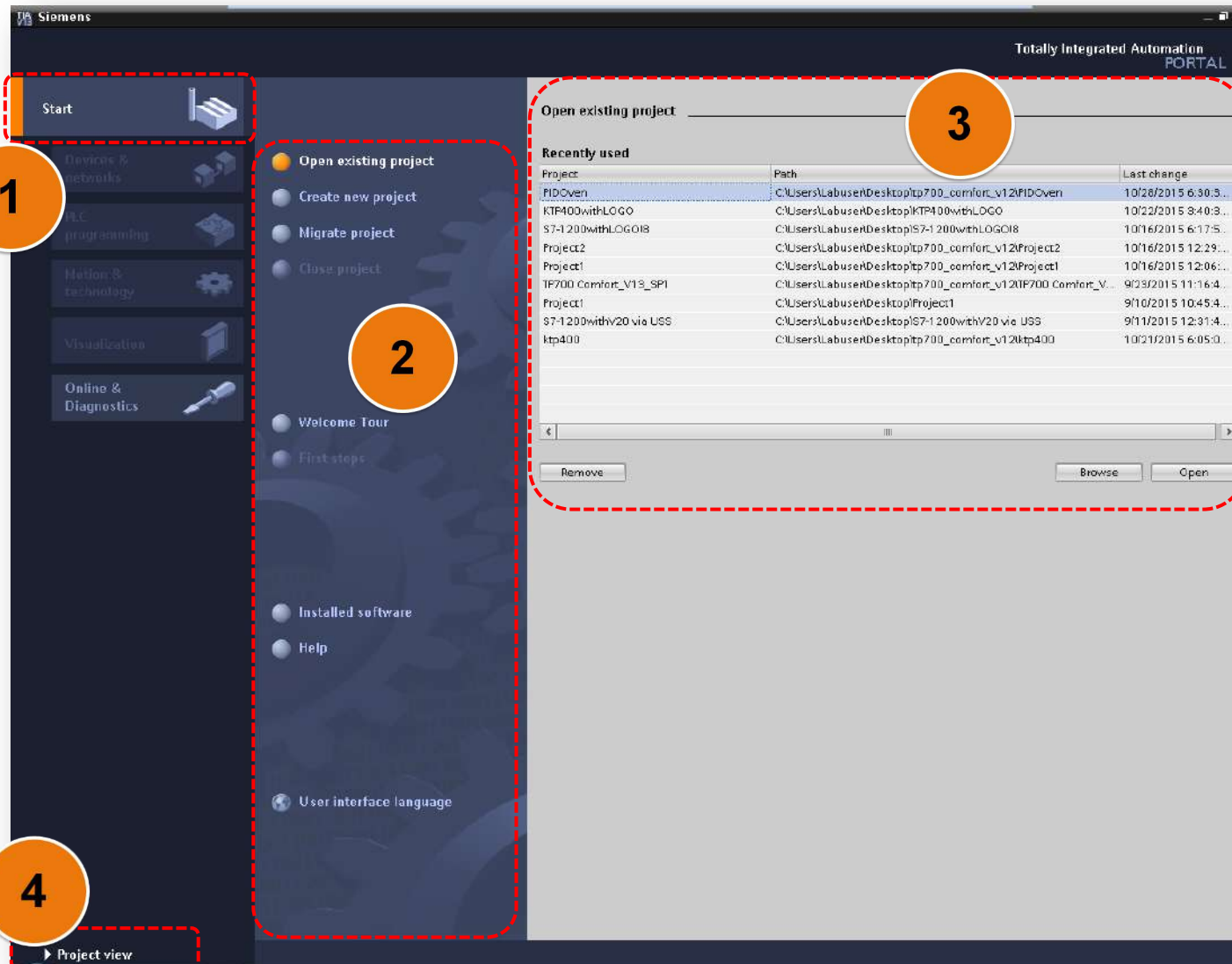


Portal View



Project View

1.2 Software Environment : Portal view



- 1) เมนูหลักเพื่อไปยังงาน (Task) ต่างๆ
- 2) งาน (Task) ที่เลือกมาจากเมนูหลัก
- 3) โปรเจกงานที่ต้องการเลือก
- 4) เปลี่ยนไปยัง มุมมองแบบโปรเจก

Project view / Portal view

1.2 Software Environment : Project view

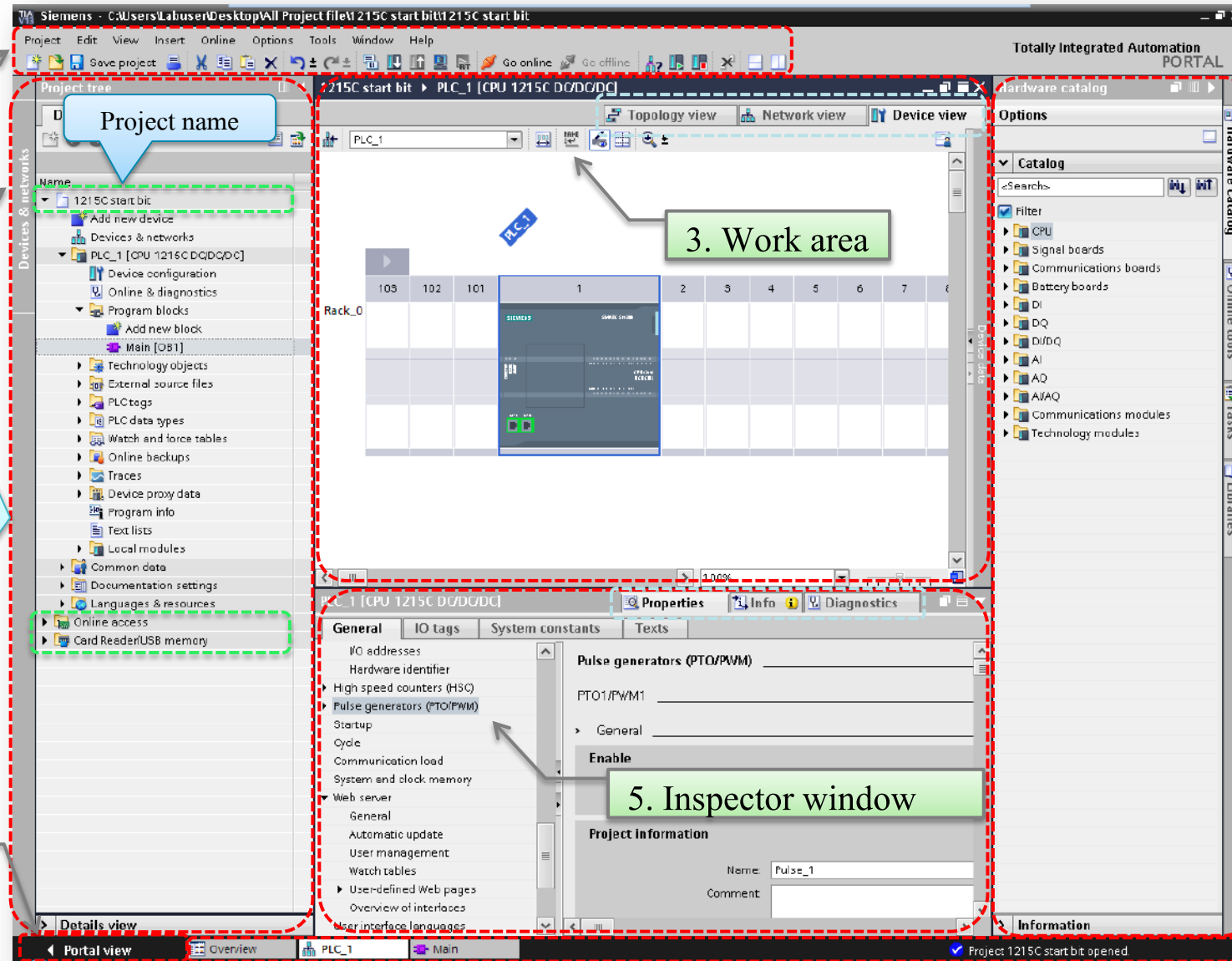
1. Menu bar &
Toolbar

2. Project navigator
(Project tree)

Project tree

- Devices
- Reference projects
- Details view

6. Changes to Portal view



4. Task cards

Task cards

- Hardware catalog
- Online tools
- Tasks
- Libraries

5. Inspector window

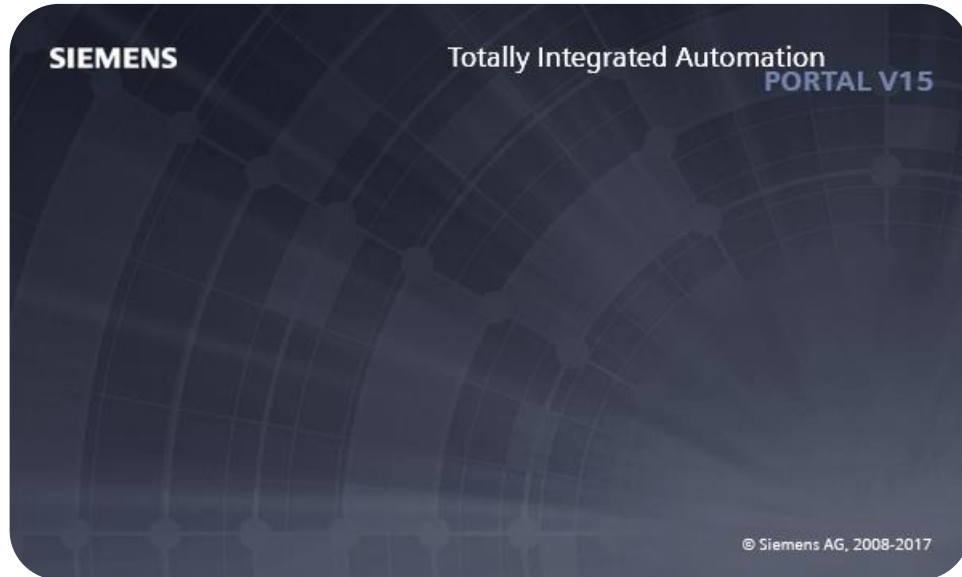
7. Editor bar

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PLC Concept & Programming Concept

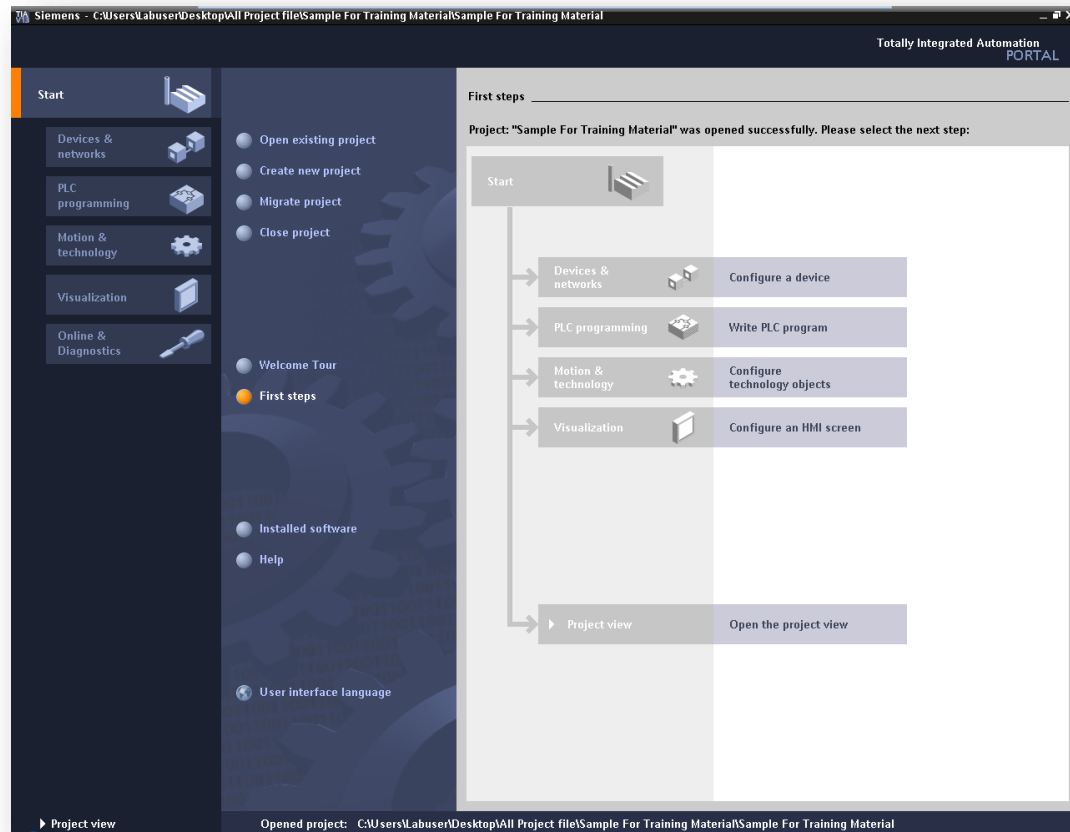
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Technical note

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Appendix

2. Programming Procedure



1. Create new project

2. Hardware Configuration

3. Declaring PLC Tags

4. Creating and Using Function Blocks

5. Compile Project

6. Download to PLC

7. Monitoring On/Off

8. Save Project

2.1 Create new project

The screenshot shows the Siemens TIA Portal interface. On the left, the 'Start' menu is visible with options like 'Devices & networks', 'PLC programming', 'Visualization', and 'Online & Diagnostics'. In the center, a list of actions includes 'Open existing project', 'Create new project' (highlighted with a red dashed box), 'Migrate project', 'Close project', 'Welcome Tour', and 'First steps'. On the right, the 'Create new project' dialog is open, showing fields for 'Project name' (Basic program), 'Path' (D:\Training project files\Basic_01), 'Author' (Chantid), and 'Comment' (Training project file for beginner). A 'Create' button is at the bottom right of the dialog. Three numbered annotations in Thai are present: 1. 'Create new project เพื่อสร้างโปรเจกใหม่ขึ้นมา' (Create new project to create a new project) pointing to the 'Create new project' button in the center menu. 2. 'ใส่ชื่อโปรเจก ที่เก็บบันทึกชื่อผู้สร้าง และคำกำกับ' (Enter project name, creator name, and remarks) pointing to the input fields in the dialog. 3. 'กด Create' (Press Create) pointing to the 'Create' button in the dialog.

1 Create new project เพื่อสร้างโปรเจกใหม่ขึ้นมา

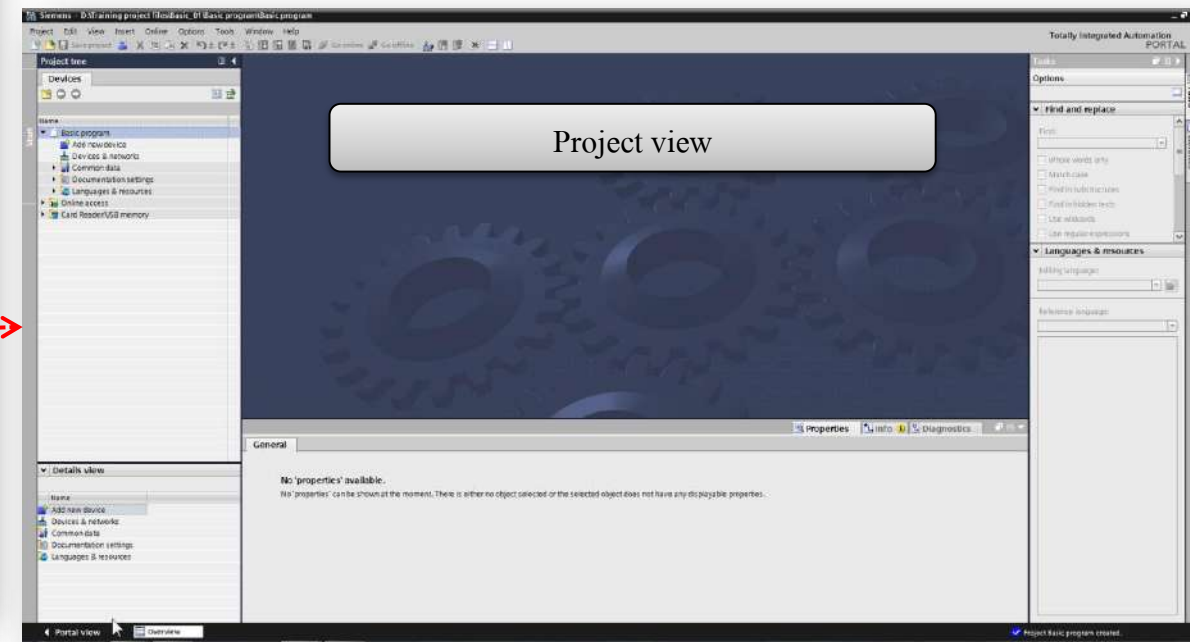
2 ใส่ชื่อโปรเจก ที่เก็บบันทึกชื่อผู้สร้าง และคำกำกับ

3 กด Create

2.1 Create new project



จากนั้นจะปรากฏหน้าต่าง First steps ขึ้นมาให้ท่าน
เลือก Open the Project View หรือ Project view ที่มุม
ล่างซ้ายของหน้าต่าง



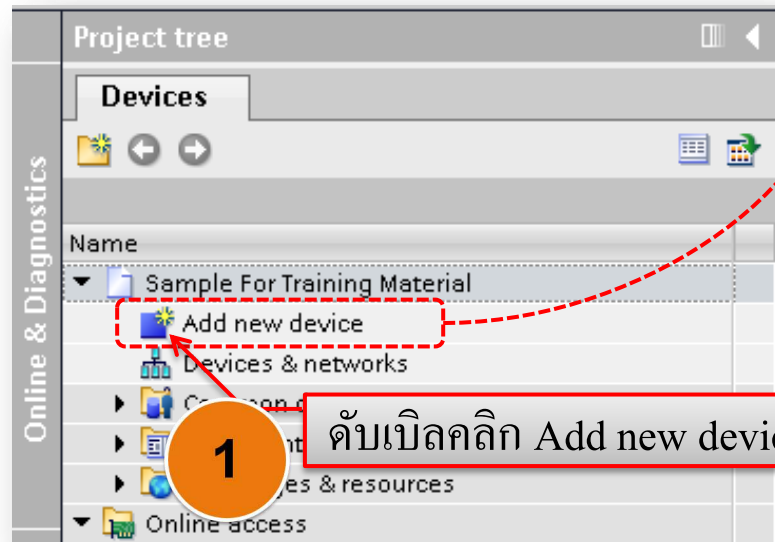
2.1 Create new project : Assign device name

Add new device

Assign device name

Select CPU model

Specify Firmware version



ดับเบิลคลิก Add new device

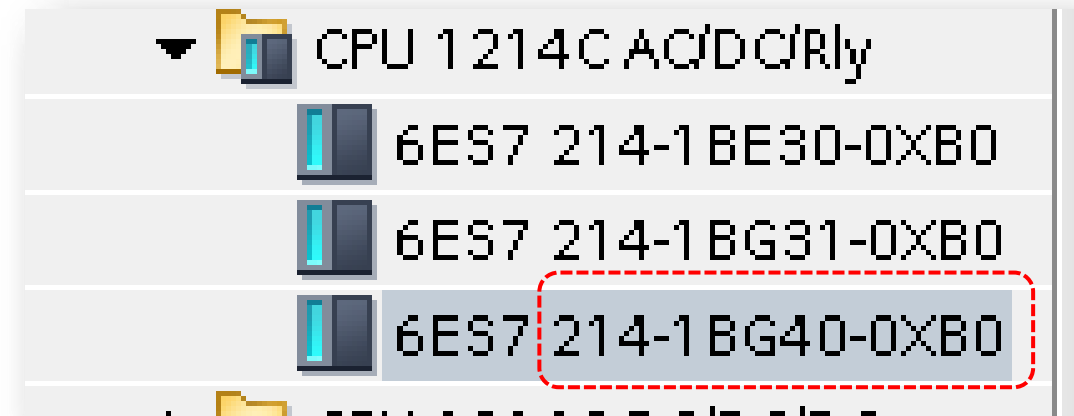
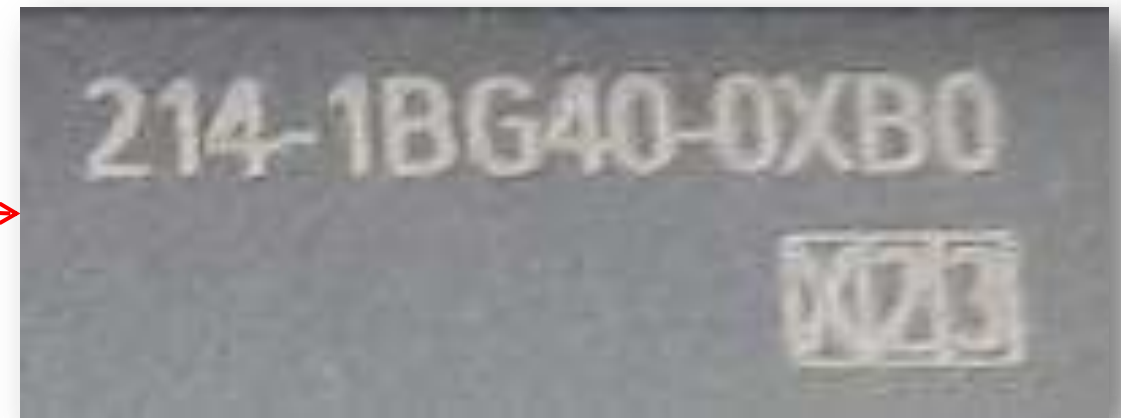
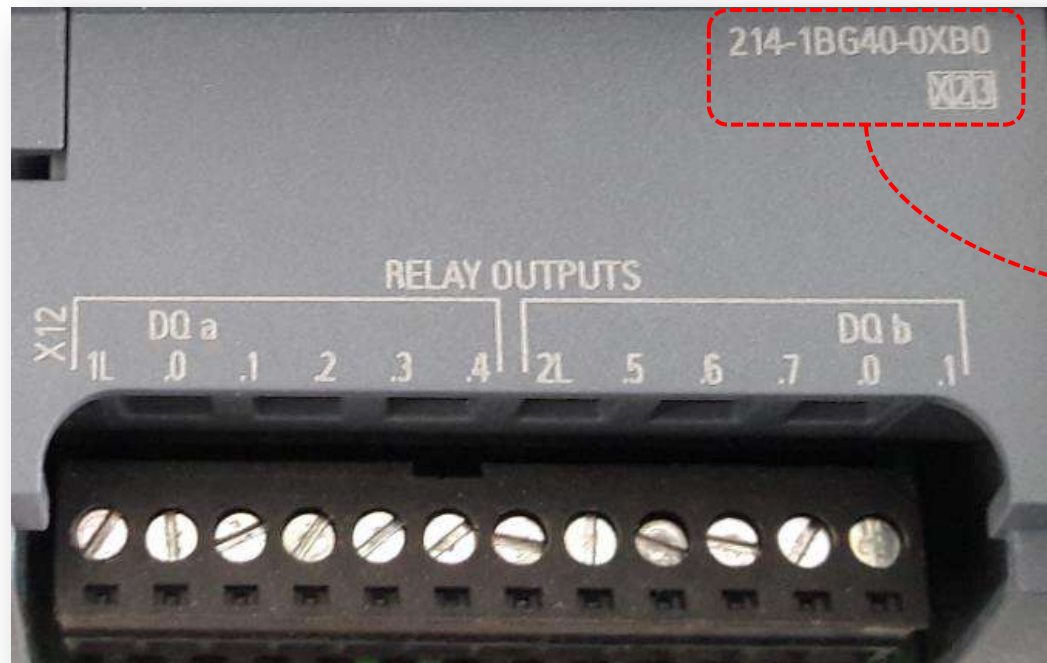


ตั้งชื่อให้แก่อุปกรณ์ (Device name)

เลือกรหัสของอุปกรณ์ให้ถูกต้อง
(ตัวอย่างการดูรหัสอุปกรณ์ในหน้าถัดไป)

2.1 Create the User Program (สร้างโปรเจกต์ไฟล์)

Device Article Number



2.2 Hardware Configuration : Using hardware catalog

คลิก Hardware catalog

6

Module	Slot	I address
PLC_Training_B01	1	
DI 8/DO 6_1	1.1	0
AI 2_1	1.2	64...67
HSC_1	1.16	1000...10...
HSC_2	1.17	1004...10...
HSC_3	1.18	1008...10...
HSC_4	1.19	1012...10...
HSC_5	1.20	1016...10...
HSC_6	1.21	1020...10...
Pulse_1	1.32	
Pulse_2	1.33	
Pulse_3	1.34	
Pulse_4	1.35	
PROFINET interface_1	1.X1	
	2	
	3	

จะปรากฏ CPU ที่ท่านเลือกขึ้นมา จากนั้นเลือกแถบ Hardware Catalog ในกรณีที่ท่านต้องการเพิ่ม module อื่นๆอีก เช่น Signal Module (SM) หรือ Communication Module (CM)

Siemens - DA Training project files\Basic_01\Basic program\Basic program

Project Edit View Insert Online Options Tools Window Help

Save project Go online Go offline Search in project

Project tree Basic program PLC_Training_801 [CPU 1212C AC/DC/Rly]

Devices

- Basic program
- Add new device
- Devices & networks
- PLC_Training_801 [CPU 1212C AC/DC/Rly]
- Device configuration
- Online & diagnostics
- Program blocks
- Technology objects
- External source files
- PLC tags
- PLC data types
- Watch and force tables
- Online backups
- Traces
- Device proxy data
- Program info
- PLC alarm text lists
- Local modules
- Ungrouped devices
- Common data
- Documentation settings
- Languages & resources
- Online access
- Card Reader/USB memory

Basic program PLC_Training_801 [CPU 1212C AC/DC/Rly]

Topology view Network view Device view

Device overview

Module	Slot	I address	Q address
PLC_Training_801	1		
DI 8/DQ 6_1	1.1	0	0
AI 2_1	1.2	64...67	
	1.3		
HSC_1	1.16	1000...10...	
HSC_2	1.17	1004...10...	
HSC_3	1.18	1008...10...	
HSC_4	1.19	1012...10...	
HSC_5	1.20	1016...10...	
HSC_6	1.21	1020...10...	
Pulse_1	1.32		
Pulse_2	1.33		
Pulse_3	1.34		
Pulse_4	1.35		
PROFINET interface_1	1.X1		
	2		
	3		

Hardware catalog

Options

Catalog

Filter Profile: <All>

- CPU
- Signal boards
- Communications boards
- Battery boards
- DI
- DQ
- DI/DQ
- AI
- AO
- AI/AO
- Communications modules
- Technology modules

Details view

Module

General Cross-references Compile Generate

Show all messages

Message

Message	Go to	Date	Time
Project Basic program created.		4/20/2017	2:58:16 PM

Portal view Overview PLC_Training...

คลิก Hide Device overview

7

Show

Hide

2.2 Hardware Configuration : Using hardware catalog

Add Signal Modules and Communications Modules

Communication module

Board

Signal module

Rack_0

103 102 101 1 2 3

SIEMENS SIMATIC 57-1000

CPU 1212C AC/DC/Rly

Hardware catalog

Options

Catalog

Search>

Filter Profile: <All>

- CPU
- Signal boards
- Communications boards
- Battery boards
- DI
- DQ
- DI/DQ
 - DI 8/DQ 8x24VDC
 - DI 16/DQ 16x24VDC
 - DI 8x24VDC/DQ 8xRelay
 - DI 16x24VDC/DQ 16xRelay
 - 6ES7 223-1PL30-0XB0
 - 6ES7 223-1PL32-0XB0
 - DI/DQ 8x120VAC/DQ 8xRelay
- AI
- AQ
- AI/AQ
- Communications modules
- Technology modules

8

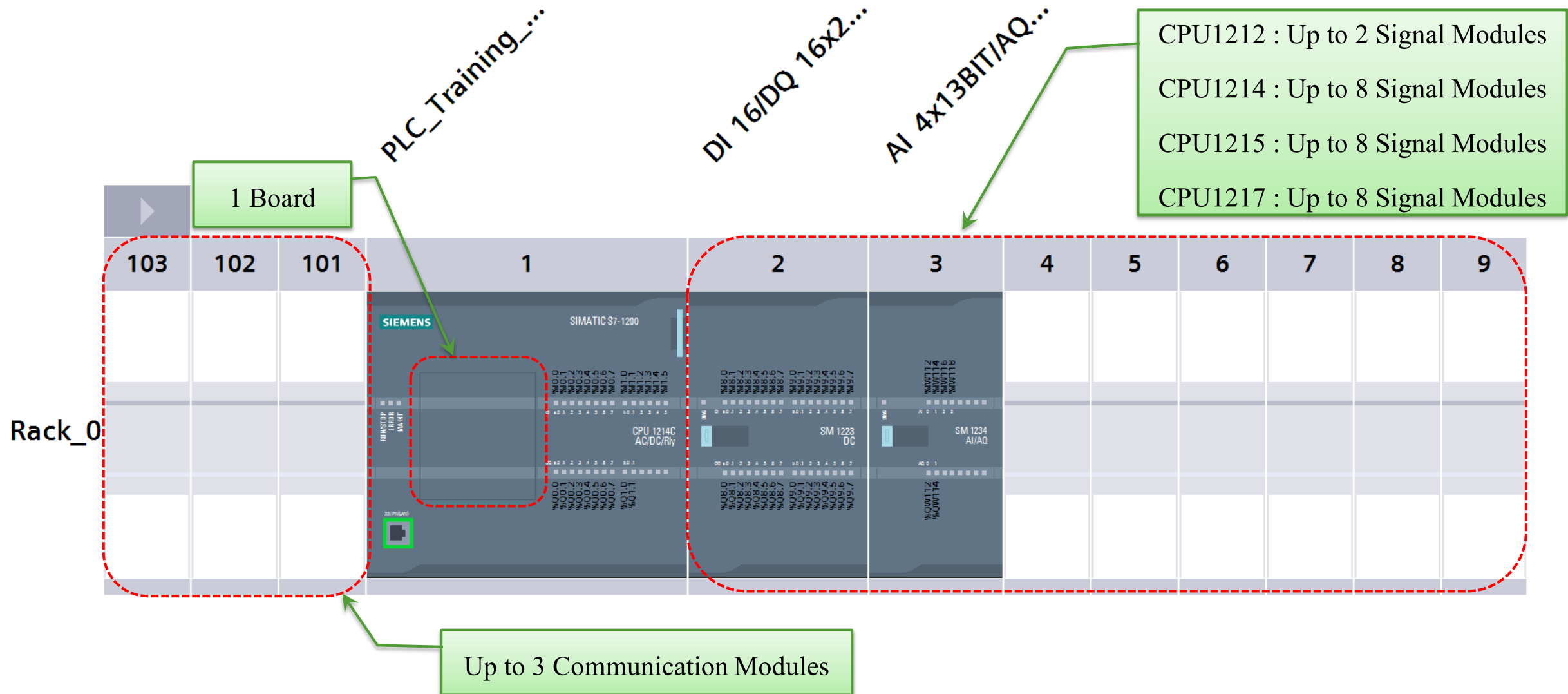
Drag & Drop

9

Drag & Drop ลงใน Device view จนครบตามที่ต้องการใช้งาน

เลือก Signal Module และ Communication Module ที่ท่านต้องการใช้งาน

2.2 Hardware Configuration



2.2 Hardware Configuration : Upload from the unit

The image illustrates the steps to create a new project and add a new device in Siemens TIA Portal. The steps are numbered 1 through 9.

Step 1: In the TIA Portal main window, click on the **Create new project** button in the left sidebar.

Step 2: In the **Create new project** dialog, enter the project name, path, author, and comment. Click the **Create** button.

Step 3: In the **Add new device** dialog, click on the **Controllers** icon in the left sidebar.

Step 4: Enter the device name **PLC_Training_B01_01** in the **Device name** field.

Step 5: In the **Controllers** tree, expand **SIMATIC 57-1200** and select **CPU**.

Step 6: In the **CPU** list, select **Unspecified CPU 1200**.

Step 7: In the **Unspecified CPU 1200** list, select the article number **6ES7 2XX-XXXXX-XXXX**.

Step 8: In the **Device** section, enter the article number **6ES7 2XX-XXXXX-XXXX** and the version **V4.2**.

Step 9: Click the **OK** button to confirm the device selection.

The final screenshot shows the **Unspecified CPU 1200** device selected in the hardware catalog, with the article number **6ES7 2XX-XXXXX-XXXX** highlighted.

2.2 Hardware Configuration : Upload from the unit

Basic program_upload hw config ▶ PLC_Training_B01_01 [Unspecific CPU 1200]

PLC_Training_B01_01

Rack_0

1

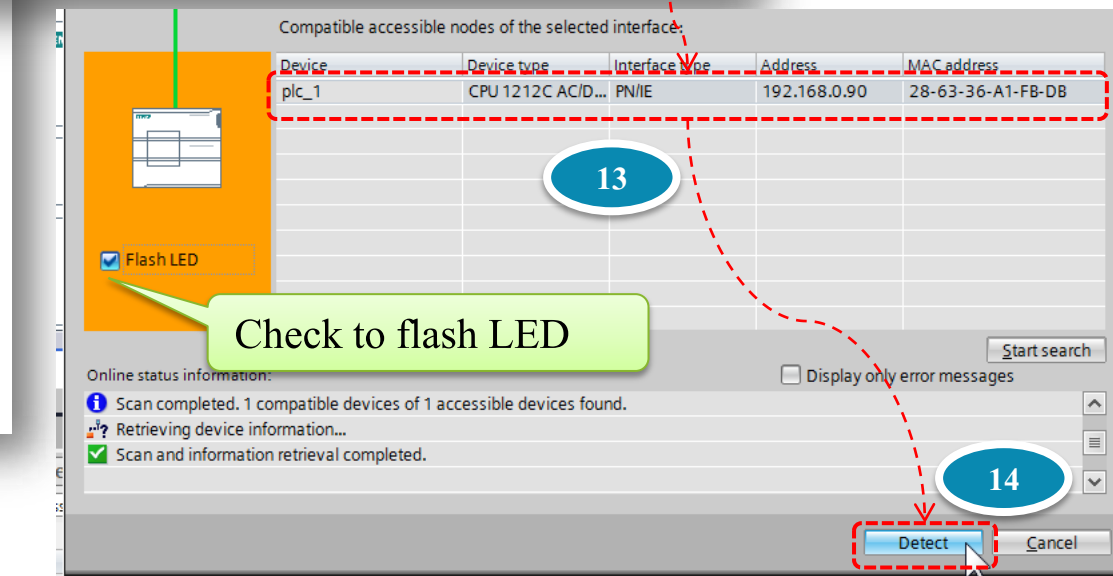
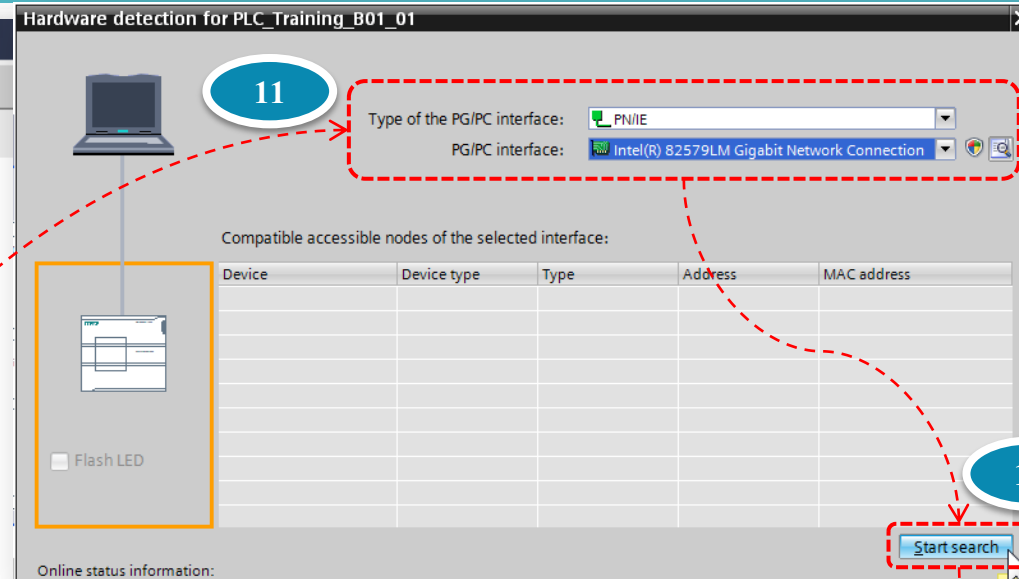
PLC_Training...

The device is not specified.

- Please use the [Hardware catalog](#) to specify the CPU,
- or **detect** the configuration of the connected device.

10

→ or **detect** t



2.2 Hardware Configuration : Setting IP address and Subnet mask

Double click to open Device view

Click ที่ CPU

ระบุ IP address ที่เราต้องการลงไปพร้อมทั้ง subnet mask

ระบุ IP address ของ Router ลงไป กรณีที่ใช้ Router

1: Double-click on the CPU in the Rack diagram to open the Properties window.

2: Select the 'Properties' tab in the Properties window.

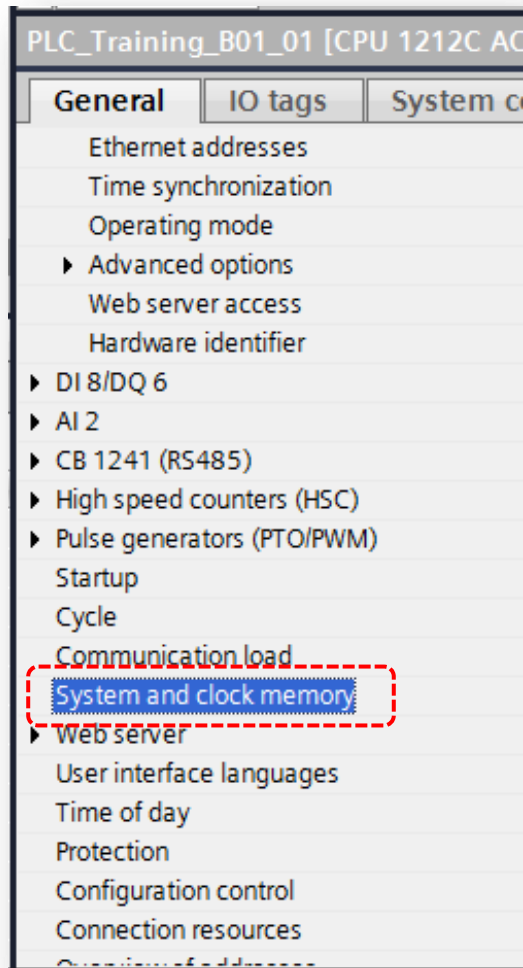
3: Select the 'General' tab in the Properties window.

4: Select the 'Ethernet addresses' section in the Properties window.

5: Set the IP address and Subnet mask in the 'Interface networked with' section.

6: Set the IP address and Subnet mask in the 'IP protocol' section.

2.2 System and clock memory



System memory bits

☒ Enable the use of system memory byte

Address of system memory byte (MBx): 1

First cycle: %M1.0 (FirstScan)

Diagnostic status changed: %M1.1 (DiagStatusUpdate)

Always 1 (high): %M1.2 (AlwaysTRUE)

Always 0 (low): %M1.3 (AlwaysFALSE)

Clock memory bits

☒ Enable the use of clock memory byte

Address of clock memory byte (MBx): 0

10 Hz clock: %M0.0 (Clock_10Hz)

5 Hz clock: %M0.1 (Clock_5Hz)

2.5 Hz clock: %M0.2 (Clock_2.5Hz)

2 Hz clock: %M0.3 (Clock_2Hz)

1.25 Hz clock: %M0.4 (Clock_1.25Hz)

1 Hz clock: %M0.5 (Clock_1Hz)

0.625 Hz clock: %M0.6 (Clock_0.625Hz)

0.5 Hz clock: %M0.7 (Clock_0.5Hz)

System and clock memory เป็น memory ที่ทำงานเป็นค่าคงที่ของระบบ เช่น M1.2 จะ ON ตลอดเวลา และ M1.3 จะ OFF ตลอดเวลาเป็นต้น ส่วน Clock memory ก็จะทำให้ Output เป็น สัญญาณนาฬิกาที่มีความถี่ต่าง ๆ กัน ดังรูป

Clock memory

Table 5- 20 Clock memory

Bit number	7	6	5	4	3	2	1	0
Tag name								
Period (s)	2.0	1.6	1.0	0.8	0.5	0.4	0.2	0.1
Frequency (Hz)	0.5	0.625	1	1.25	2	2.5	5	10

Because clock memory runs asynchronously to the CPU cycle, the status of the clock memory can change several times during a long cycle.

10 Hz clock:	%M0.0 (Clock_10Hz)
5 Hz clock:	%M0.1 (Clock_5Hz)
2.5 Hz clock:	%M0.2 (Clock_2.5Hz)
2 Hz clock:	%M0.3 (Clock_2Hz)
1.25 Hz clock:	%M0.4 (Clock_1.25Hz)
1 Hz clock:	%M0.5 (Clock_1Hz)
0.625 Hz clock:	%M0.6 (Clock_0.625Hz)
0.5 Hz clock:	%M0.7 (Clock_0.5Hz)

10 Hz	100 ms
5 Hz	200 ms
2.5 Hz	400 ms
2 Hz	500 ms
1.25 Hz	800 ms
1 Hz	1 Sec
0.625 Hz	1.6 Sec
0.5 Hz	2 Sec

2.3 Declaring PLC Tags

ท่านสามารถกำหนดชื่อตัวแปร Tags ต่างๆ ในแถบ IO tags ของ
ตัวอุปกรณ์ต่างๆ

Basic program ▶ PLC_Training_B01_01 [CPU 1212C AC/DC/Rly]

Topology view Network view Device view

PLC_Training_... DI 16x24VDCI... AI 4x13BIT/AQ...

Rack_0

103 102 101 1 2 3

SIEMENS SIMATIC 37-1300

CPU 1212C AC/DC Rly

DI 16x24VDCI

AI 4x13BIT/AQ

1

Type

Tag table

Absolute address

Symbolic name

ระบุนาม / symbol
ของแต่ละ I/O

2

General IO tags System constants Texts

Name	Type	Address	Tag table	Comment
CMD_Start_M1	Bool	%I0.0	Default tag table	
	Bool	%I0.1		
	Bool	%I0.2		
CMD_Stop_M1	Bool	%I0.3	Default tag table	
	Bool	%I0.4		
	Bool	%I0.5		
	Bool	%I0.6		
	Bool	%I0.7		
Run_M1	Bool	%Q0.0	Default tag table	
	Bool	%Q0.1		
	Bool	%Q0.2		
	Bool	%Q0.3		
	Bool	%Q0.4		
	Bool	%Q0.5		

PLC_Training_... DI 16x24VDCI... AI 4x13BIT/AQ...

Rack_0

103 102 101 1 2 3 4

SIEMENS SIMATIC 37-1300

CPU 1212C AC/DC Rly

DI 16x24VDCI

AI 4x13BIT/AQ

DI 16x24VDCI_1 [Module]

Name	Type	Address	Tag table
	Bool	%I0.0	
	Bool	%I0.1	
	Bool	%I0.2	
	Bool	%I0.3	
	Bool	%I0.4	
	Bool	%I0.5	
	Bool	%I0.6	
	Bool	%I0.7	
	Bool	%I0.8	
	Bool	%I0.9	
	Bool	%I1.0	
	Bool	%I1.1	
	Bool	%I1.2	
	Bool	%I1.3	
	Bool	%I1.4	
	Bool	%I1.5	
	Bool	%I1.6	
	Bool	%I1.7	
	Bool	%I1.8	
	Bool	%I1.9	
	Bool	%I2.0	
	Bool	%I2.1	
	Bool	%I2.2	
	Bool	%I2.3	
	Bool	%I2.4	
	Bool	%I2.5	
	Bool	%I2.6	
	Bool	%I2.7	
	Bool	%I2.8	
	Bool	%I2.9	
	Bool	%I3.0	
	Bool	%I3.1	
	Bool	%I3.2	
	Bool	%I3.3	
	Bool	%I3.4	
	Bool	%I3.5	
	Bool	%I3.6	
	Bool	%I3.7	
	Bool	%I3.8	
	Bool	%I3.9	
	Bool	%I4.0	
	Bool	%I4.1	
	Bool	%I4.2	
	Bool	%I4.3	
	Bool	%I4.4	
	Bool	%I4.5	
	Bool	%I4.6	
	Bool	%I4.7	
	Bool	%I4.8	
	Bool	%I4.9	
	Bool	%I5.0	
	Bool	%I5.1	
	Bool	%I5.2	
	Bool	%I5.3	
	Bool	%I5.4	
	Bool	%I5.5	
	Bool	%I5.6	
	Bool	%I5.7	
	Bool	%I5.8	
	Bool	%I5.9	
	Bool	%I6.0	
	Bool	%I6.1	
	Bool	%I6.2	
	Bool	%I6.3	
	Bool	%I6.4	
	Bool	%I6.5	
	Bool	%I6.6	
	Bool	%I6.7	
	Bool	%I6.8	
	Bool	%I6.9	
	Bool	%I7.0	
	Bool	%I7.1	
	Bool	%I7.2	
	Bool	%I7.3	
	Bool	%I7.4	
	Bool	%I7.5	
	Bool	%I7.6	
	Bool	%I7.7	
	Bool	%I7.8	
	Bool	%I7.9	
	Bool	%I8.0	
	Bool	%I8.1	
	Bool	%I8.2	
	Bool	%I8.3	
	Bool	%I8.4	
	Bool	%I8.5	
	Bool	%I8.6	
	Bool	%I8.7	
	Bool	%I8.8	
	Bool	%I8.9	
	Bool	%I9.0	
	Bool	%I9.1	
	Bool	%I9.2	
	Bool	%I9.3	
	Bool	%I9.4	
	Bool	%I9.5	
	Bool	%I9.6	
	Bool	%I9.7	
	Bool	%I9.8	
	Bool	%I9.9	

AI 4x13BIT/AQ_1 [Module]

Name	Type	Address	Tag table
	Int	%AI0.12	
	Int	%AI0.14	
	Int	%AI0.16	
	Int	%AI0.18	
	Int	%AI0.20	
	Int	%AI0.22	
	Int	%AI0.24	
	Int	%AI0.26	
	Int	%AI0.28	
	Int	%AI0.30	
	Int	%AI0.32	
	Int	%AI0.34	
	Int	%AI0.36	
	Int	%AI0.38	
	Int	%AI0.40	
	Int	%AI0.42	
	Int	%AI0.44	
	Int	%AI0.46	
	Int	%AI0.48	
	Int	%AI0.50	
	Int	%AI0.52	
	Int	%AI0.54	
	Int	%AI0.56	
	Int	%AI0.58	
	Int	%AI0.60	
	Int	%AI0.62	
	Int	%AI0.64	
	Int	%AI0.66	
	Int	%AI0.68	
	Int	%AI0.70	
	Int	%AI0.72	
	Int	%AI0.74	
	Int	%AI0.76	
	Int	%AI0.78	
	Int	%AI0.80	
	Int	%AI0.82	
	Int	%AI0.84	
	Int	%AI0.86	
	Int	%AI0.88	
	Int	%AI0.90	
	Int	%AI0.92	
	Int	%AI0.94	
	Int	%AI0.96	
	Int	%AI0.98	
	Int	%AI1.00	
	Int	%AI1.02	
	Int	%AI1.04	
	Int	%AI1.06	
	Int	%AI1.08	
	Int	%AI1.10	
	Int	%AI1.12	
	Int	%AI1.14	
	Int	%AI1.16	
	Int	%AI1.18	
	Int	%AI1.20	
	Int	%AI1.22	
	Int	%AI1.24	
	Int	%AI1.26	
	Int	%AI1.28	
	Int	%AI1.30	
	Int	%AI1.32	
	Int	%AI1.34	
	Int	%AI1.36	
	Int	%AI1.38	
	Int	%AI1.40	
	Int	%AI1.42	
	Int	%AI1.44	
	Int	%AI1.46	
	Int	%AI1.48	
	Int	%AI1.50	
	Int	%AI1.52	
	Int	%AI1.54	
	Int	%AI1.56	
	Int	%AI1.58	
	Int	%AI1.60	
	Int	%AI1.62	
	Int	%AI1.64	
	Int	%AI1.66	
	Int	%AI1.68	
	Int	%AI1.70	
	Int	%AI1.72	
	Int	%AI1.74	
	Int	%AI1.76	
	Int	%AI1.78	
	Int	%AI1.80	
	Int	%AI1.82	
	Int	%AI1.84	
	Int	%AI1.86	
	Int	%AI1.88	
	Int	%AI1.90	
	Int	%AI1.92	
	Int	%AI1.94	
	Int	%AI1.96	
	Int	%AI1.98	
	Int	%AI2.00	
	Int	%AI2.02	
	Int	%AI2.04	
	Int	%AI2.06	
	Int	%AI2.08	
	Int	%AI2.10	
	Int	%AI2.12	
	Int	%AI2.14	
	Int	%AI2.16	
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	Int	%AI2.24	
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	Int	%AI2.92	
	Int	%AI2.94	
	Int	%AI2.96	
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	Int	%AI3.04	
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	Int	%AI3.40	
	Int	%AI3.42	
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	Int	%AI3.46	
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	Int	%AI3.52	
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	Int	%AI3.84	
	Int	%AI3.86	
	Int	%AI3.88	
	Int	%AI3.90	
	Int	%AI3.92	
	Int	%AI3.94	
	Int	%AI3.96	
	Int	%AI3.98	
	Int	%AI4.00	
	Int	%AI4.02	
	Int	%AI4.04	
	Int	%AI4.06	
	Int	%AI4.08	
	Int	%AI4.10	
	Int	%AI4.12	
	Int	%AI4.14	
	Int	%AI4.16	
	Int	%AI4.18	
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	Int	%AI4.26	
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	Int	%AI4.34	
	Int	%AI4.36	
	Int	%AI4.38	
	Int	%AI4.40	
	Int	%AI4.42	
	Int	%AI4.44	
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	Int	%AI4.48	
	Int	%AI4.50	
	Int	%AI4.52	
	Int	%AI4.54	
	Int	%AI4.56	
	Int	%AI4.58	
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	Int	%AI4.62	
	Int	%AI4.64	
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	Int	%AI4.68	
	Int	%AI4.70	
	Int	%AI4.72	
	Int	%AI4.74	
	Int	%AI4.76	
	Int	%AI4.78	
	Int	%AI4.80	
	Int	%AI4.82	
	Int	%AI4.84	
	Int	%AI4.86	
	Int	%AI4.88	
	Int	%AI4.90	
	Int	%AI4.92	
	Int	%AI4.94	
	Int	%AI4.96	
	Int	%AI4.98	
	Int	%AI5.00	
	Int	%AI5.02	
	Int	%AI5.04	
	Int	%AI5.06	
	Int	%AI5.08	
	Int	%AI5.10	
	Int	%AI5.12	
	Int	%AI5.14	
	Int	%AI5.16	
	Int	%AI5.18	
	Int	%AI5.20	
	Int	%AI5.22	
	Int	%AI5.24	
	Int	%AI5.26	
	Int	%AI5.28	
	Int	%AI5.30	
	Int	%AI5.32	
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	Int	%AI5.44	
	Int	%AI5.46	
	Int	%AI5.48	
	Int	%AI5.50	
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	Int	%AI5.58	
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	Int	%AI6.06	
	Int	%AI6.08	
	Int	%AI6.10	
	Int	%AI6.12	
	Int	%AI6.14	

PLC Tags table

Approved
Partner

Factory
Automation

SIEMENS

Siemens - D:\Training project files\Basic_01\Basic program\Basic program

Project Edit View Insert Online Options Tools Window Help

Save project <Search in project>

Project tree

Basic program ▶ PLC_Training_B01_01 [CPU 1212C AC/DC/Rly] ▶ PLC tags ▶ Default tag table [34]

Devices

PLC programming

Basic program

- Add new device
- Devices & networks
- PLC_Training_B01 [CPU 1212C AC/DC/R]
- PLC_Training_B01_01 [CPU 1212C AC/DC/R]

 - Device configuration
 - Online & diagnostics
 - Program blocks
 - Add new block
 - Main [OB1]
 - Technology objects
 - Energy objects
 - External source files
 - PLC tags
 - Show all tags
 - Add new tag table
 - Default tag table [34]
 - PLC data types

Default tag table

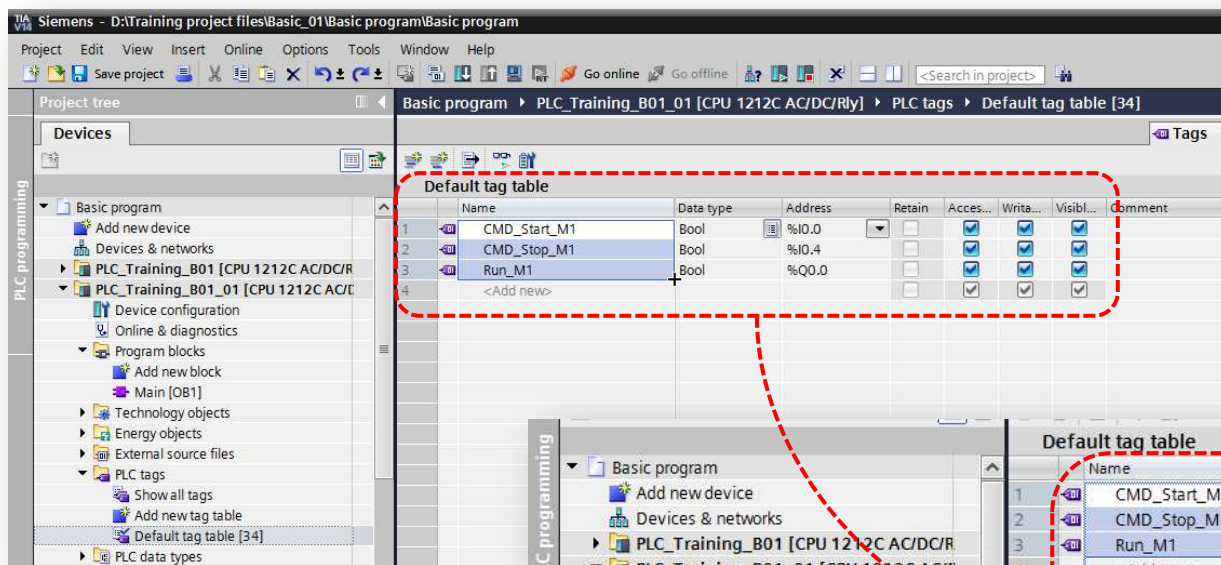
	Name	Data type	Address	Retain	Acces...	Writa...	Visibl...	Comment
1	CMD_Start_M1	Bool	%I0.0		☒	☒	☒	
2	CMD_Stop_M1	Bool	%I0.4		☒	☒	☒	
3	Run_M1	Bool	%Q0.0		☒	☒	☒	
4	<Add new>				☒	☒	☒	

PLC Tags table : Auto Fill

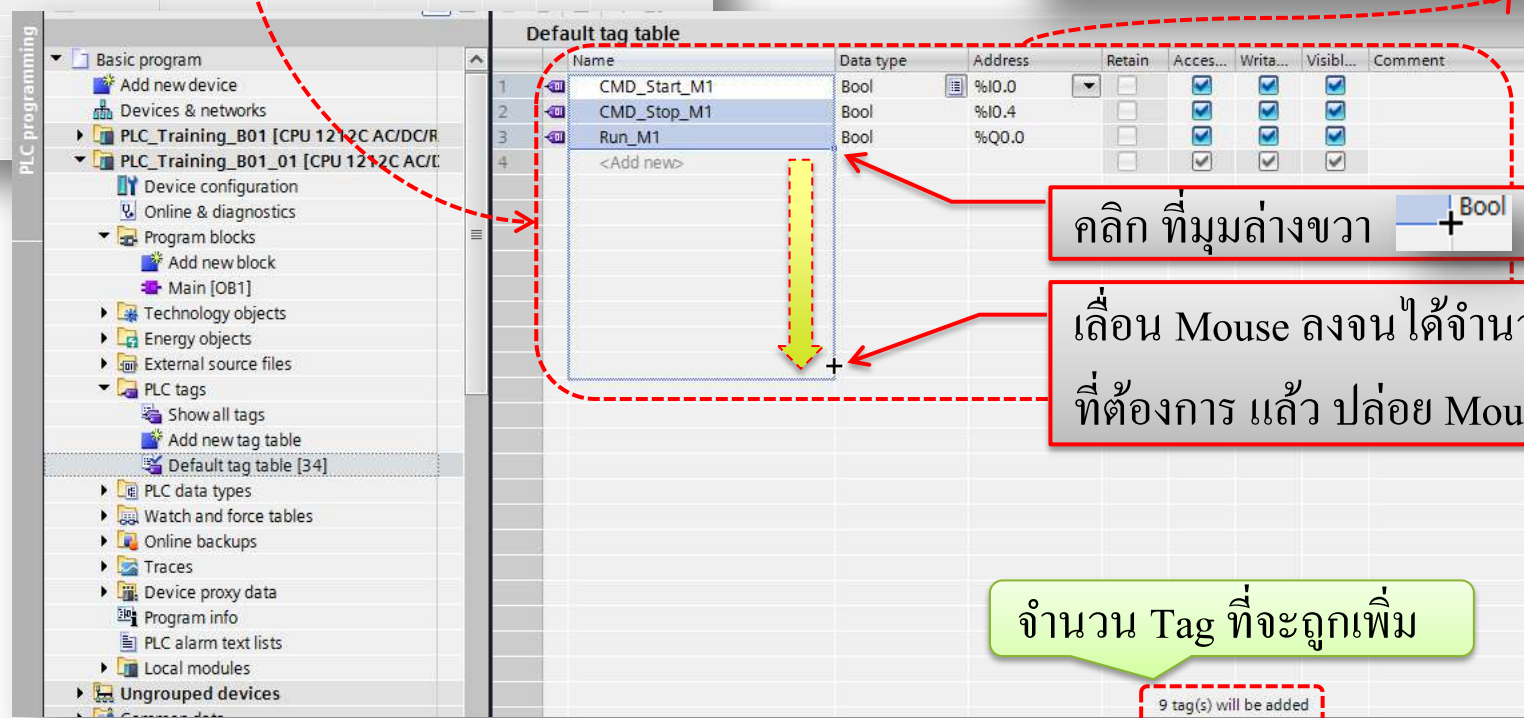
Approved
Partner

Factory
Automation

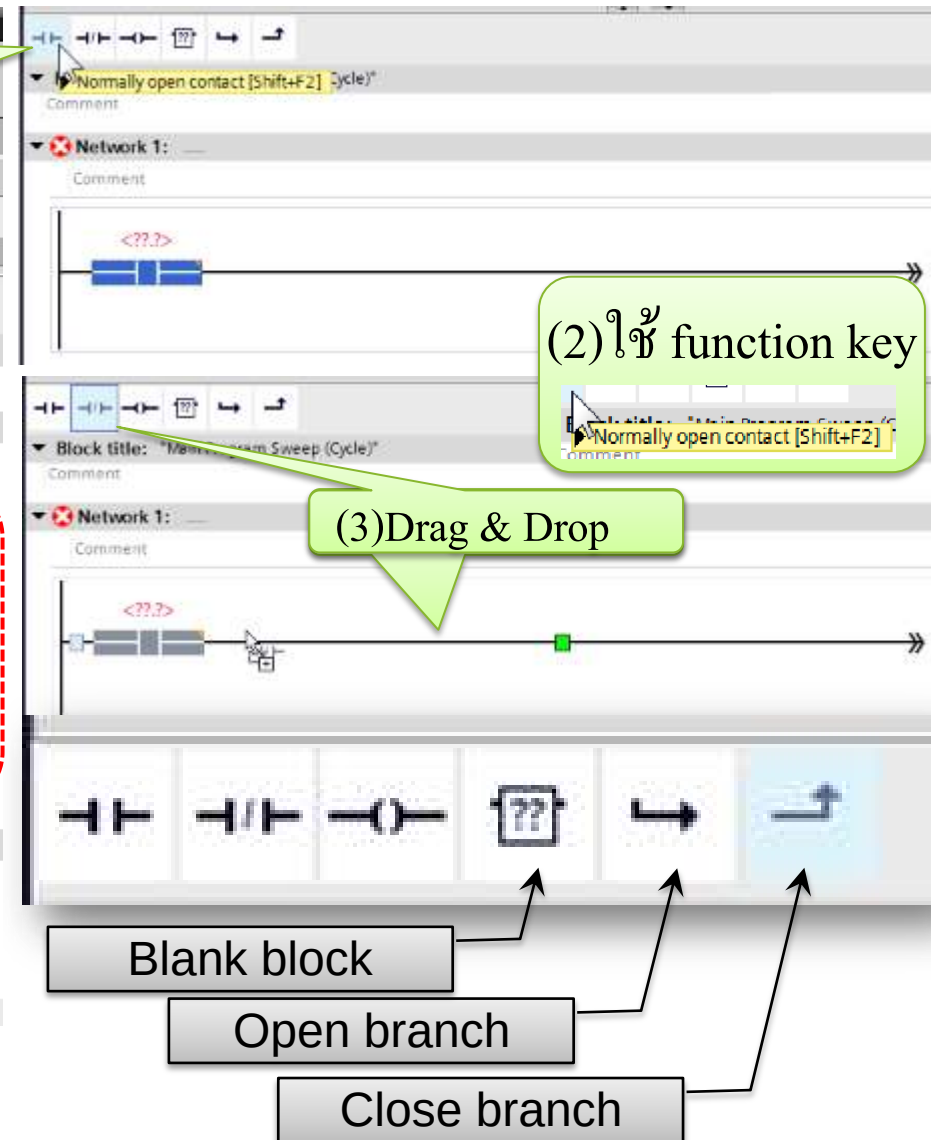
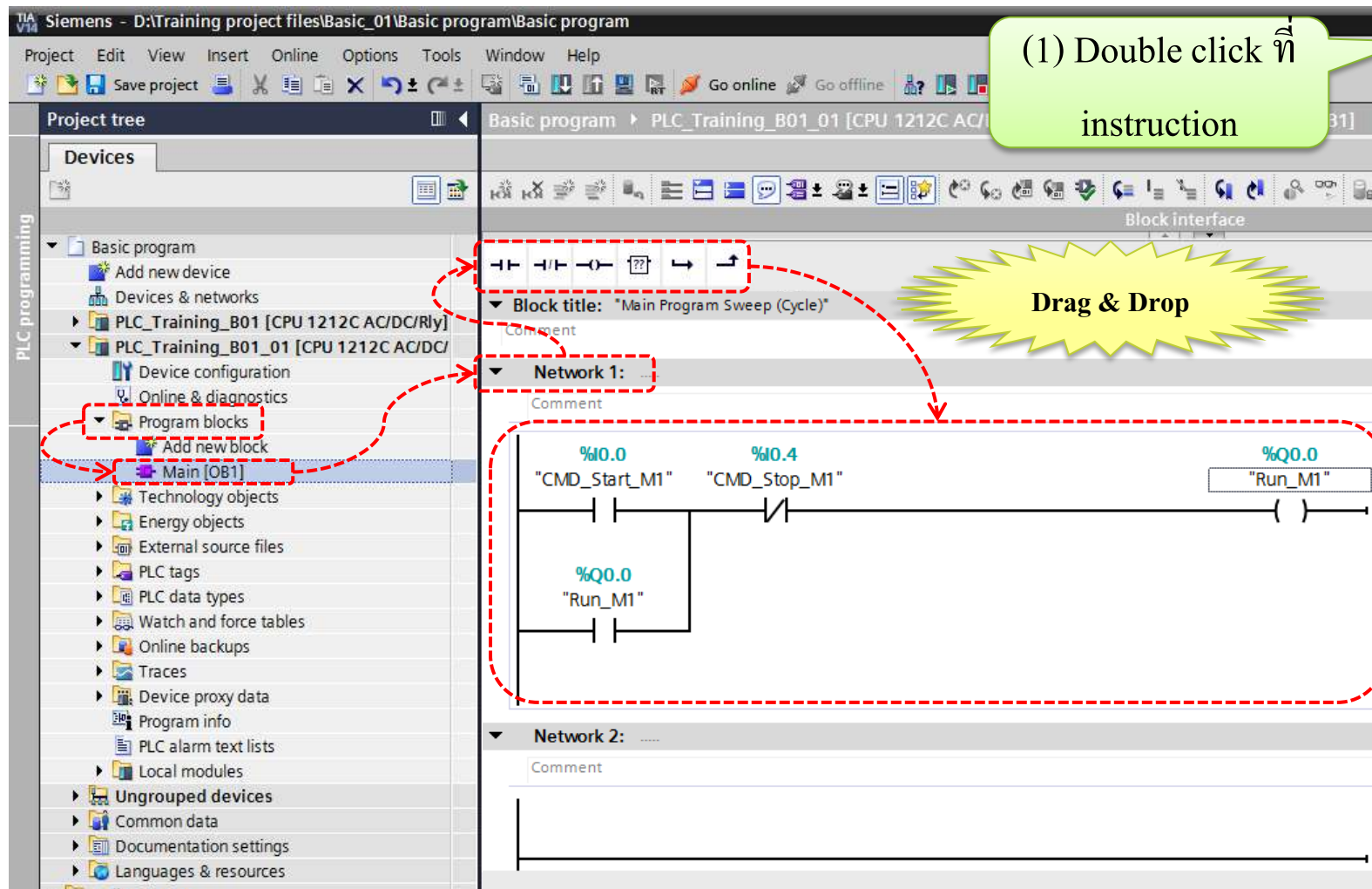
SIEMENS



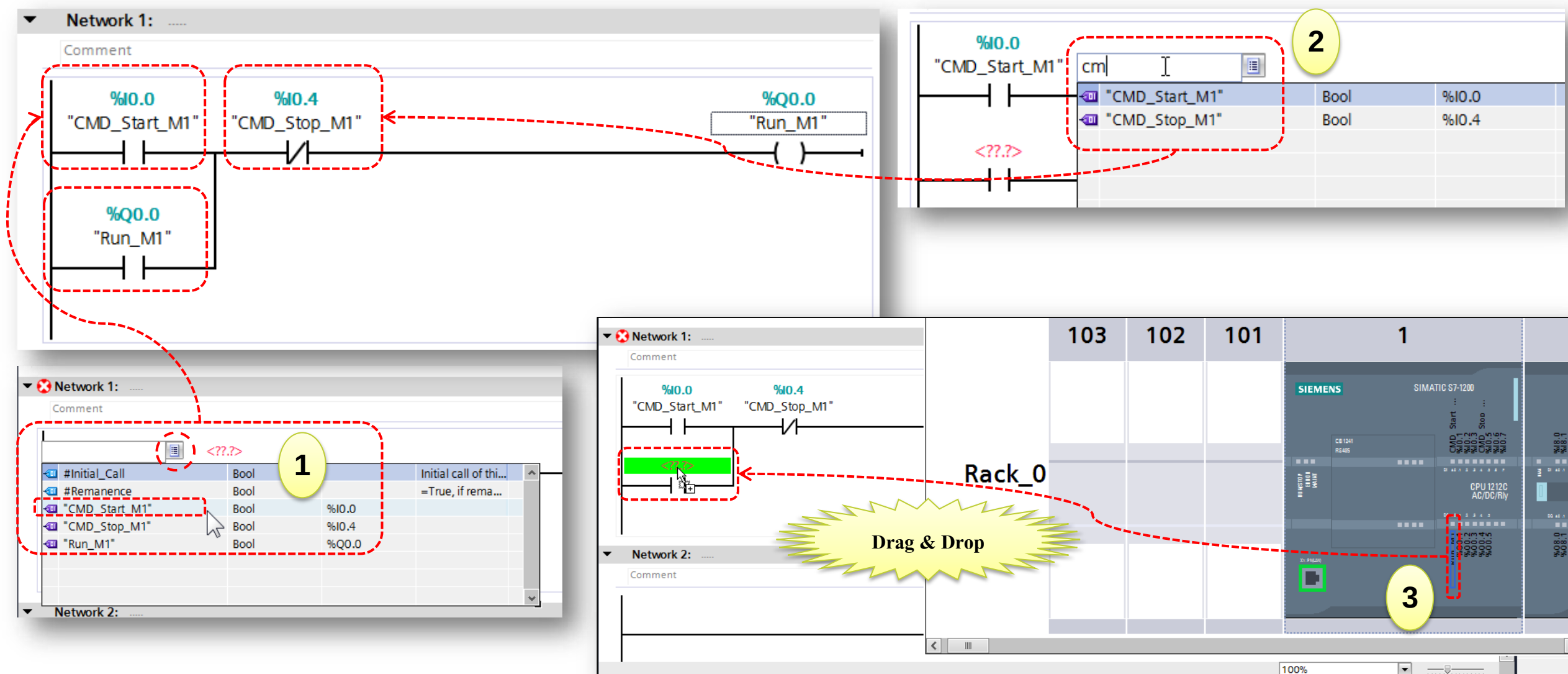
	Name	Data type	Address	Retain	Acces...	Writa...	Visibl...	C
1	CMD_Start_M1	Bool	%I0.0		☒	☒	☒	
2	CMD_Stop_M1	Bool	%I0.4		☒	☒	☒	
3	Run_M1	Bool	%Q0.0		☒	☒	☒	
4	CMD_Start_M2	Bool	%I0.1		☒	☒	☒	
5	CMD_Stop_M2	Bool	%I0.5		☒	☒	☒	
6	Run_M2	Bool	%Q0.1		☒	☒	☒	
7	CMD_Start_M3	Bool	%I0.2		☒	☒	☒	
8	CMD_Stop_M3	Bool	%I0.6		☒	☒	☒	
9	Run_M3	Bool	%Q0.2		☒	☒	☒	
10	CMD_Start_M4	Bool	%I0.3		☒	☒	☒	
11	CMD_Stop_M4	Bool	%I0.7		☒	☒	☒	
12	Run_M4	Bool	%Q0.3		☒	☒	☒	
13	<Add new>				☒	☒	☒	



2.4 Programming



Sample 1 : Self holding



2.5 Compile Project

Project Edit View Insert Online Options Tools Window Help

 Save project          Go online Go offline

กด Compile Project

1

แต่ถ้าเกิดข้อผิดพลาดขึ้น ท่านสามารถไปยังข้อผิดพลาดนั้นเพื่อแก้ไขได้โดย ให้กดลูกศร  ที่แถบ Go to เพราะถ้าไม่ทำการแก้ไขข้อผิดพลาด จะทำให้ไม่สามารถดาวน์โหลดโปรเจกต์ไฟล์ไปยัง CPU ได้

3

General  Cross-references Compile Syntax

   Show all messages

Compiling completed (errors: 0; warnings: 0)

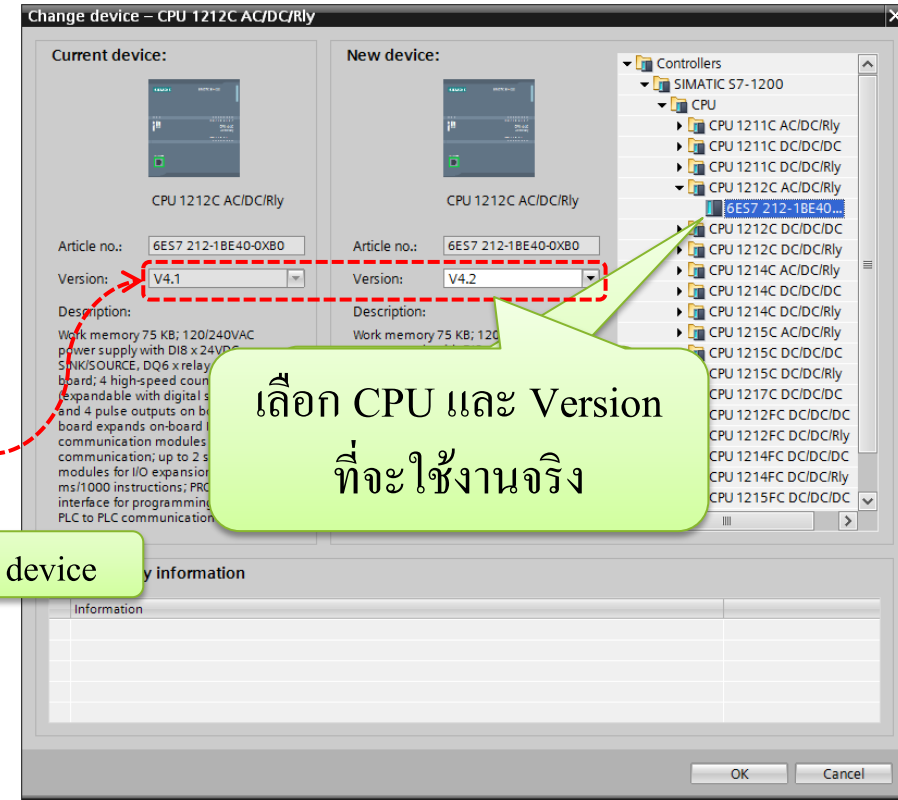
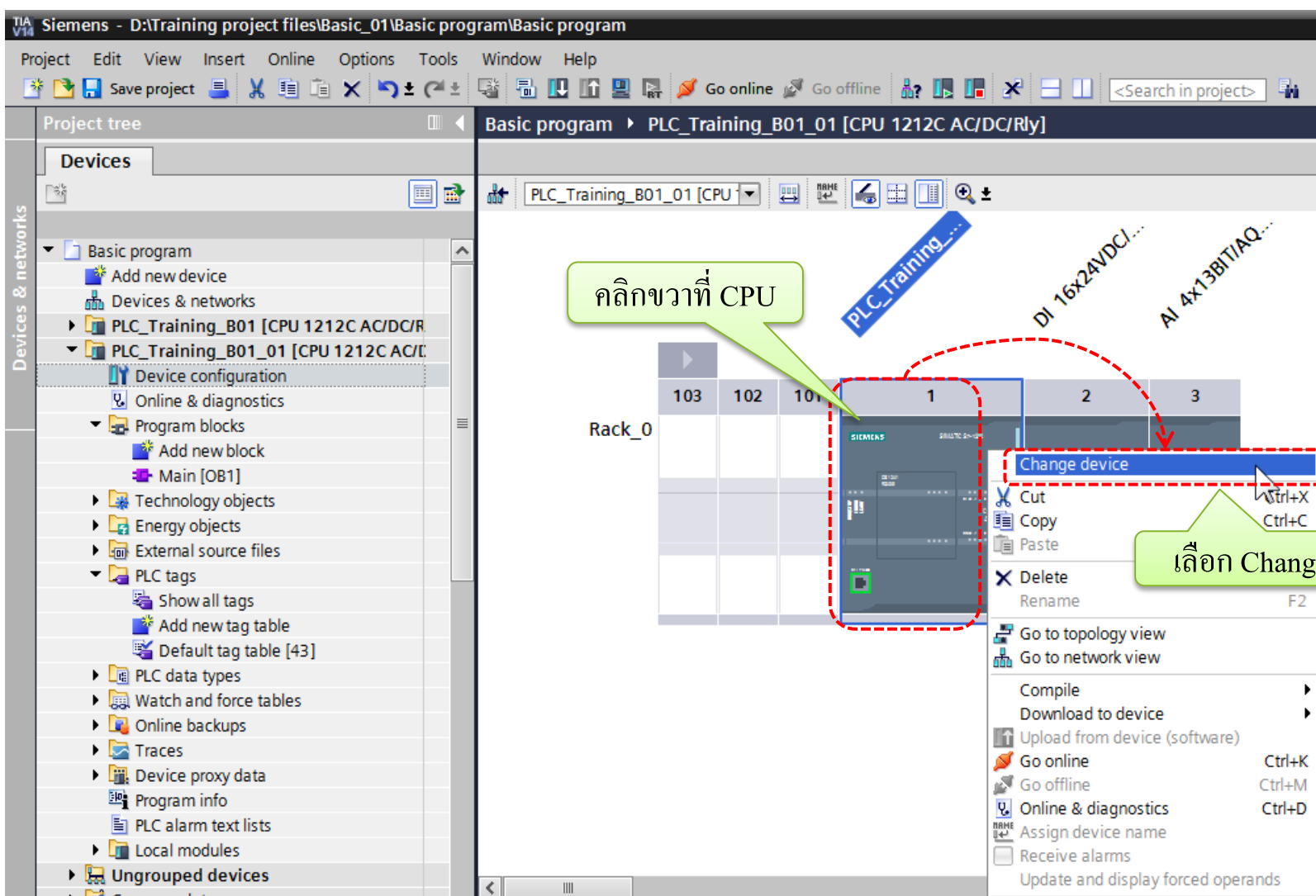
Path	Description	Go to	Errors	Warnings	Time
PLC_Training_BO1			0	0	1:58:14 PM
Program blocks			0	0	1:58:14 PM
Main (OB1)	Block was successfully compiled.				1:58:14 PM
	Compiling completed (errors: 0; warnings: 0)				1:58:16 PM

aining... Main Default tag t...

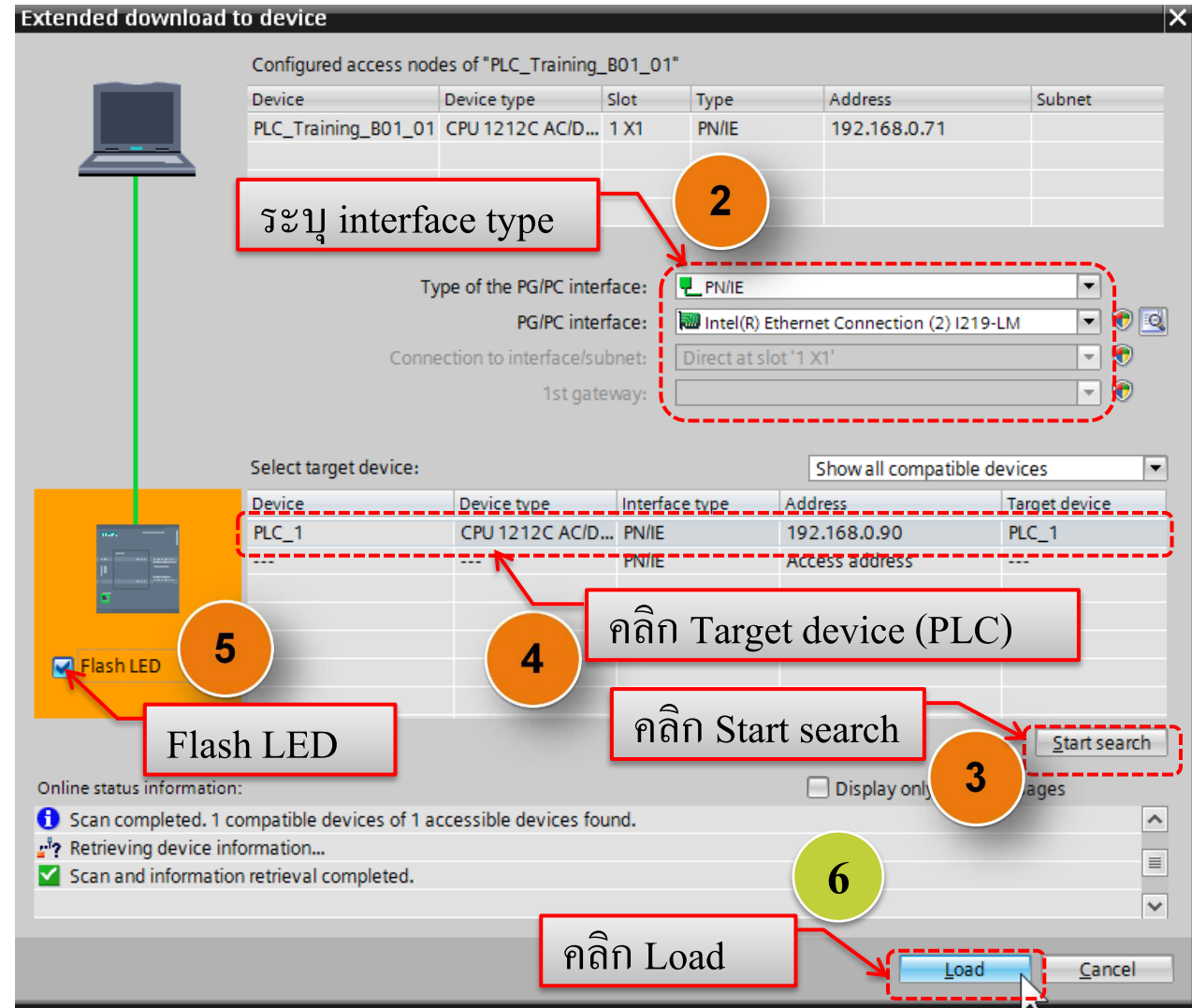
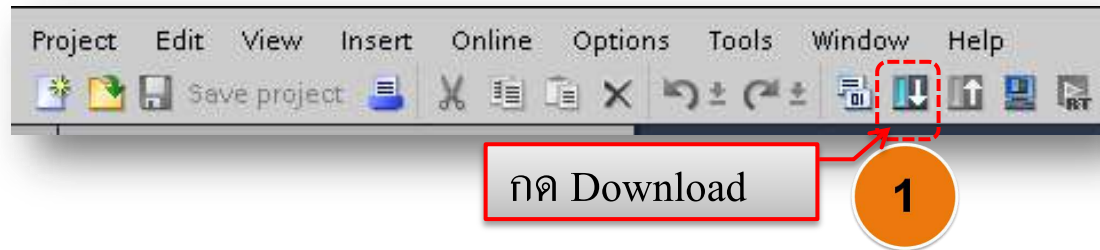
2

หลังจากกด Compile Project แล้วโปรแกรมจะประมวลผลว่าลอจิกที่เราเขียนนั้นมีข้อผิดพลาดอะไรหรือไม่ เมื่อประมวลผลเสร็จจะแสดงรายละเอียดในแถบ Info ในตัวอย่างลอจิกที่เขียนไม่พบข้อผิดพลาดอะไร

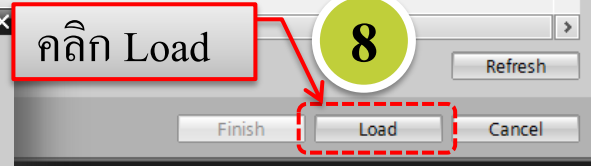
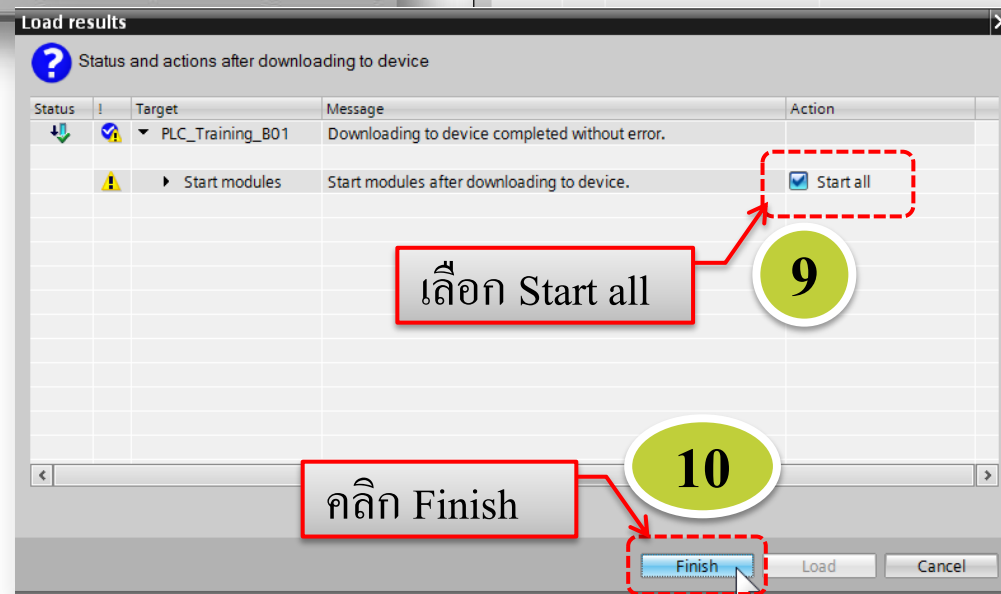
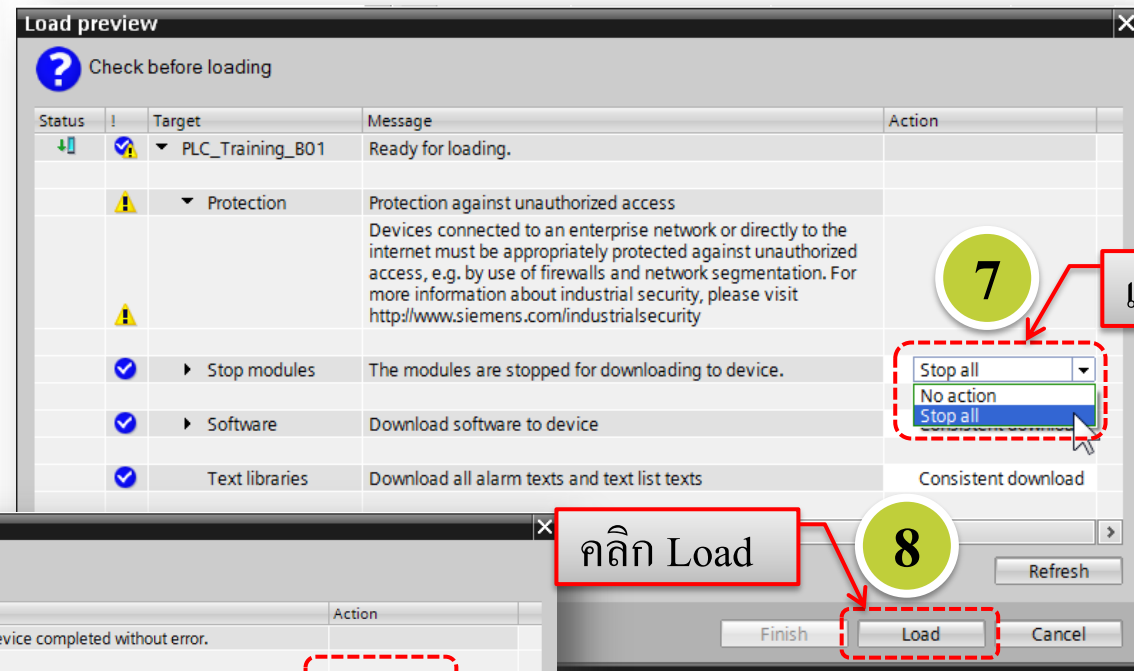
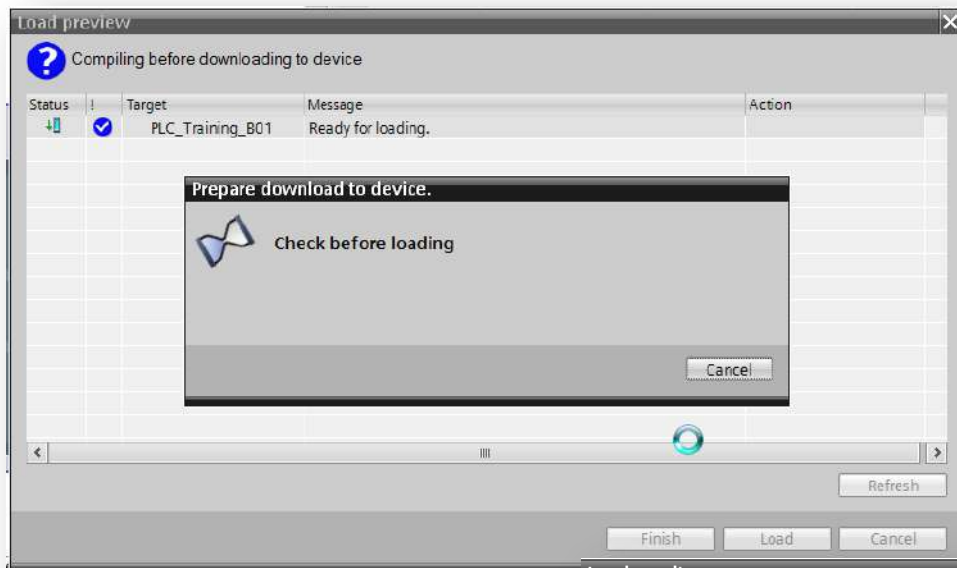
Change CPU device



2.6 Download to PLC



2.6 Download to PLC



2.7 Monitoring Online

Project Edit View Insert Online Options Tools Window Help

Save project Go online Go offline

1

Siemens - D:\Training project files\Basic_01\Basic program\Basic program

Project Edit View Insert Online Options Tools Window Help

Save project Go online Go offline

Project tree

Devices

Basic program

- Add new device
- Devices & networks
- PLC_Training_B01 [CPU 1212C AC/D
- PLC_Training_B01_01 [CPU 1212C A
- Device configuration
- Online & diagnostics
- Program blocks
 - Add new block
 - Main [OB1]
 - Technology objects
 - Energy objects
 - External source files
 - PLC tags
 - Show all tags
 - Add new tag table
 - Default tag table [43]
 - PLC data types
 - Watch and force tables
 - Online backups
 - Traces
 - Device proxy data
 - Program info
 - PLC alarm text lists
 - Local modules
 - Ungrouped devices
 - Common data

Basic program > PLC_Training_B01_01 [CPU 1212C AC/DC/Rly] > Program blocks > Main [OB1]

Block interface

Network 1:

Comment

%I0.0 "CMD_Start_M1" %I0.4 "CMD_Stop_M1" %Q0.0 "Run_M1"

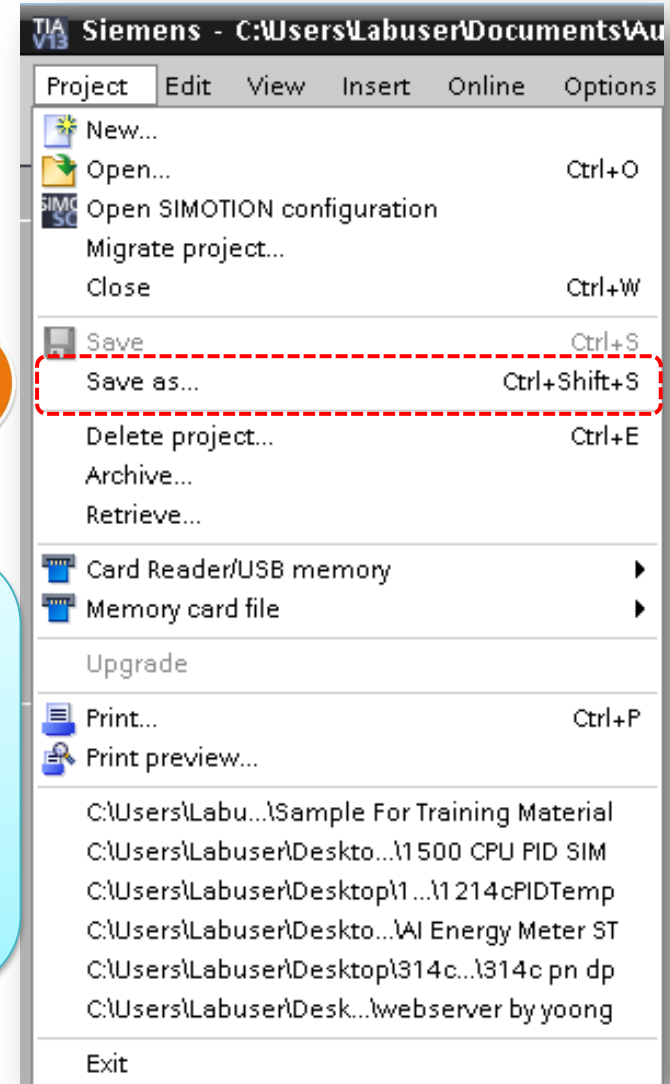
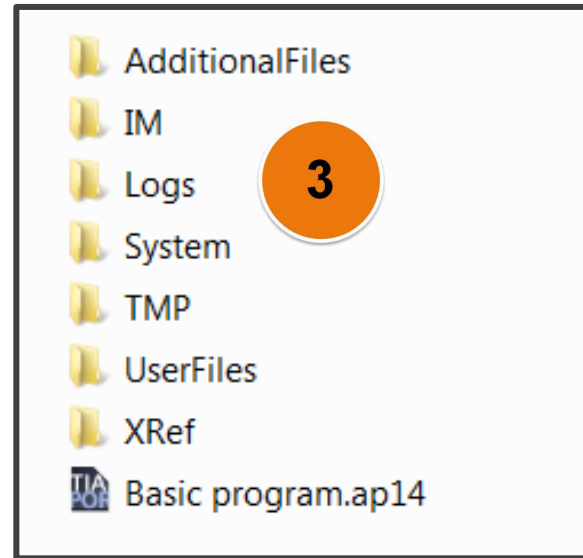
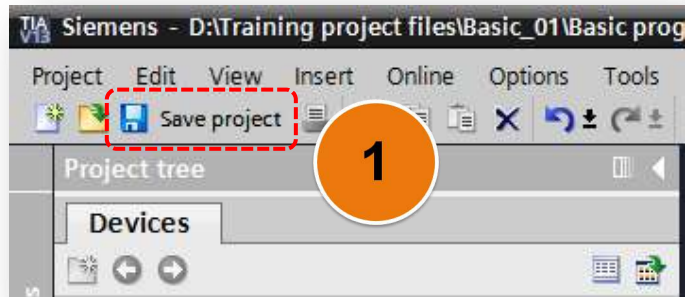
%Q0.0 "Run_M1"

Network 2:

Comment

2

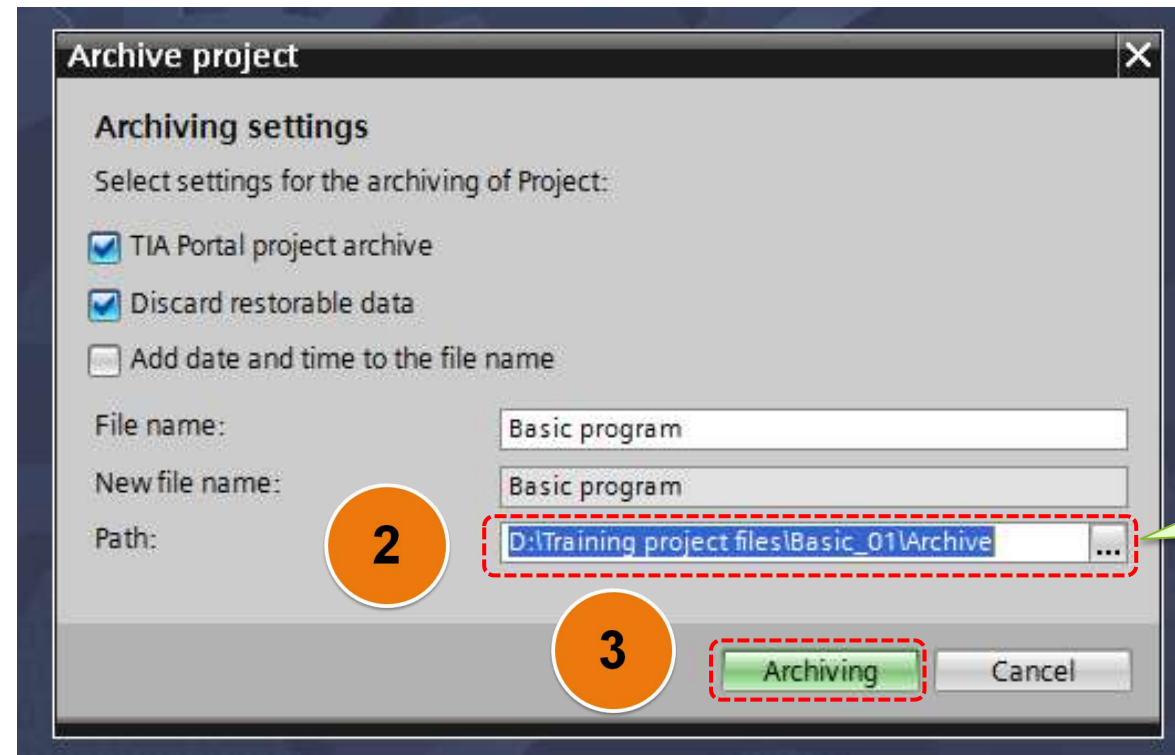
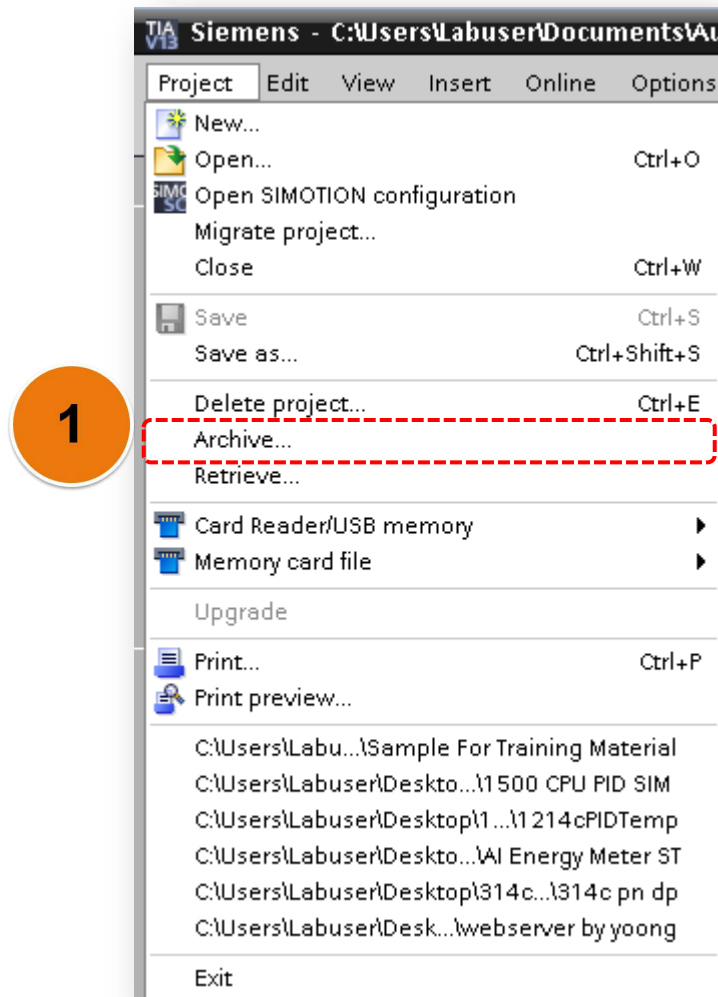
2.8 Save Project



1. Save as เป็นการบันทึกโปรเจกไฟล์ของเราใหม่
2. Save Project เป็นการบันทึกการเปลี่ยนแปลงเมื่อเราได้ทำการแก้ไขลอจิกภายหลังจากที่เราได้ Save as
3. จากนั้นก็จะได้โปรเจกไฟล์ ซึ่งจะมี backup ไฟล์มาด้วยทุกครั้ง และในโปรเจกไฟล์ก็จะมีไฟล์ต่างๆดังรูป ถ้าท่านจะนำไปใช้เครื่องอื่น ควรคัดลอกโฟลเดอร์โปรเจกไฟล์ไปทั้งหมด ไม่ใช่คัดลอกไปแค่ไฟล์นามสกุล .ap14

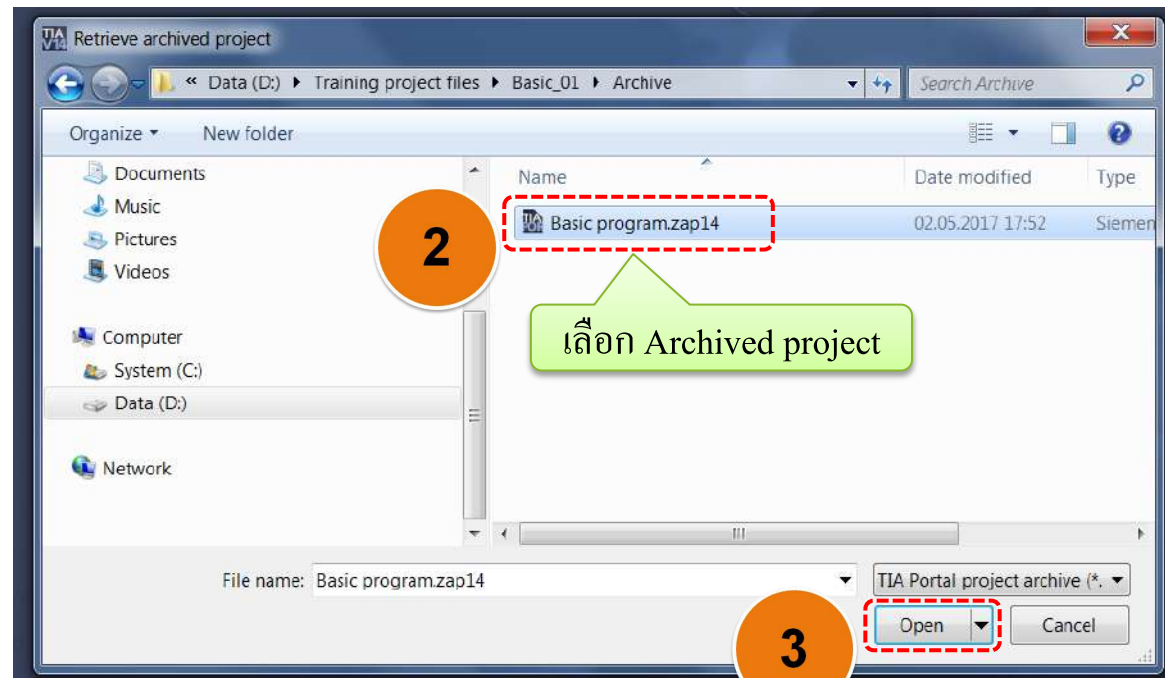
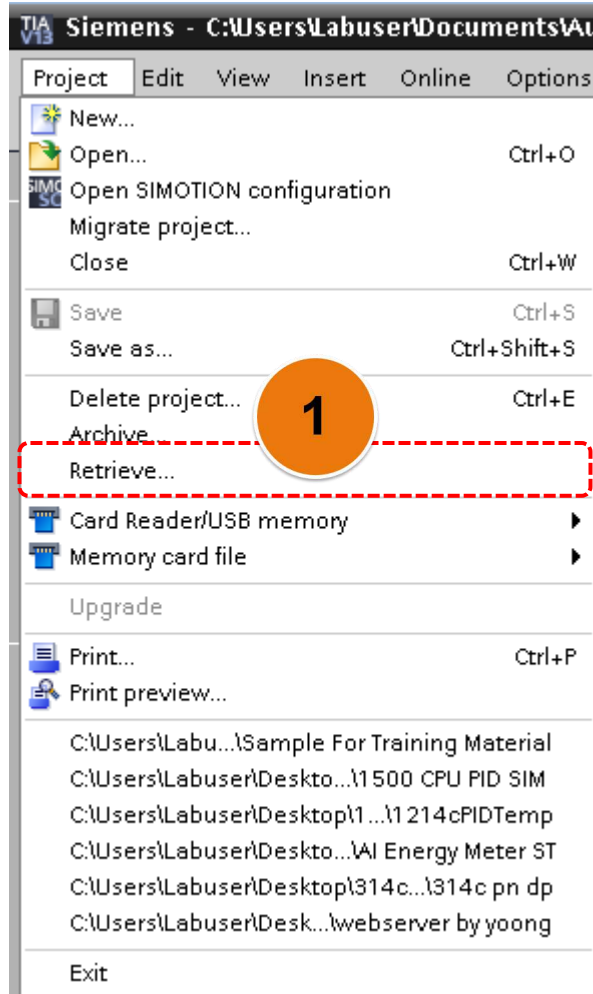
กรณีใช้ TIA Portal V15 หรือ V16 ไฟล์นามสกุล .ap15 หรือ .ap16

A-1 Archiving and Retrieving Project

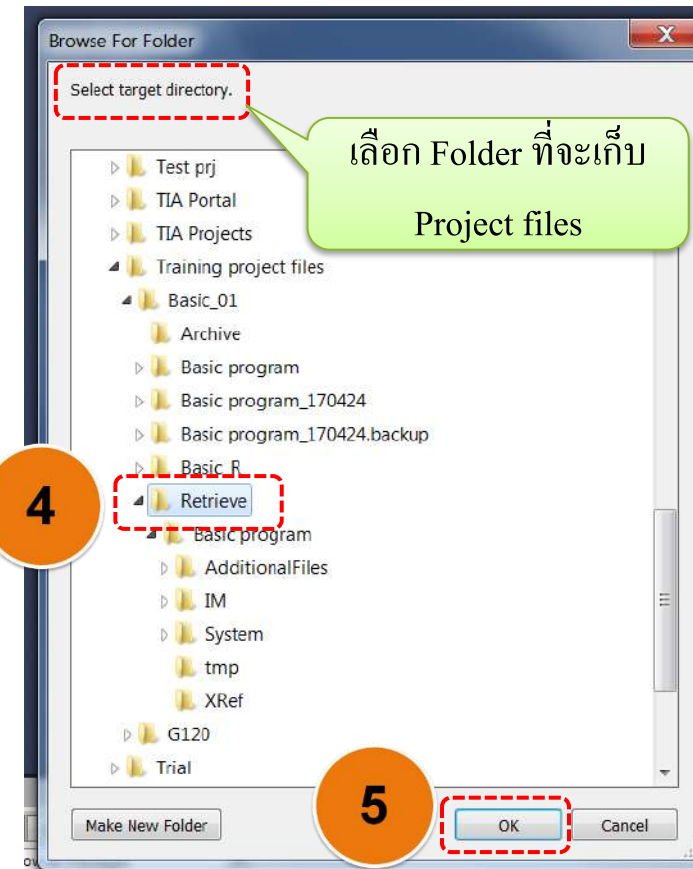


1. กด Archive การ Archive เป็นการบันทึกโปรเจกไฟล์แบบบีบอัดเป็นไฟล์เดียวซึ่งจะสะดวกต่อการโอนย้ายไฟล์
2. เลือกที่อยู่ของไฟล์ว่าจะบันทึกไว้ที่ใด ไฟล์จะมีนามสกุลเป็น .zap14
3. กด Save

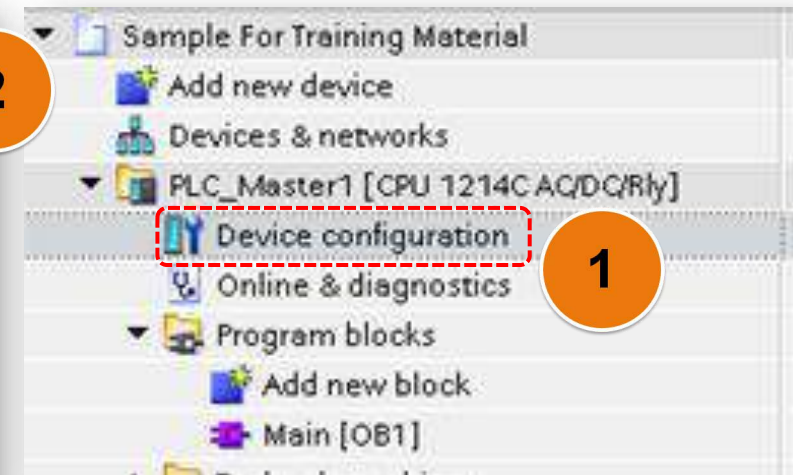
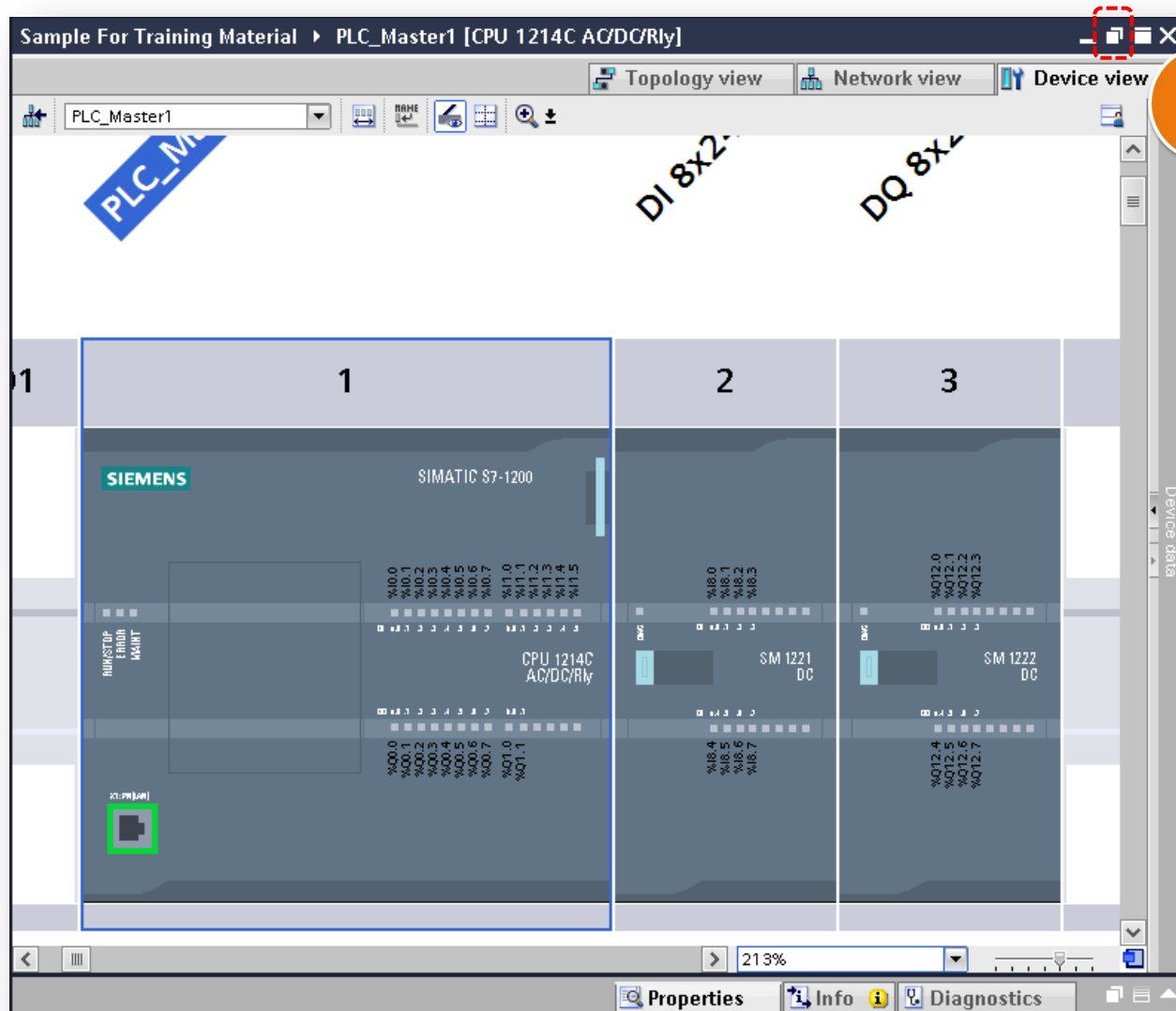
A-1 Archiving and Retrieving Project



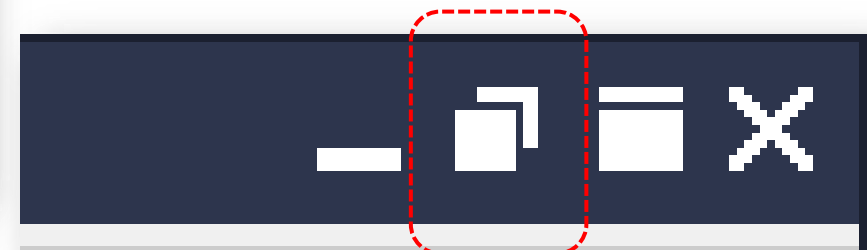
1. กด Retrieve
2. เลือกที่อยู่ของไฟล์ Archive ที่จะเปิดว่าจะบันทึกไว้ที่ใด ไฟล์จะมีนามสกุลเป็น .zap14
3. กด Open
4. เลือกที่อยู่ที่จะแยกไฟล์ออก
5. กด OK



A-2 Convenient tools : Floating working area



1. ไปที่ Device Configuration
2. กดเลือก Float ท่านจะได้หน้าต่าง PLC_Master1 แยกออกมา



A-2 Convenient tools : Floating working area

The image displays two windows from the Siemens TIA Portal software. The left window, titled 'Sample For Training Material > PLC_Master1 [CPU 1214C AC/DC/Rly] > Program blocks > Main [OB1]', shows the 'Block interface' for the 'Main Program Sweep (Cycle)' block. It contains two networks: 'Network 1' with a ladder logic diagram involving 'Start', 'Stop', and 'Light' variables, and 'Network 2' which is currently empty. The right window, titled 'Sample For Training Material > PLC_Master1 [CPU 1214C AC/DC/Rly]', shows the 'Topology view' of the PLC rack. It displays a SIMATIC S7-1200 system with a CPU 1214C AC/DC/Rly and two SM 1221 DC modules. The rack is divided into three sections labeled 1, 2, and 3. A green box highlights the CPU module in section 1. The status bar at the bottom of the right window shows '13%'.

เช่นเดียวกัน ให้ท่านกด Float ที่ Main OB ด้วยเพื่อที่จะทำการ Drag & Drop I/O

A-2 Convenient tools : Drag & Drop

Sample For Training Material ▶ PLC_Master1 [CPU 1214C AC/DC/Rly] ▶ Program blocks ▶ Main [OB1]

Block interface

Block title: "Main Program Sweep (Cycle)"

Comment

Network 1:

Comment

Start %I0.0, Stop %I1.0, Light %Q0.0

2

Drag & Drop

1

400%

Totally Integrated Automation PORTAL

- ขยายขนาดด้วยการซูม ประมาณ 400 % ตัว Infographic ของ CPU และ Module ต่างๆจะปรากฏ Address ขึ้นมา
- ให้ท่านเลือก Drag&Drop I/O ที่อยู่ในหน้าต่างของ Main OB ไปยัง CPU

A-2 Convenient tools : Split editor space

The screenshot illustrates the 'Split editor space' feature in Siemens SIMATIC Manager. The interface is divided into two main sections:

- Top Section (Topology view):** Shows the PLC rack configuration for 'PLC_1 [CPU 1214C AC/DC/Rly]'. The 'Module' list on the right includes DI 1..., AI 2_1, HSC_1, HSC_2, HSC_3, HSC_4, HSC_5, and HSC_6. A red dashed box highlights the 'Split editor' icon in the top toolbar.
- Bottom Section (Program blocks):** Shows the 'Main [OB1]' program editor. A red dashed box highlights a tag 'Tag1' in the program editor. A yellow arrow labeled 'Drag & Drop' points from this tag to the rack configuration.

Red dashed arrows indicate the workflow: from the 'Split editor' icon to the 'Module' list, and from the 'Tag1' in the program editor to the rack configuration.

สามารถ Drag and drop จาก Program blocks ไปยัง Device configuration ได้เลย

A-2 Convenient tools : Change version

The screenshot displays the Siemens TIA Portal interface. The main window shows a ladder logic network with a 'PID_3Step' block. The 'Instructions' window is open on the right, showing the 'Options' tab. The 'Technology' section is expanded, and the 'Compact PID' instruction is selected. The 'Version' dropdown menu is open, showing a list of versions: V1.1, V1.2, V2.0, V3.0, V4.0, and V5.0. The 'V5.0' version is highlighted. The background shows a ladder logic network with a 'PID_3Step' block.

..._1 [CPU 1214C AC/DC/Rly] ▶ Program blocks ▶ Main [OB1]

Block interface

Comment

%DB2
"PID_3Step_1"

PID_3Step

EN ENO

0.0 Setpoint ScaledInput ...

0.0 Input ScaledFeedback ...

0 Input_PER k ...

FALSE Actuator_H Output_UP ...

FALSE Actuator_L Output_DN ...

0.0 Feedback Output_PER ...

0 Feedback_PER SetpointLimit_H ...

0.0 Disturbance SetpointLimit_L ...

FALSE ManualEnable InputWarning_H ...

0.0 ManualValue InputWarning_L ...

FALSE Manual_UP State ...

FALSE Manual_DN Error ...

FALSE ErrorAck ErrorBits ...

FALSE Reset

FALSE ModeActivate

... Mode

Instructions

Options

Favorites

Basic instructions

Extended instructions

Technology

Counting V1.1

PID Control

Compact PID

PID_Compact Universal PID controller

PID_3Step PID controller with inte

PID_Temp PID controller for temp

Motion Control

V5.0

V1.2

V2.0

V3.0

V4.0

V5.0

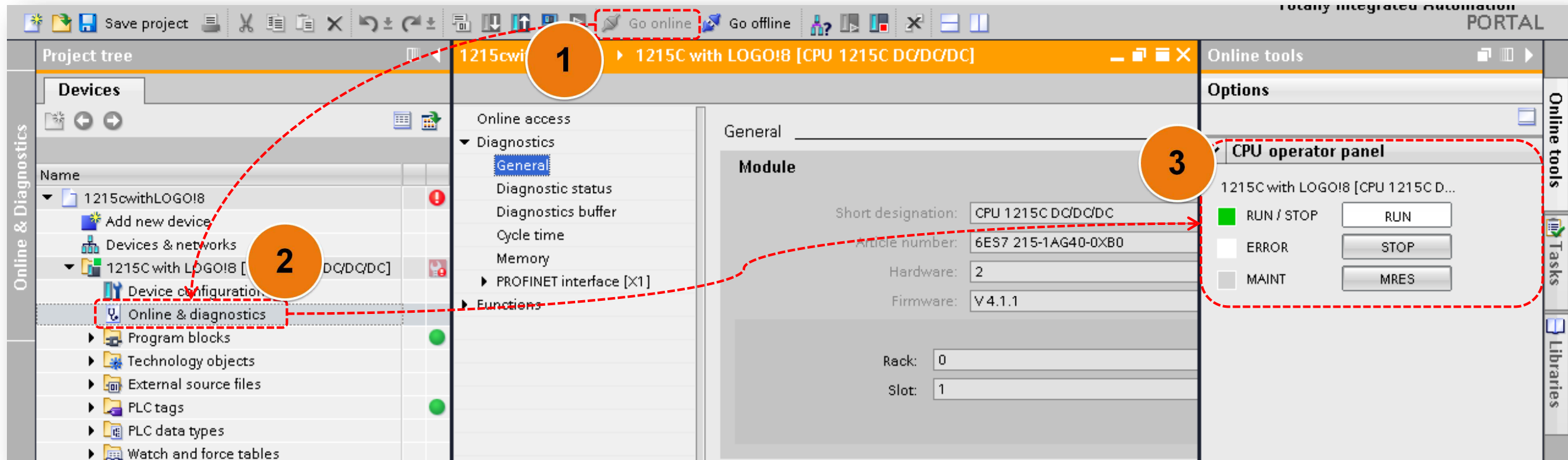
สามารถ เปลี่ยน Version ของ Instructions ได้ เพื่อให้เลือกใช้งานว่า
จะใช้ Version ใดสำหรับการเขียนโปรแกรมและการ Migration

Easy-to-use tools

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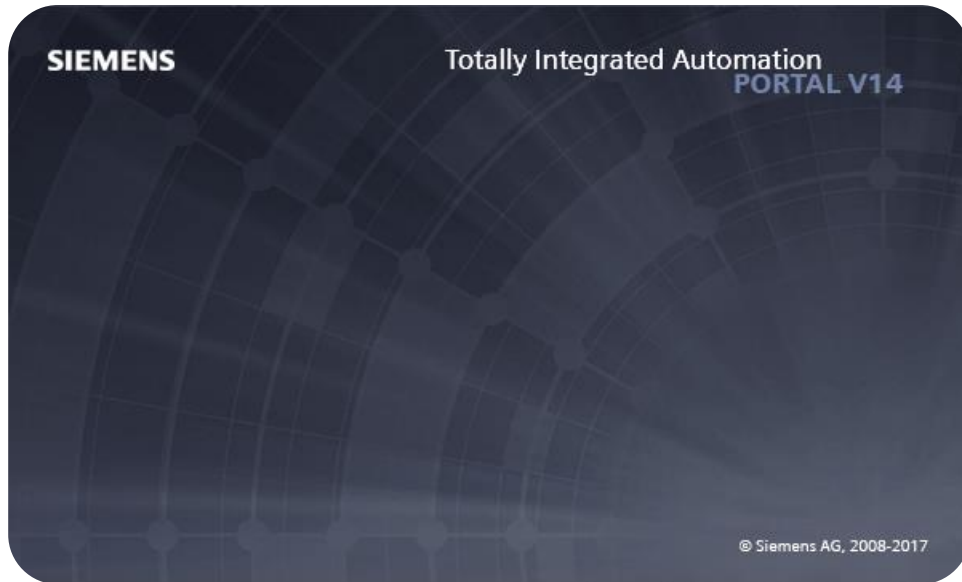
ใน CPU S7-1200 นั้นจะไม่มี Physical switch ที่จะ STOP – RUN เหมือนรุ่นอื่นดังนั้นสามารถใช้งาน STOP - RUN ได้ใน CPU Operator Panel เท่านั้น

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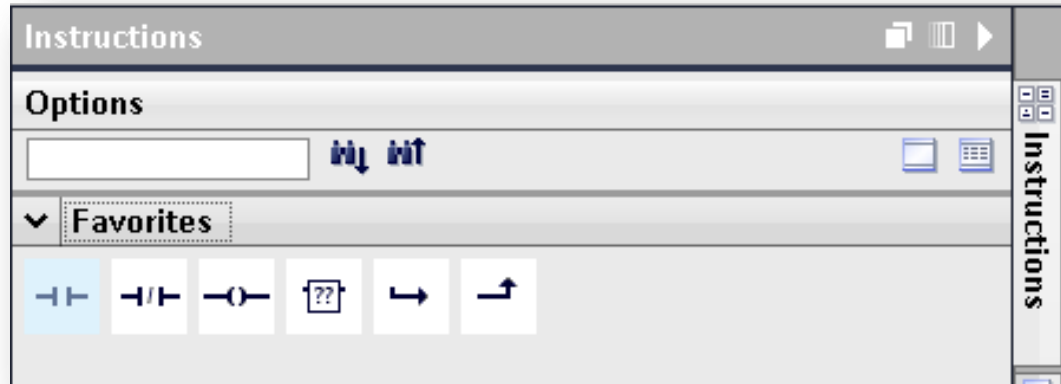
6

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Appendix

3. Basic programming 1 : Bit operant



1

NO

2

NC

3

AND

4

OR

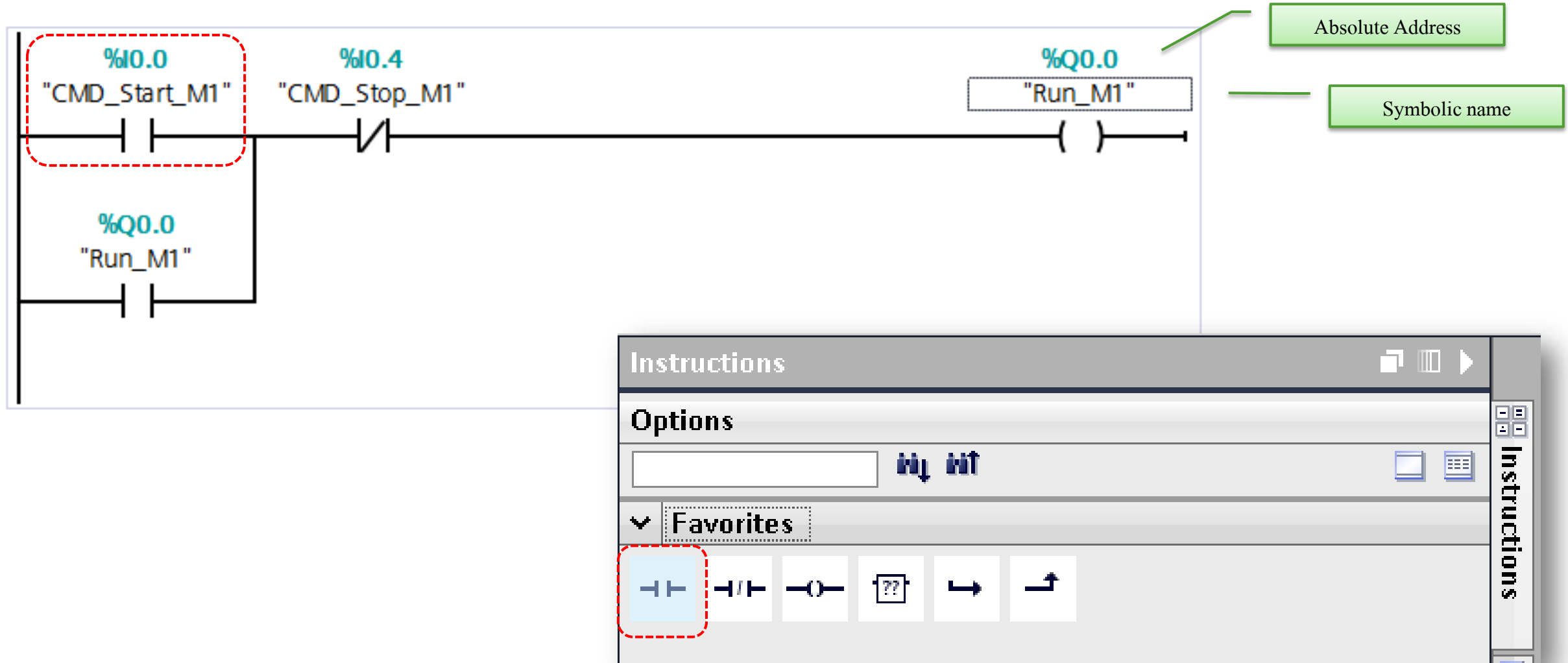
5

Set

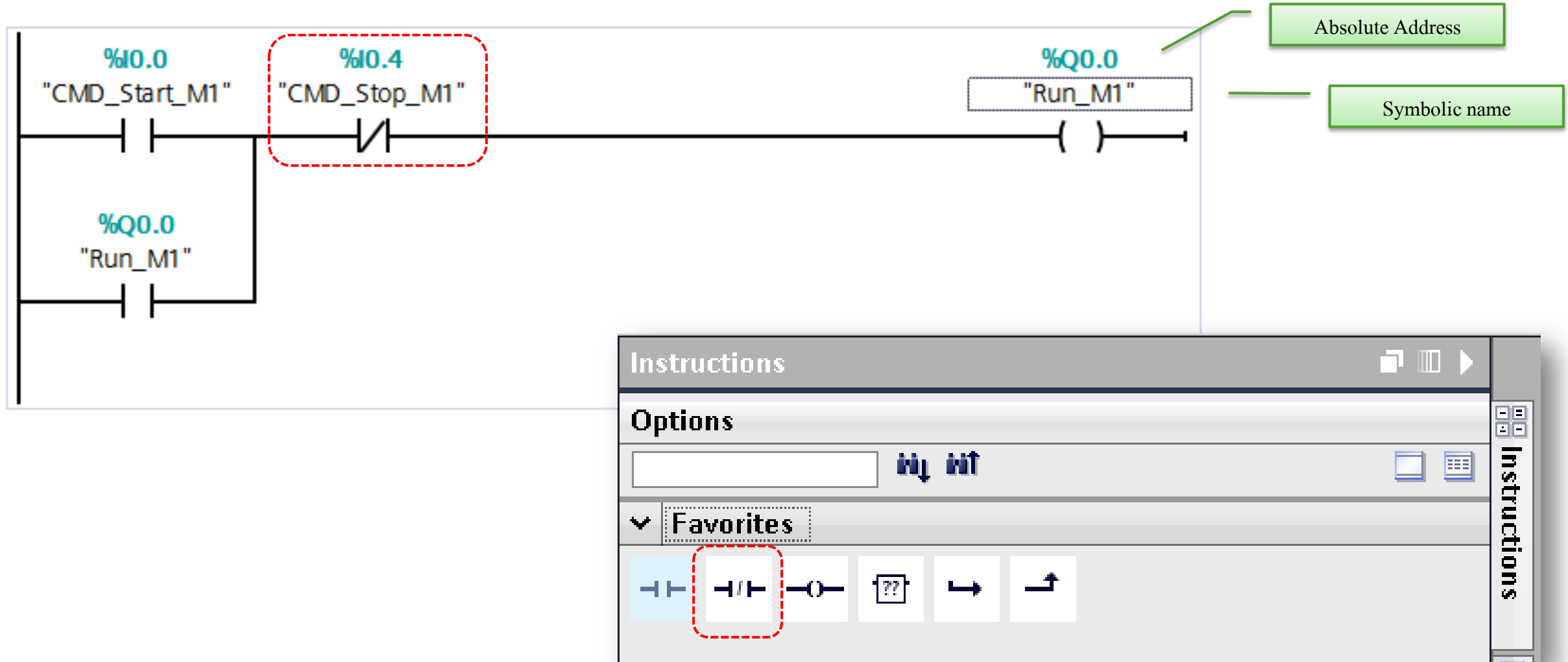
6

Ptrig

3.1 NO (Normally Open)



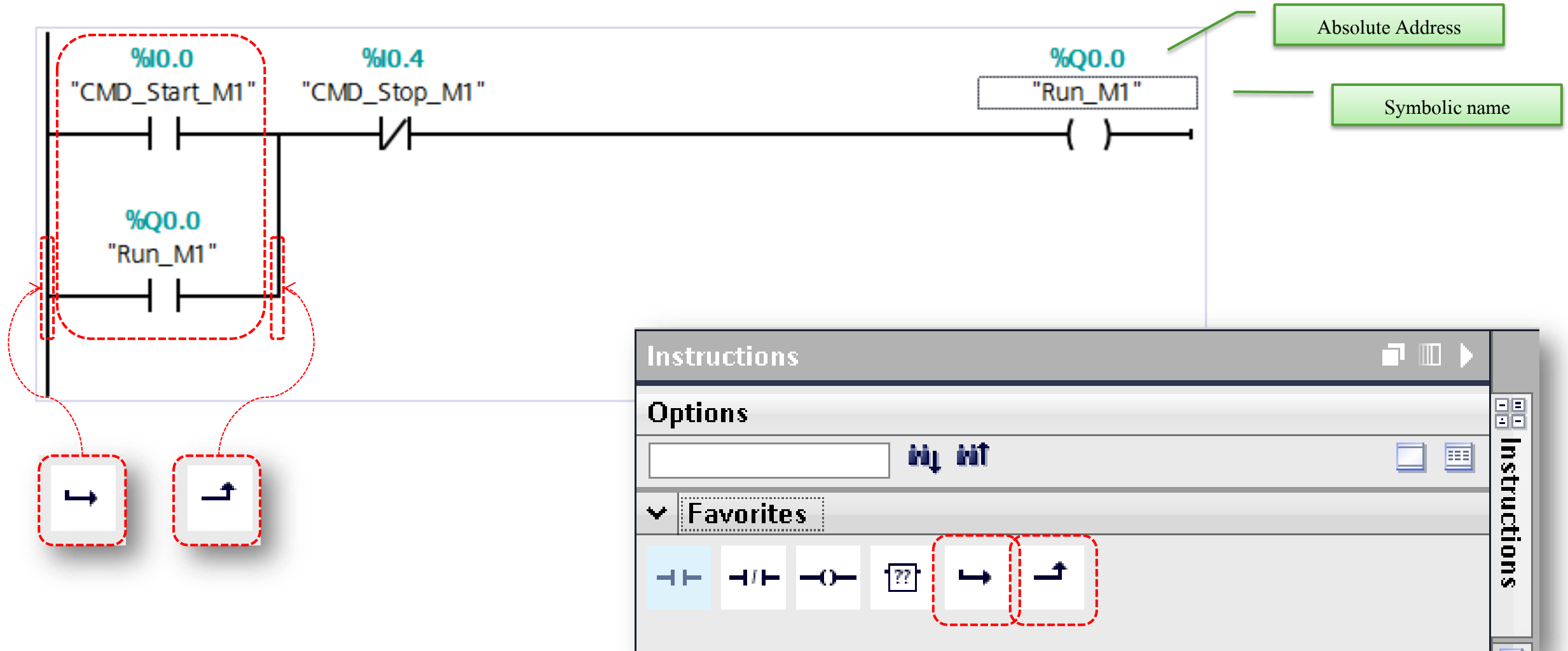
3.2 NC (Normally Close)



3.3 AND



3.4 OR

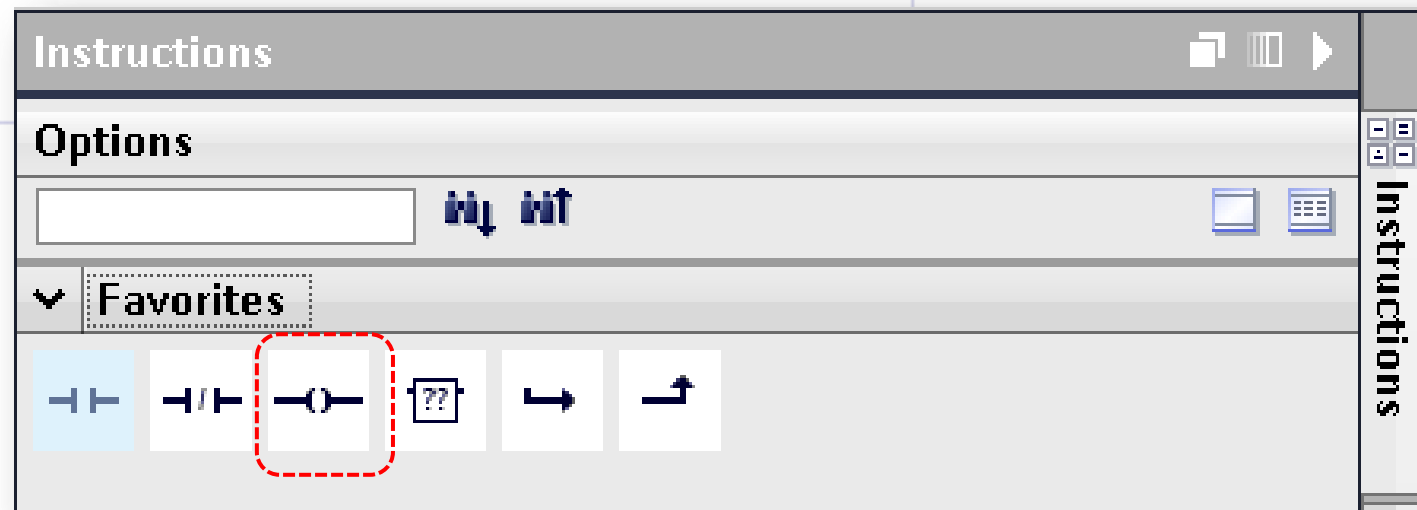
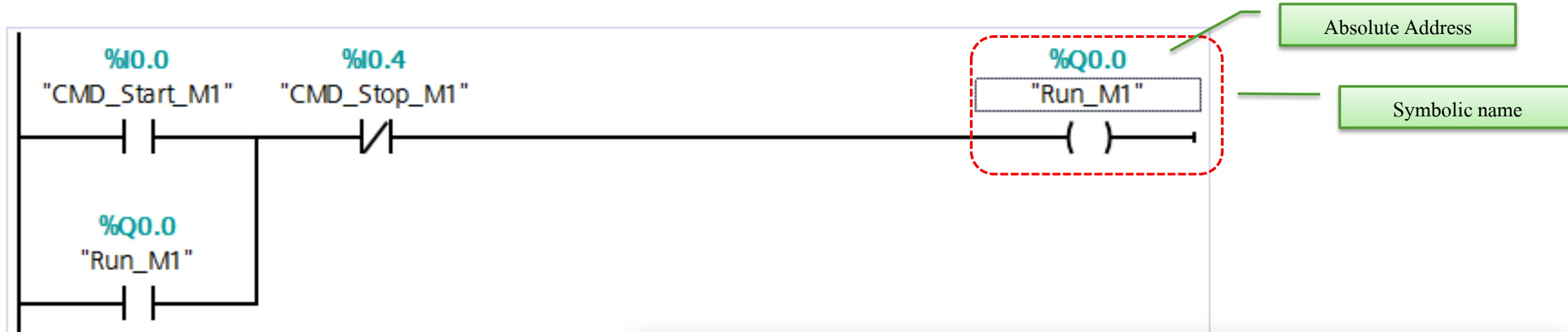


3.4 SET

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3.6 Trig : Ptrig & Ntrig

Network 2:

Comment

%I0.0
"CMD_Start_M1"

| P |

%M2.0
"Ptrig_01"

Network 3:

Comment

Basic instructions

Name	Description	Version
General		
Bit logic operations		
-- I --	Normally open contact [Shift+F2]	
-- /I --	Normally closed contact [Shift+F3]	
-- NOT --	Invert RLO	
--()--	Assignment [Shift+F7]	
--(I)--	Negate assignment	
--(R)--	Reset output	
--(S)--	Set output	
SET_BF	Set bit field	
RESET_BF	Reset bit field	
SR	Set/reset flip-flop	
RS	Reset/set flip-flop	
-- P --	Scan operand for positive signal edge	
-- N --	Scan operand for negative signal edge	
--(P)--	Set operand on positive signal edge	
--(N)--	Set operand on negative signal edge	
P_TRIG	Scan RLO for positive signal edge	
N_TRIG	Scan RLO for negative signal edge	
R_TRIG	Detect positive signal edge	

Network 2:

Comment

%I0.0
"CMD_Start_M1"

--|P|--

%M2.0
"Ptrig_01"

%I0.0
"CMD_Start_M1"

| P |

%M2.0
"Ptrig_01"

%I0.0
"CMD_Start_M1"

| N |

%M2.0
"Ptrig_01"

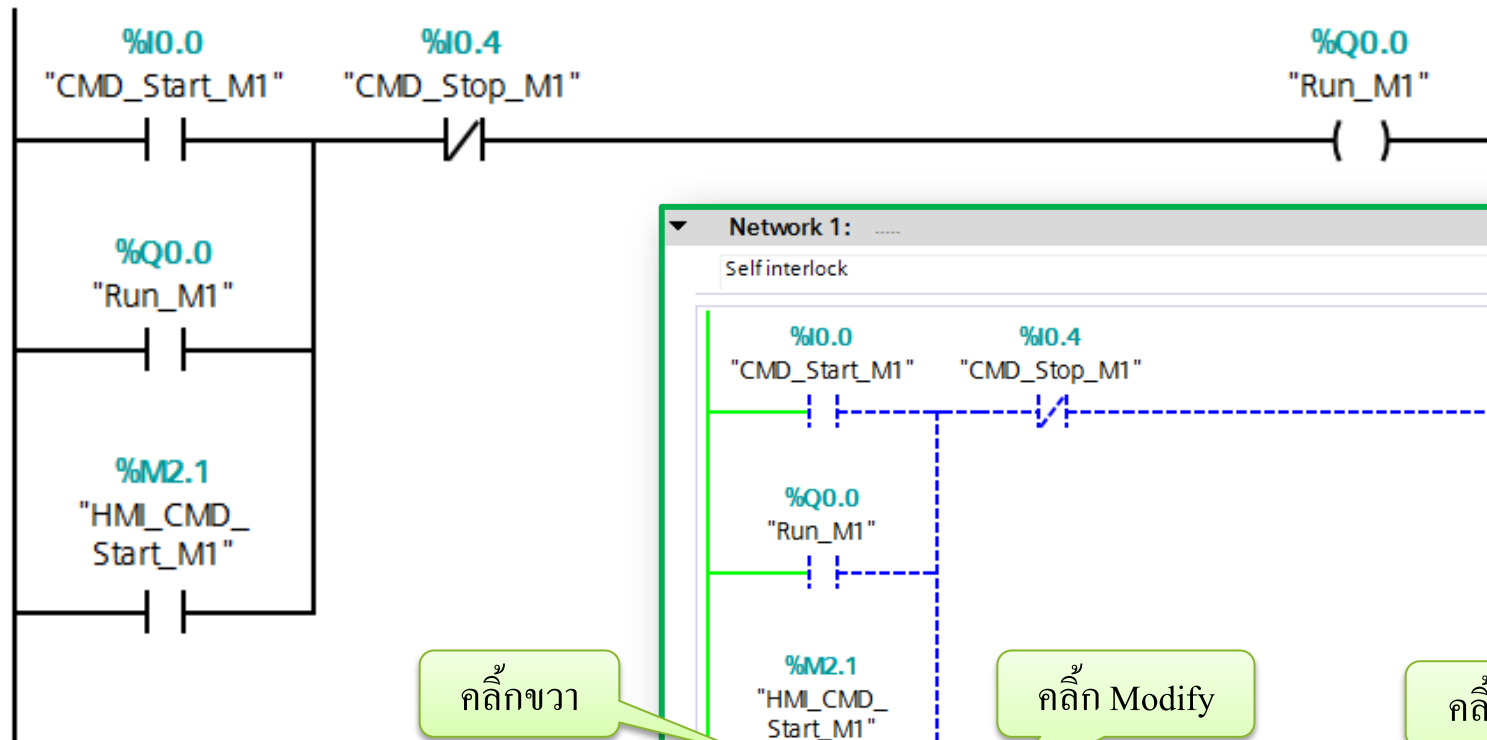
Trig จะมี 2 รูปแบบคือ Positive Trig (ขอบขาขึ้น)
และ Negative Trig (ขอบขาลง)

Exercise - 1

Comment

▼ Network 1:

Self interlock



คลิกขวา

คลิก Modify

คลิก Modify to 1

▼ Network 2:

The screenshot shows the same ladder logic network as above, but with a context menu open over the **%M2.1** contact. The menu options are:

- Modify (highlighted)
- Monitor
- Display format
- Define tag... (Ctrl+Shift+I)
- Rename tag (Ctrl+Shift+T)

The **Modify** option is further expanded, showing the following sub-options:

- Modify to 0 (Ctrl+F3)
- Modify to 1 (Ctrl+F2, highlighted by a mouse cursor)
- Modify operand ... (Ctrl+Shift+2)

Note :

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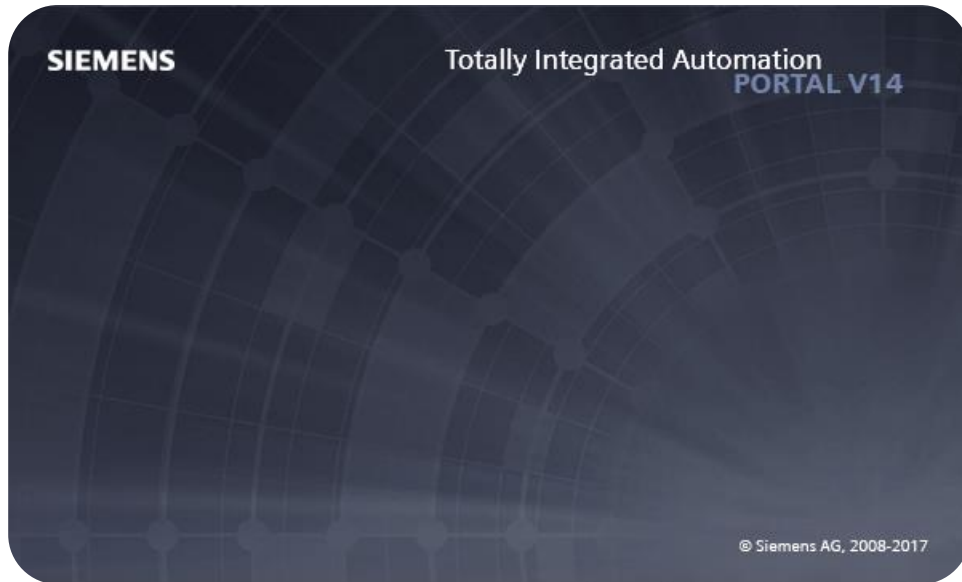
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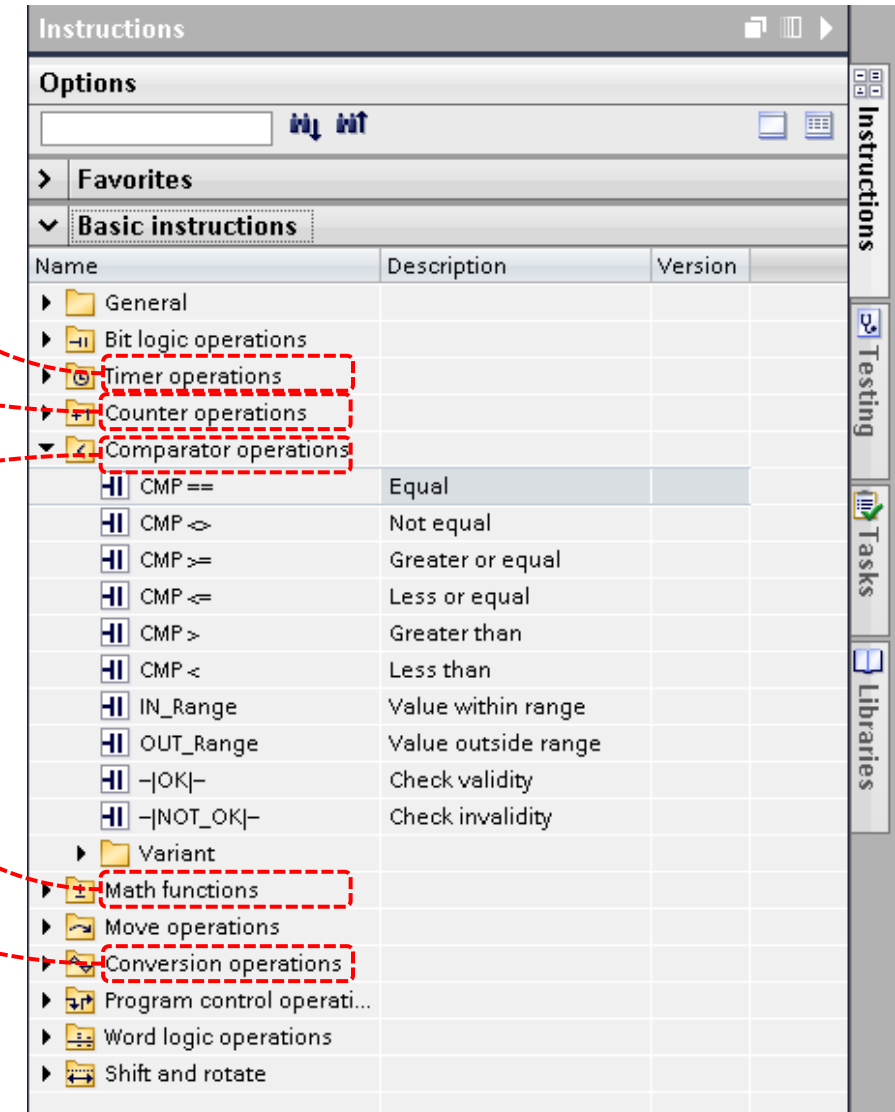
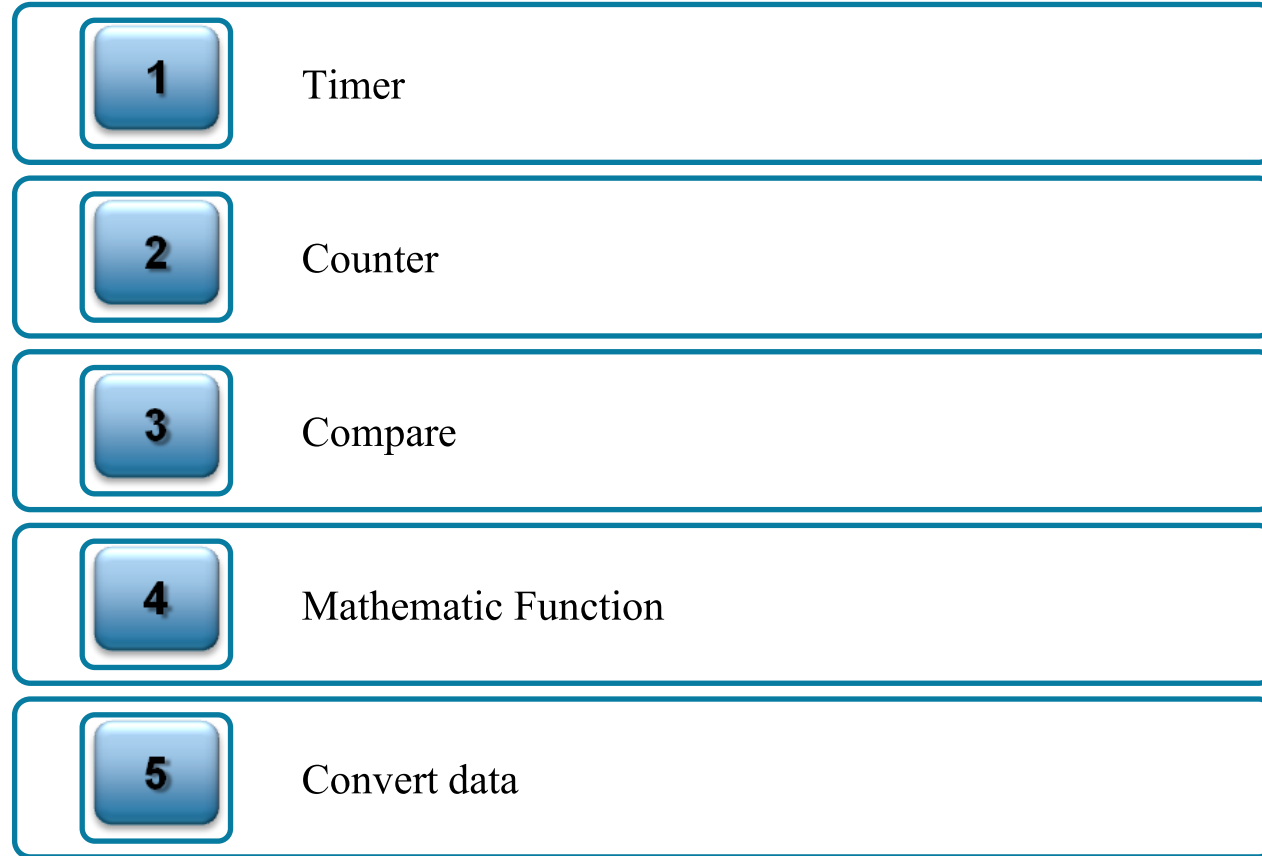
6

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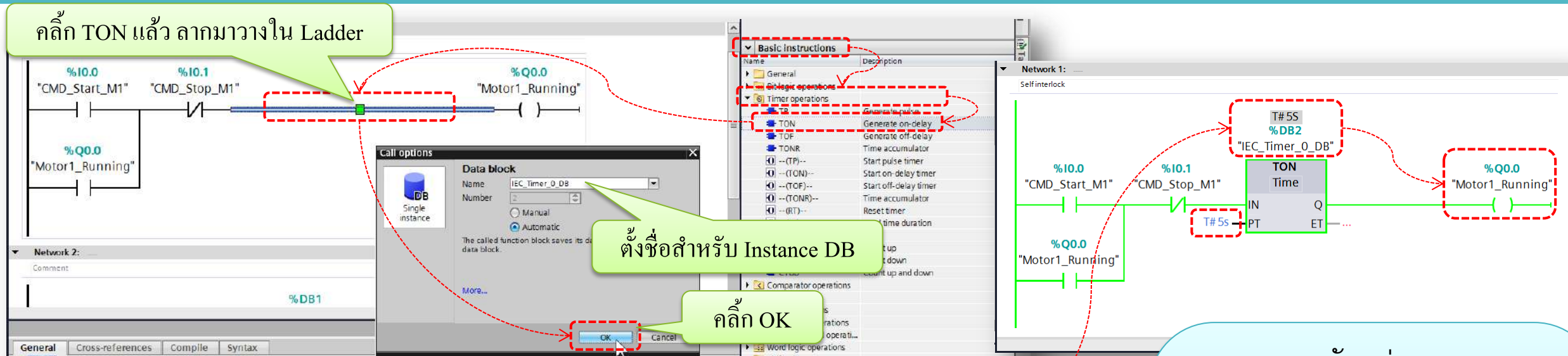
Appendix

4. Basic programming 2 : Word operant



4.1 Timer

คลิก TON แล้ว ลากมาวางใน Ladder



ตั้งชื่อสำหรับ Instance DB

คลิก OK

การกำหนดค่า เวลาสำหรับ PT นั้น อาจกำหนดเป็นค่าคงที่หรือผ่านตัวแปรชนิด Time

ตัวอย่าง

กำหนดค่า PT แบบคงที่

PT = 700 ms → T#700ms

PT = 5 Sec → T#5s

PT = 10 Minute → T#10m

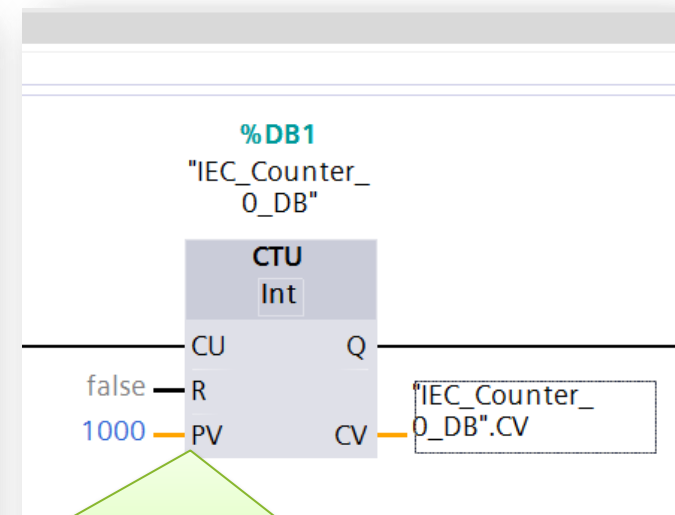
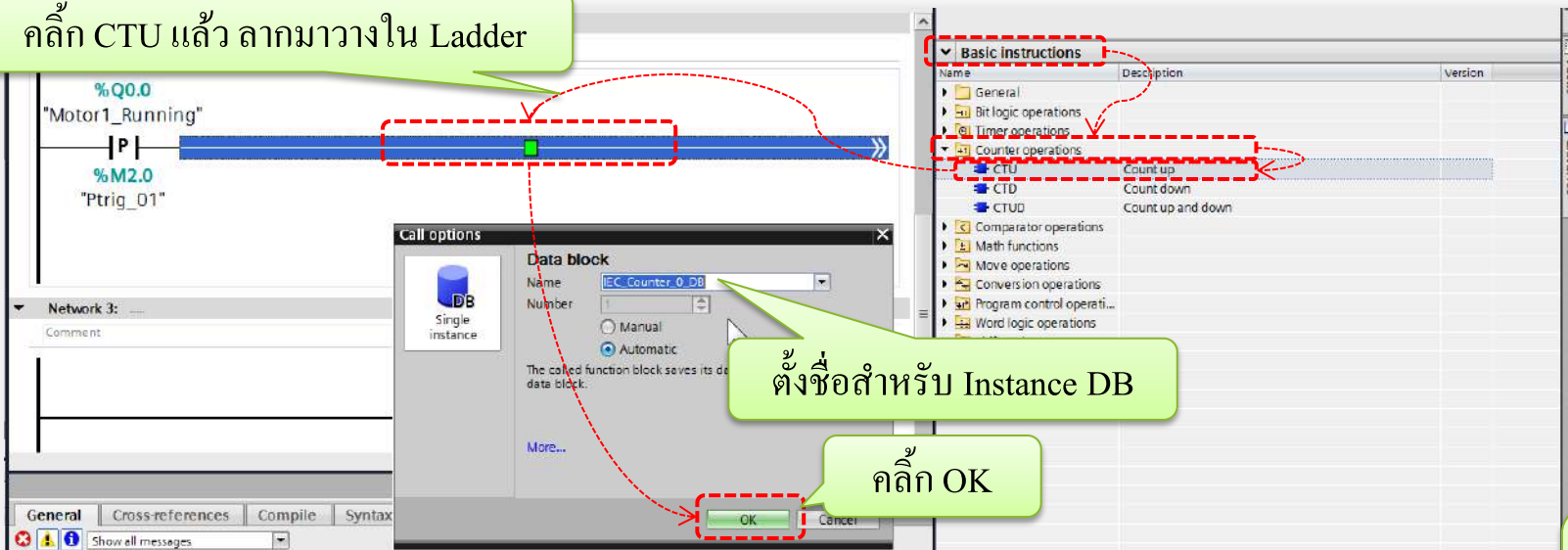
PT = 8 Hour → T#8h

PT = 5 Day → T#5d

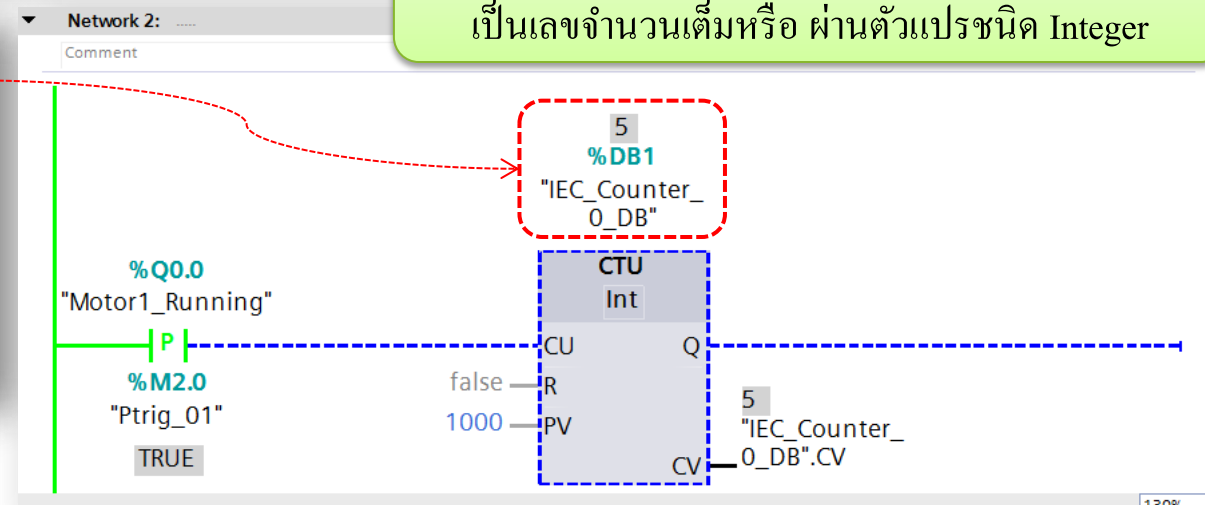
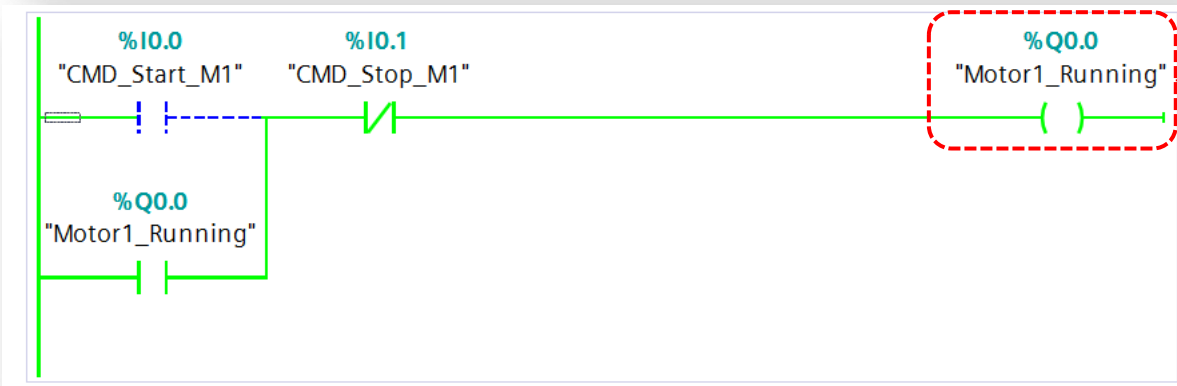
PT = 1hr, 45min, 30sec, 15ms
→ T#1h_45m_30s_15ms

4.2 Counter

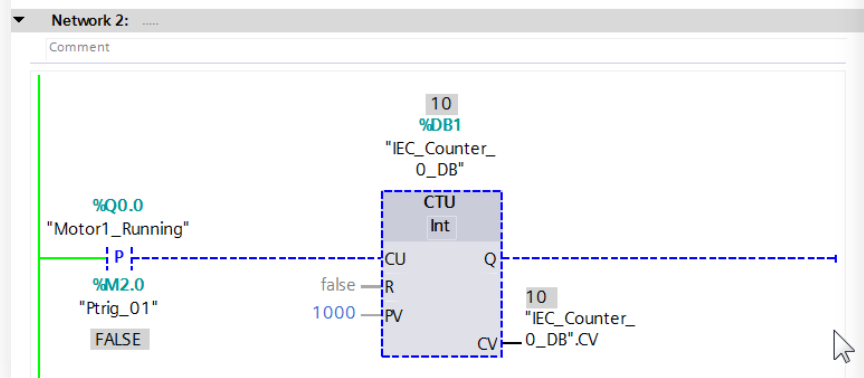
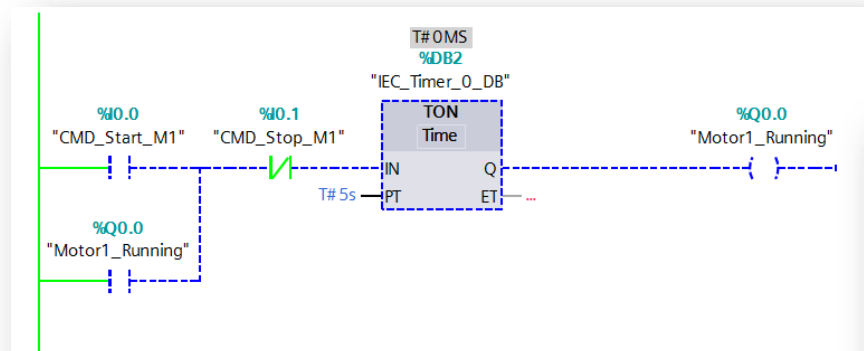
คลิก CTU แล้ว ลากมาวางใน Ladder



การกำหนดค่า สำหรับ PV นั้น อาจกำหนดเป็นค่าคงที่เป็นเลขจำนวนเต็มหรือ ผ่านตัวแปรชนิด Integer

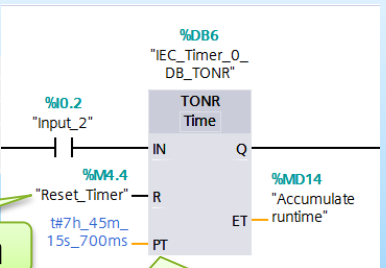


Example : Timer & Counter



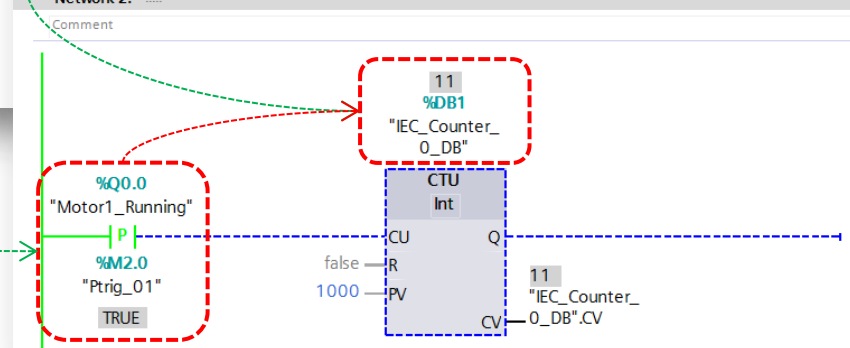
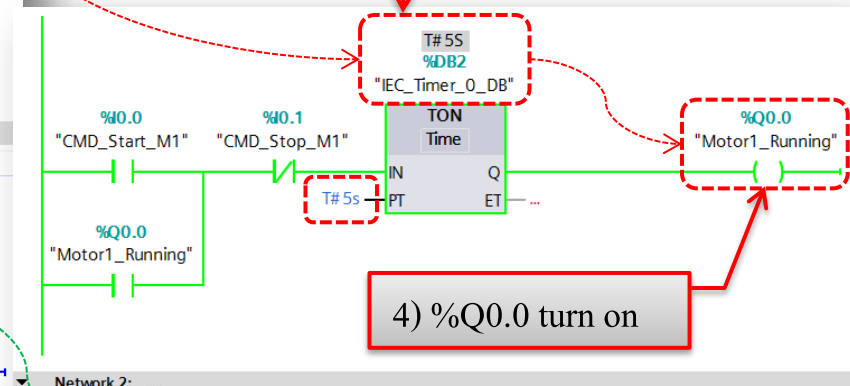
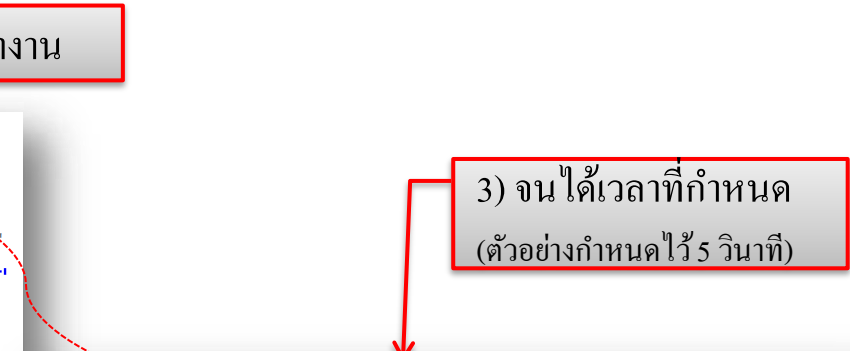
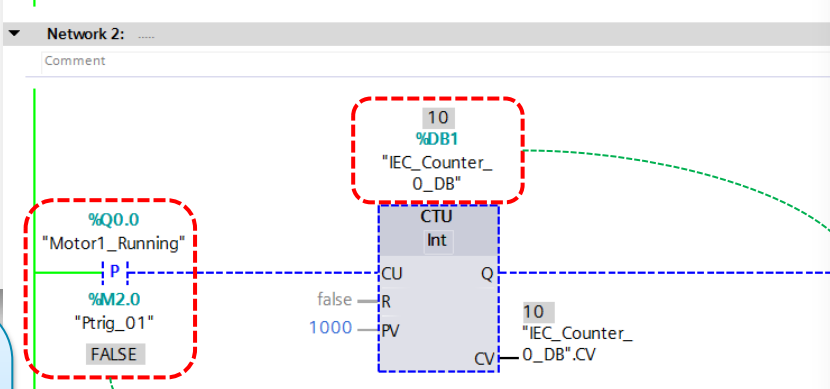
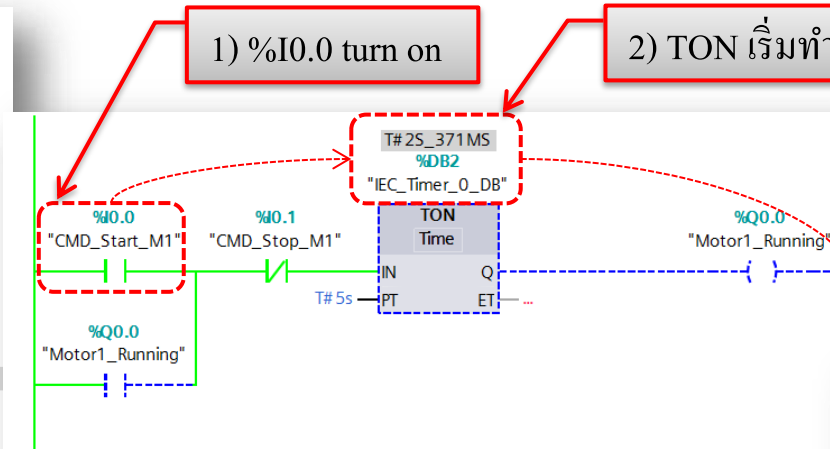
Timer แบบ Accumulator

Timer operations		V1.0
TP	Generate pulse	V1.0
TON	Generate on-delay	V1.0
TOF	Generate off-delay	V1.0
TONR	Time accumulator	V1.0



กำหนด Tag สำหรับ reset เวลา

การกำหนดรอบเวลาสำหรับ PT



1) %I0.0 turn on

2) TON เริ่มทำงาน

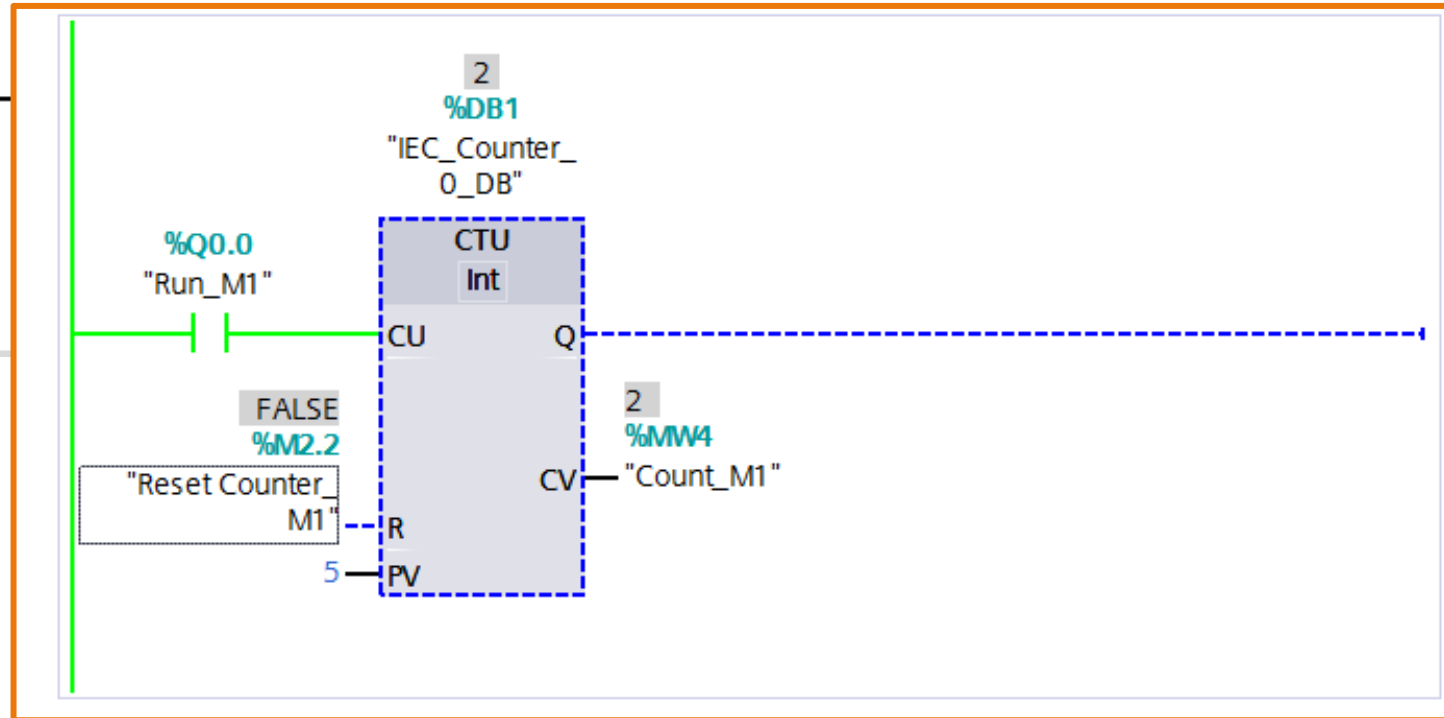
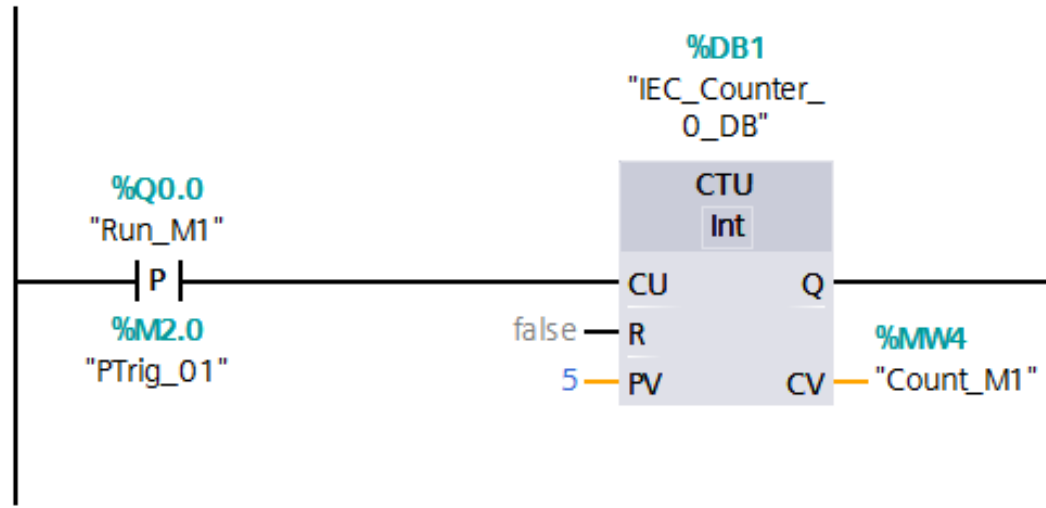
3) จนได้เวลาที่กำหนด
(ตัวอย่างกำหนดไว้ 5 วินาที)

4) %Q0.0 turn on

Exercise – 2 : Counter

Network 2:

Comment

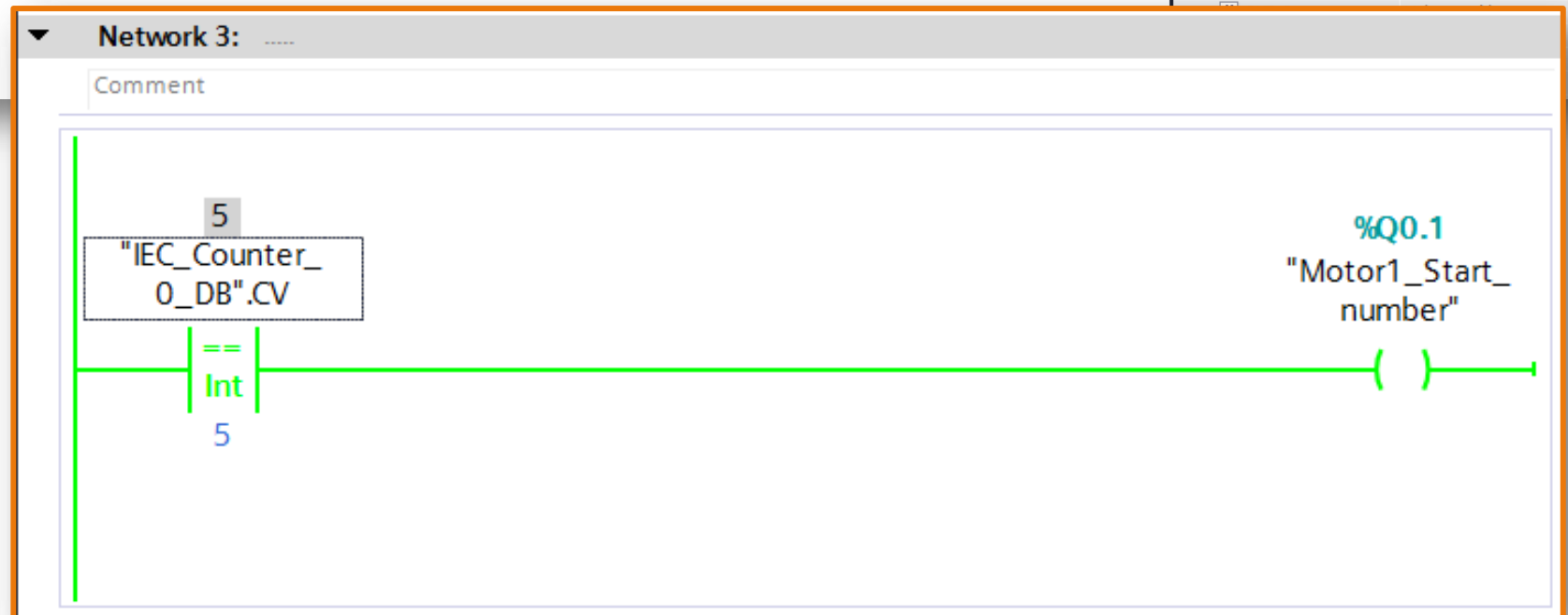
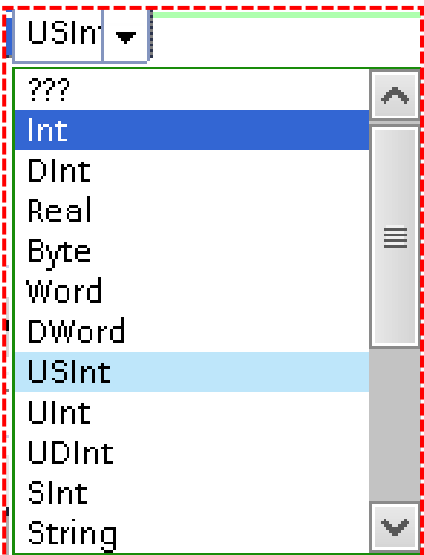
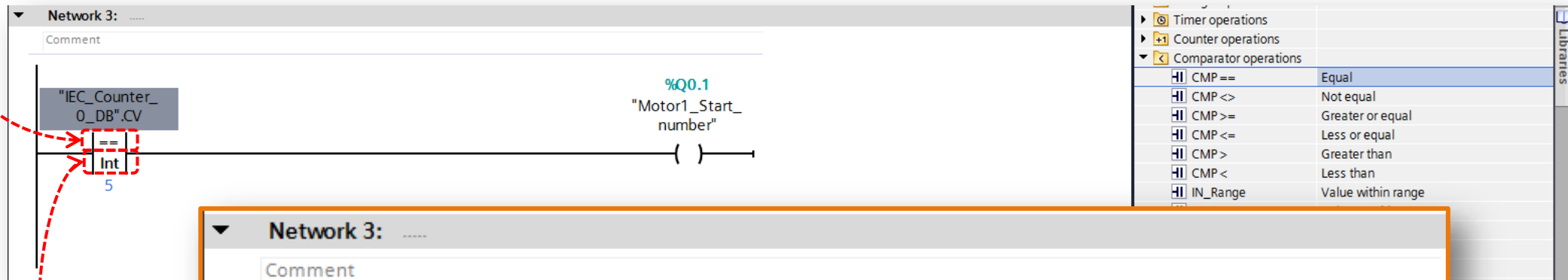


Blocks, timers, and counters supported by S7-1200

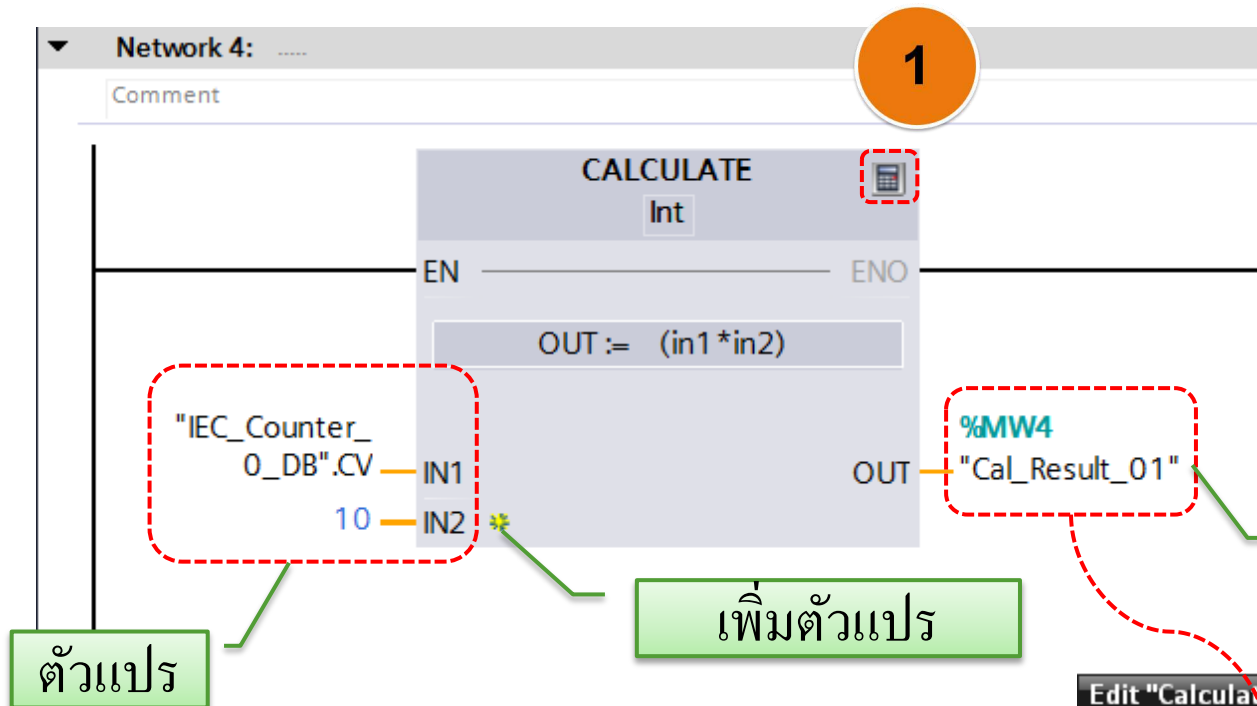
Element		Description
Blocks	Type	OB, FB, FC, DB
	Size	50 Kbytes (CPU 1211C) 75 Kbytes (CPU 1212C) 100 Kbytes (CPU 1214C) 125 Kbytes (CPU 1215C) 150 Kbytes (CPU 1217C)
	Quantity	Up to 1024 blocks total (OBs + FBs + FCs + DBs)
	Nesting depth	16 from the program cycle or startup OB; 6 from any interrupt event OB
	Monitoring	Status of 2 code blocks can be monitored simultaneously
OBs	Program cycle	Multiple
	Startup	Multiple
	Time-delay interrupts	4 (1 per event)
	Cyclic interrupts	4 (1 per event)
	Hardware interrupts	50 (1 per event)
	Time error interrupts	1
	Diagnostic error interrupts	1
	Pull or plug of modules	1
	Rack or station failure	1
	Time of day	Multiple
	Status	1
	Update	1
	Profile	1
Timers	Type	IEC
	Quantity	Limited only by memory size
	Storage	Structure in DB, 16 bytes per timer
Counters	Type	IEC
	Quantity	Limited only by memory size
	Storage	Structure in DB, size dependent upon count type • SInt, USInt: 3 bytes • Int, UInt: 6 bytes • DInt, UDInt: 12 bytes

4.3 Compare

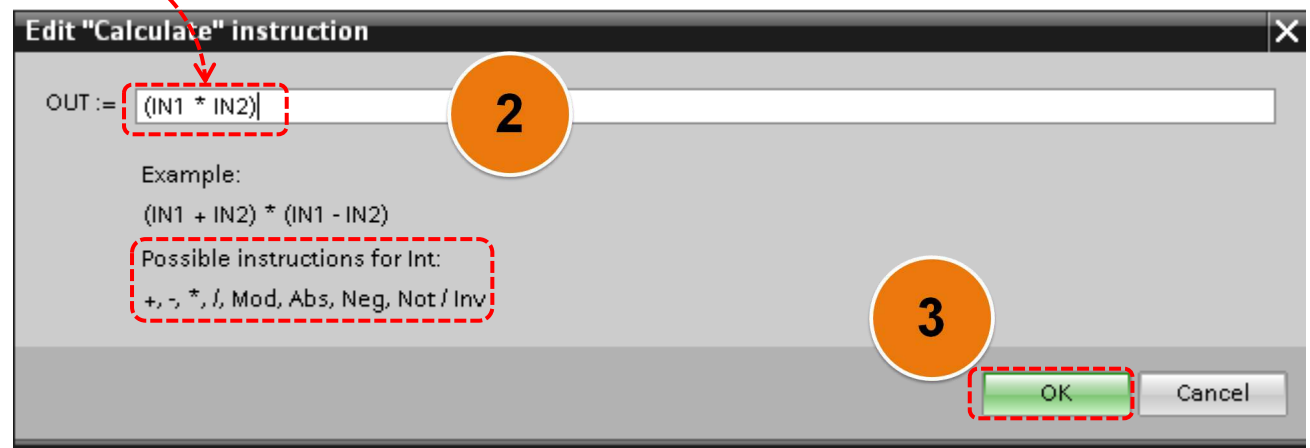
Compare เป็นฟังก์ชันการเปรียบเทียบค่าสองค่าโดยเมื่อตรงตามเงื่อนไขแล้วก็จะ ON



4.4 Mathematic Function



CALCULATE เป็นฟังก์ชันการคำนวณค่าข้อมูลทางตัวเลข โดยการเขียนคำสั่งต่างๆด้วยตัวเอง ทำให้สามารถลดการใช้ block ในการคำนวณลงได้เหลือเพียง block เดียว



Exercise – 3 : Calculation

Network 3: ...

Comment

CALCULATE
???
EN — ENO
OUT := <??>
IN1 <??>
IN2 <??>
OUT <??>

Click ✖ เพื่อ
เพิ่มตัวแปร

กำหนด Block
เป็น Real

Real

Int

Real

Int

Convert Int → Real

%MW10
"Data_02" — IN2

100.0
%MD6
"Data_01" — IN1

0
%MW10
"Data_02" — IN2

10.0
%MD12
"Data_03" — IN3

0
%MW16
"Data_04" — IN4

1000.0
%MD18
OUT — "Data_05"

Real

แปลงจาก Int
เป็น Real

Define tag

Name	Section	Address	Data type	PLC tag table	Comment
Data_02	Global Memory	%MW10	Int	Default tag table	
Data_03	Global Memory	%MD12	Real	Default tag table	
Data_04	Global Memory	%MW16	Int	Default tag table	
Data_05	Global Memory	%MD18	Real	Default tag table	

Edit "Calculate" instruction

OUT := (IN1+IN2)*(IN3-IN4)

Example:
(IN1 + IN2) * (IN1 - IN2)

Possible instructions for Real:
+, -, /, Abs, Neg, Exp, **, Frac, Ln, Sin, ASin, Cos, ACos, Tan, ATan, Sqr, Sqrt, Round, Ceil, Floor, Trunc

OK Cancel

Edit "Calculate" instruction

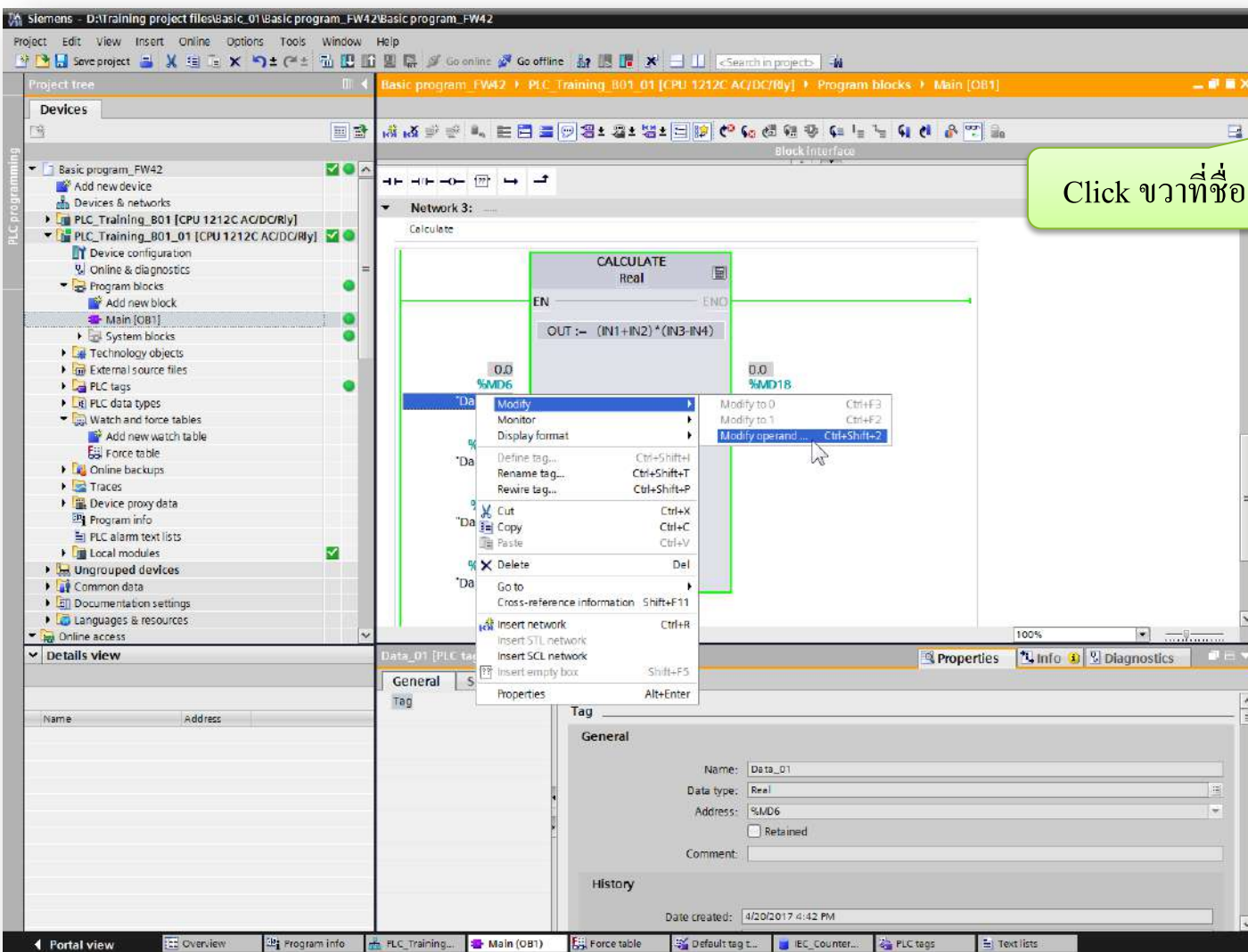
OUT := (IN1+IN2)*(IN3-IN4)

Monitor on/off

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Click ขวาที่ชื่อ Tag

Click Modify

Click Modify Operand

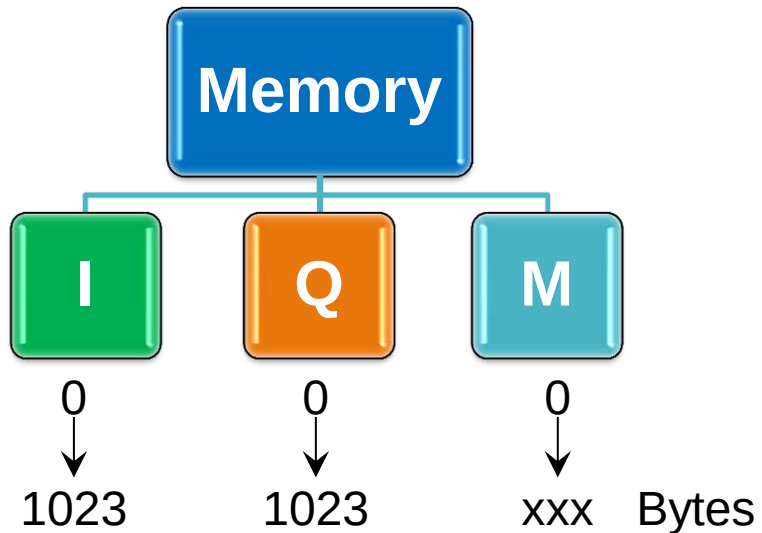
ป้อนค่าที่ต้องการ

Click OK

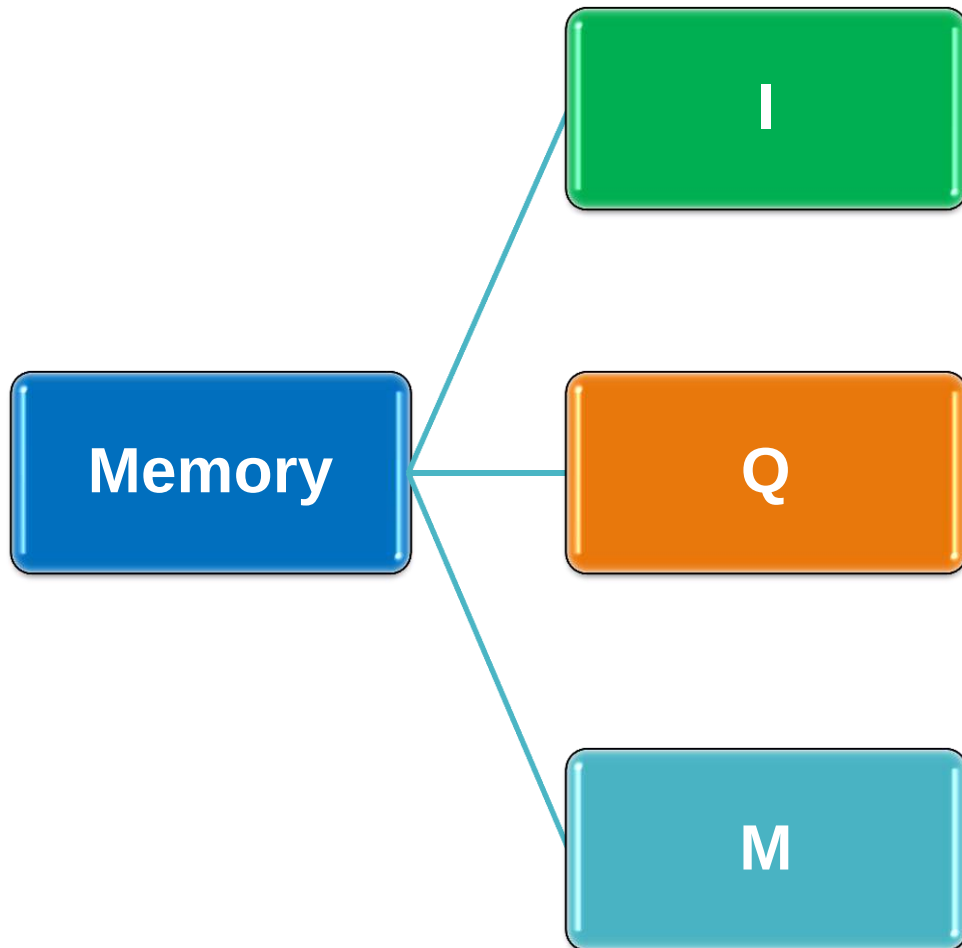
$$(100.0 + 10.0) \times (15.0 - 5.0) = 1100$$

Data type

I3.7	I3.6	I3.5	I3.4	I3.3	I3.2	I3.1	I3.0	I2.7	I2.6	I2.5	I2.4	I2.3	I2.2	I2.1	I2.0	I1.7	I1.6	I1.5	I1.4	I1.3	I1.2	I1.1	I1.0	I0.7	I0.6	I0.5	I0.4	I0.3	I0.2	I0.1	I0.0
BYTE 3								BYTE 2								BYTE 1								(1 byte data) BYTE 0							
IB3								IB2								IB1								IB0							
WORD 2																(2 bytes data) WORD 0															
IW2																IW0															
(4 bytes data) DWORD 0																															
ID0																															



IO.7	IO.6	IO.5	IO.4	IO.3	IO.2	IO.1	IO.0	0	IB0	IW0	ID0
I1.7	I1.6	I1.5	I1.4	I1.3	I1.2	I1.1	I1.0	1	IB1		
I2.7	I2.6	I2.5	I2.4	I2.3	I2.2	I2.1	I2.0	2	IB2	IW2	
I3.7	I3.6	I3.5	I3.4	I3.3	I3.2	I3.1	I3.0	3	IB3		
I4.7	I4.6	I4.5	I4.4	I4.3	I4.2	I4.1	I4.0	4	IB4	IW4	ID4
I5.7	I5.6	I5.5	I5.4	I5.3	I5.2	I5.1	I5.0	5	IB5		
I6.7	I6.6	I6.5	I6.4	I6.3	I6.2	I6.1	I6.0	6	IB6	IW6	
I7.7	I7.6	I7.5	I7.4	I7.3	I7.2	I7.1	I7.0	7	IB7		



CPU 1211C / 1212 C

- I : 0...1023 (1024 bytes)
- Q : 0...1023 (1024 bytes)
- M : 0...4095 (4096 bytes)

CPU 1214C / 1215C / 1217C

- I : 0...1023 (1024 bytes)
- Q : 0...1023 (1024 bytes)
- M : 0...8191 (8192 bytes)

Data type

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Table 5- 29 Bit and bit sequence data types

Data type	Bit size	Number type	Number range	Constant examples	Address examples
Bool	1	Boolean	FALSE or TRUE	TRUE	I1.0 Q0.1 M50.7 DB1.DBX2.3 Tag_name
		Binary	2#0 or 2#1	2#0	
		Unsigned integer	0 or 1	1	
		Octal	8#0 or 8#1	8#1	
		Hexadecimal	16#0 or 16#1	16#1	
Byte	8	Binary	2#0 to 2#1111_1111	2#1000_1001	IB2 MB10 DB1.DBB4 Tag_name
		Unsigned integer	0 to 255	15	
		Signed integer	-128 to 127	-63	
		Octal	8#0 to 8#377	8#17	
		Hexadecimal	B#16#0 to B#16#FF, 16#0 to 16#FF	B#16#F, 16#F	
Word	16	Binary	2#0 to 2#1111_1111_1111_1111	2#1101_0010_1001_0110	MW10 DB1.DBW2 Tag_name
		Unsigned integer	0 to 65535	61680	
		Signed integer	-32768 to 32767	72	
		Octal	8#0 to 8#177_777	8#170_362	
		Hexadecimal	W#16#0 to W#16#FFFF, 16#0 to 16#FFFF	W#16#F1C0, 16#A67B	
DWord	32	Binary	2#0 to 2#1111_1111_1111_1111_1111_1111_1111_1111	2#1101_0100_1111_1110_1000_1100	MD10 DB1.DBD8 Tag_name
		Unsigned integer*	0 to 4_294_967_295	15_793_935	
		Signed integer*	-2_147_483_648 to 2_147_483_647	-400000	
		Octal	8#0 to 8#37_777_777_777	8#74_177_417	
		Hexadecimal	DW#16#0000_0000 to DW#16#FFFF_FFFF, 16#0000_0000 to 16#FFFF_FFFF	DW#16#20_F30A, 16#B_01F6	

* The underscore "_" is a thousands separator to enhance readability for numbers greater than eight digits.

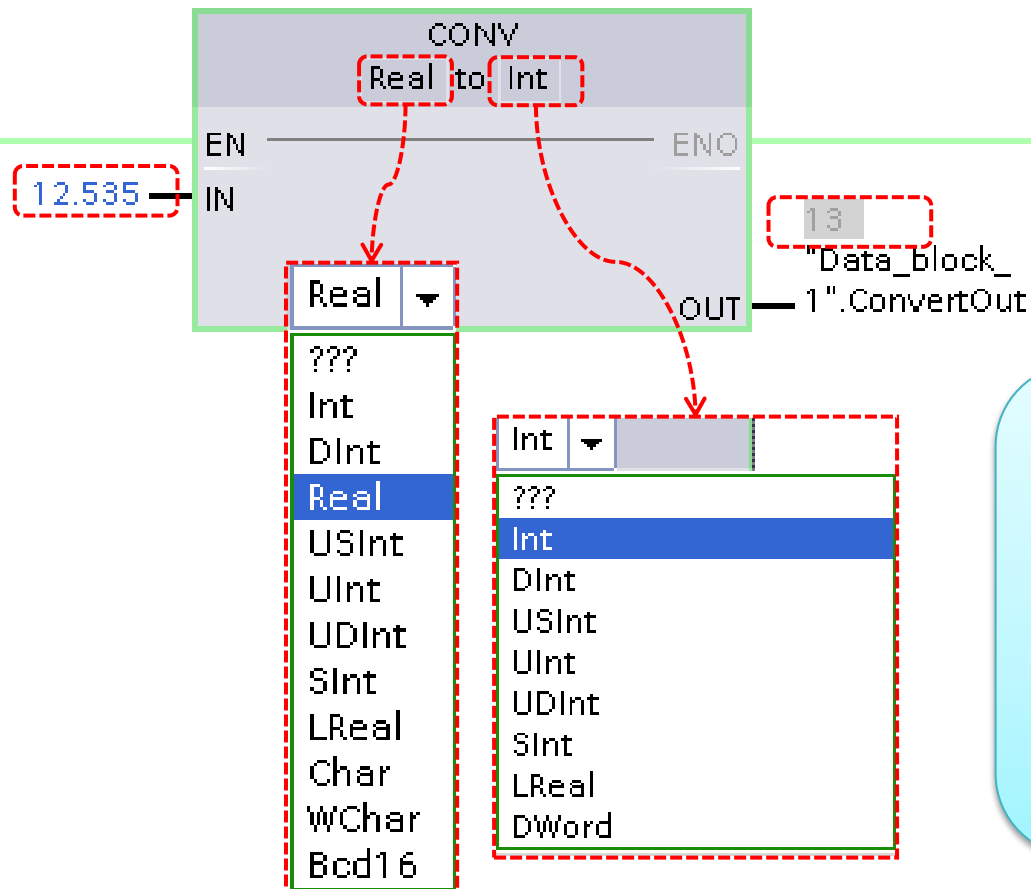
Table 5- 30 Integer data types (U = unsigned, S = short, D= double)

Data type	Bit size	Number Range	Constant examples	Address examples
USInt	8	0 to 255	78, 2#01001110	MB0, DB1.DBB4, Tag_name
SInt	8	-128 to 127	+50, 16#50	
UInt	16	0 to 65,535	65295, 0	MW2, DB1.DBW2, Tag_name
Int	16	-32,768 to 32,767	30000, +30000	
UDInt	32	0 to 4,294,967,295	4042322160	MD6, DB1.DBD8, Tag_name
DInt	32	-2,147,483,648 to 2,147,483,647	-2131754992	

Table 5- 31 Floating-point real data types (L=Long)

Data type	Bit size	Number range	Constant Examples	Address examples
Real	32	-3.402823e+38 to -1.175 495e-38, ±0, +1.175 495e-38 to +3.402823e+38	123.456, -3.4, 1.0e-5	MD100, DB1.DBD8, Tag_name
LReal	64	-1.7976931348623158e+308 to -2.2250738585072014e-308, ±0, +2.2250738585072014e-308 to +1.7976931348623158e+308	12345.123456789e40, 1.2E+40	DB_name.var_name Rules: - No direct addressing support - Can be assigned in an OB, FB, or FC block interface table

4.5 Convert Data



Convert เป็นการแปลงค่าข้อมูลตัวเลขแบบหนึ่งไปเป็นอีกแบบหนึ่ง เช่นตัวอย่างแปลงจาก Real เป็น Integer ซึ่งแปลงจากเลขทศนิยมเป็นเลขจำนวนเต็ม จาก 12.535 จึงปัดเศษทศนิยมได้เป็น เลขจำนวนเต็ม 13

4.6 Move

คลิก MOVE แล้ว ลากมาวางใน Ladder

The screenshot shows the Siemens STEP 7 software interface. The main window displays a Ladder Logic network (Network 5) with a MOVE instruction. The instruction is highlighted with a red dashed box. A red arrow points from the 'Basic instructions' library on the right to the MOVE instruction. The library lists various instructions including MOVE, MOVE_BLK, and MOVE_BLK_VARIANT. The MOVE instruction is described as 'Move value'.

Network 5:

Comment

EN — ENO

IN 100

OUT1

%M2.2 "Trig_03"

%MW4 "Cal_Result_01"

Basic instructions

Name	Description
General	
Bit logic operations	
Timer operations	
Counter operations	
Comparator operations	
Math functions	
Move operations	
MOVE	Move value
Deserialize	Deserialize
Serialize	Serialize
MOVE_BLK	Move block
MOVE_BLK_VARIANT	Move block
UMOVE_BLK	Move block uninterruptible
FILL_BLK	Fill block
UFILL_BLK	Fill block uninterruptible
SWAP	Swap
Variant	
Legacy	
Conversion operations	
Program control operati...	
Word logic operations	
Shift and rotate	

Extended instructions

Technology

Communication

Optional packages

Properties Info Diagnostics

General Cross-references Compile Syntax

Show all messages

Message

Message	Go to	?	Date	Time
Connected to PLC_Training_B01, address IP=192.168.0.71.			5/16/2016	8:08:20 PM
Connection to PLC_Training_B01 terminated.			5/16/2016	8:13:10 PM

Default tag t... Main Online & dia...

Connection to PLC_Training_B01 termi...

4.7 Processing of analog values

Analog signal module จะอ่านค่าจาก input หรือส่ง output ที่มีช่วง $\pm 10\text{ V}$, $\pm 5\text{ V}$, $\pm 2.5\text{ V}$, $0 - 20\text{ mA}$, or $4 - 20\text{ mA}$ ซึ่งการคำนวณก็จะนำค่า Analog input value มา scaling กับ Engineering units ซึ่ง Analog input value นั้นมีช่วงระหว่าง -27648 ถึง 27648 ซึ่งตัวอย่างด้านล่างได้แสดงการ scaling อุณหภูมิ โดยที่ Analog input value มีช่วงระหว่าง 0-27648 Engineering units มีช่วงระหว่าง 50-100 องศาเซลเซียส

4 mA ↓ 20 mA	Analog input value	Engineering units	0% ↓ 100%
	0	50 °C	
	6192	62.5 °C	
	12384 X	75 °C Y	
	18576	87.5 °C	
	27648	100 °C	

$$\frac{(X - 0)}{(27,648 - 0)}$$

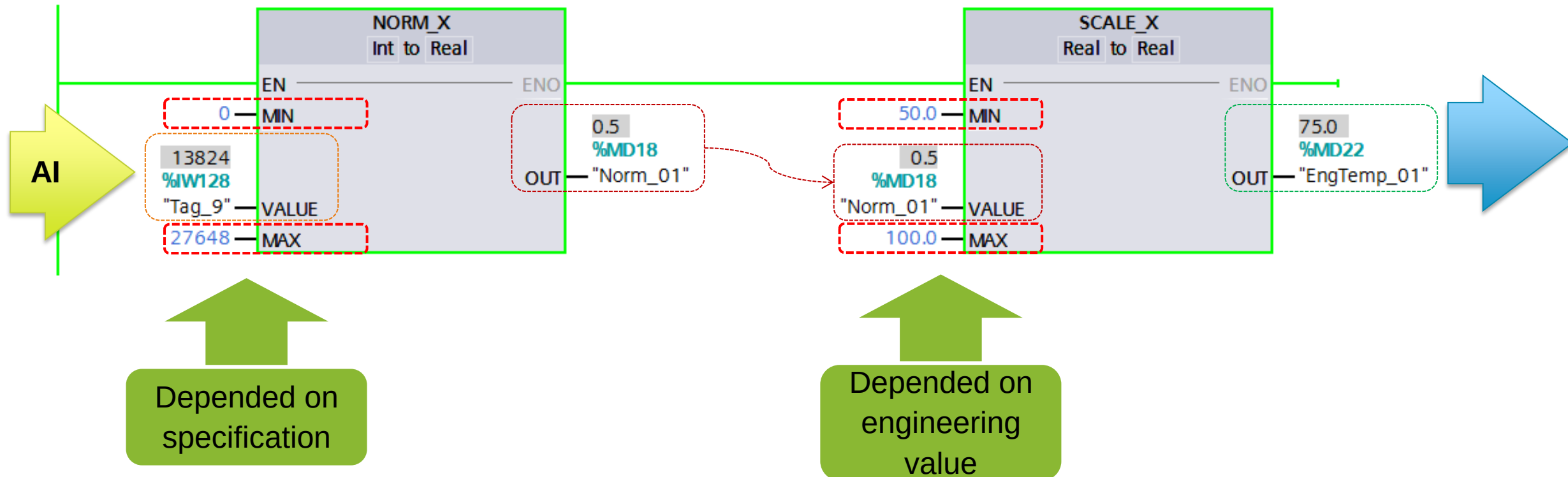
=

$$\frac{(Y - 50)}{(100 - 50)}$$

$$Y = 50 + \frac{(X-0) * (100-50)}{(27,648-0)}$$

4.7 Processing of analog values

ตัวอย่างการใช้งาน Block NORM_X และ SCALE_X เพื่อการ scale analog value
(ค่า NORM out ที่ออกมา นั้นจะมีค่าระหว่าง 0.0-1.0 เท่านั้น)



4.7 Processing of analog values

Range	Raw value (16 bits) of the PLC tag		Module output [mA]			Measuring range [%]
	Decimal	Hexadecimal	0 .. 20 (unipolar)	-20 .. +20 (bipolar)	4 .. 20 (life zero)	
Overflow	32767	7FFF	> 23.515	> 23.515	> 22.810	> 117.593
Overrange	32511	7EFF	23.515	23.515	22.810	117.593
	...	...	...	...	...	...
	27649	6C01	20.001	20.001	20.001	100.004
Nominal range (unipolar / life zero)	27648	6C00	20		20	100
	...	...	...		...	...
	0	0000	0		4	0

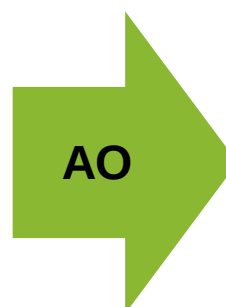
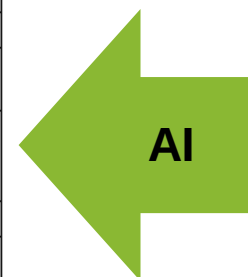
Analog Input / Output for CPU unit

Technical data	Description
Number of inputs	2
Type	Voltage (single-ended)
Full-scale range	0 to 10 V
Full-scale range (data word)	0 to 27648
Overshoot range	10.001 to 11.759 V
Overshoot range (data word)	27649 to 32511
Overflow range	11.760 to 11.852 V
Overflow range (data word)	32512 to 32767
Resolution	10 bits
Maximum withstand voltage	35 V DC

Technical data	Description
Number of outputs	2
Type	Current
Full-scale range	0 to 20 mA
Full-scale range (data word)	0 to 27648
Overshoot range	20.01 to 23.52 mA
Overshoot range (data word)	27649 to 32511
Overflow range	see footnote ¹
Overflow range data word	32512 to 32767
Resolution	10 bits

Analog Input / Output for Signal module (SM 1231, SM 1232)

Model	SM 1231 AI 4 x 13 bit	SM 1231 AI 8 x 13 bit	SM 1231 AI 4 x 16 bit
Number of inputs	4	8	4
Type	Voltage or current (differential): Selectable in groups of 2		Voltage or current (differential)
Range	± 10 V, ± 5 V, ± 2.5 V, 0 to 20 mA, or 4 mA to 20 mA		± 10 V, ± 5 V, ± 2.5 V, ± 1.25 V, 0 to 20 mA or 4 mA to 20 mA
Full scale range (data word)	-27648 to 27648 voltage / 0 to 27648 current		
Overshoot/undershoot range (data word)	Voltage: 32511 to 27649 / -27649 to -32,512 Current: 32511 to 27649 / 0 to -4864		
Refer to the section on analog input ranges for voltage and current (Page 1471).			
Overflow/underflow (data word)	Voltage: 32767 to 32512 / -32513 to -32768 Current 0 to 20 mA: 32767 to 32512 / -4865 to -32768 Current 4 to 20 mA: 32767 to 32512 (values below -4864 indicate open wire)		
Refer to the section on input ranges for voltage and current (Page 1471).			
Resolution1	12 bits + sign bit		15 bits + sign bit
Maximum withstand voltage/current	± 35 V / ± 40 mA		



Technical data	SM 1232 AQ 2 x 14 bit	SM 1232 AQ 4 x 14 bit
Number of outputs	2	4
Type	Voltage or current	
Range	± 10 V, 0 to 20 mA, or 4 mA to 20 mA	
Resolution	Voltage: 14 bits Current: 13 bits	
Full scale range (data word)	Voltage: -27,648 to 27,648 ; Current: 0 to 27,648 Refer to the output ranges for voltage and current (Page 1472).	
Accuracy (25 °C / -20 to 60 °C)	$\pm 0.3\%$ / $\pm 0.6\%$ of full scale	
Settling time (95% of new value)	Voltage: 300 μ s (R), 750 μ s (1 uF) Current: 600 μ s (1 mH), 2 ms (10 mH)	
Load impedance	Voltage: $\geq 1000 \Omega$ Current: $\leq 600 \Omega$	
Maximum output short circuit current	Voltage mode: ≤ 24 mA Current mode: ≥ 38.5 mA	
Behavior on RUN to STOP	Last value or substitute value (default value 0)	

Analog Input / Output for Signal module (SM 1234)

Model	SM 1234 AI 4 x 13 bit / AQ 2 x 14 bit
Number of inputs	4
Type	Voltage or Current (differential): Selectable in groups of 2
Range	± 10 V, ± 5 V, ± 2.5 V, 0 to 20 mA, or 4 mA to 20 mA
Full scale range (data word)	-27648 to 27648
Overshoot/undershoot range (data word)	Voltage: 32511 to 27649 / -27649 to -32512 Current: 32511 to 27649 / 0 to -4864 Refer to the section on input ranges for voltage and current (Page 1471).
Overflow/underflow (data word)	Voltage: 32767 to 32512 / -32513 to -32768 Current: 32767 to 32512 / -4865 to -32768 Refer to the section on input ranges for voltage and current (Page 1471).
Resolution	12 bits + sign bit
Maximum withstand voltage/current	± 35 V / ± 40 mA

Technical data	SM 1234 AI 4 x 13 bit / AQ 2 x 14 bit
Number of outputs	2
Type	Voltage or current
Range	± 10 V or 0 to 20 mA
Resolution	Voltage: 14 bits ; Current: 13 bits
Full scale range (data word)	Voltage: -27648 to 27648; Current: 0 to 27648 Refer to the section on output ranges for voltage and current (Page 1472).
Accuracy (25 °C / -20 to 60 °C)	$\pm 0.3\%$ / $\pm 0.6\%$ of full scale
Settling time (95% of new value)	Voltage: 300 μ s (R), 750 μ s (1 μ F) Current: 600 μ s (1 mH), 2 ms (10 mH)
Load impedance	Voltage: $\geq 1000 \Omega$ Current: $\leq 600 \Omega$
Maximum output short circuit current	Voltage mode: ≤ 24 mA Current mode: ≥ 38.5 mA

Note : Using analog

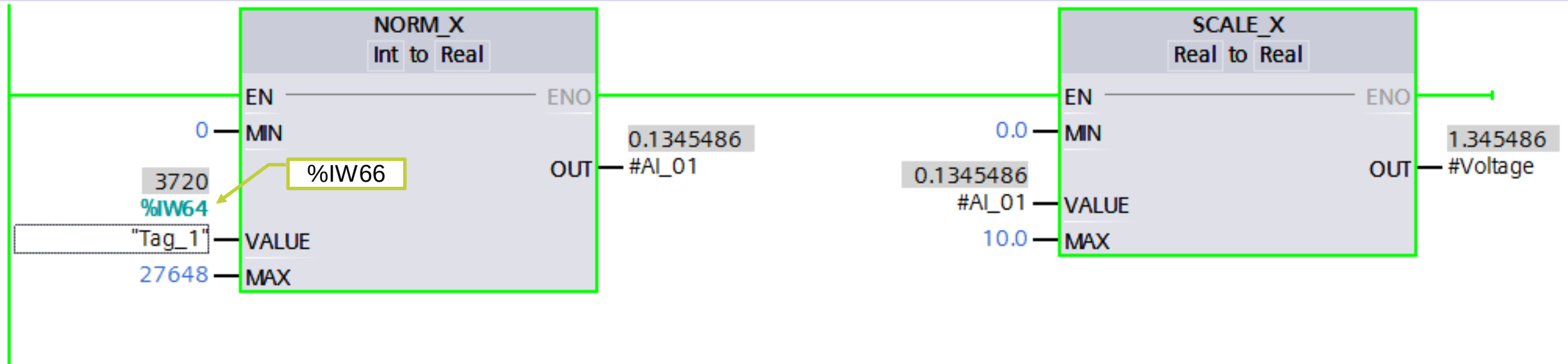
MASS_99 [CPU 1212C DC/DC/DC]

General	IO tags	System constants	Texts
Name	Type	Address	Tag table
	Int	%IW64	
	Int	%IW66	

Channel 0 (AI 0) → %IW64

Channel 1 (AI 1) → %IW66 : 0..10 Volt

Value	Voltage
0	0
xxxx	3
27648	10

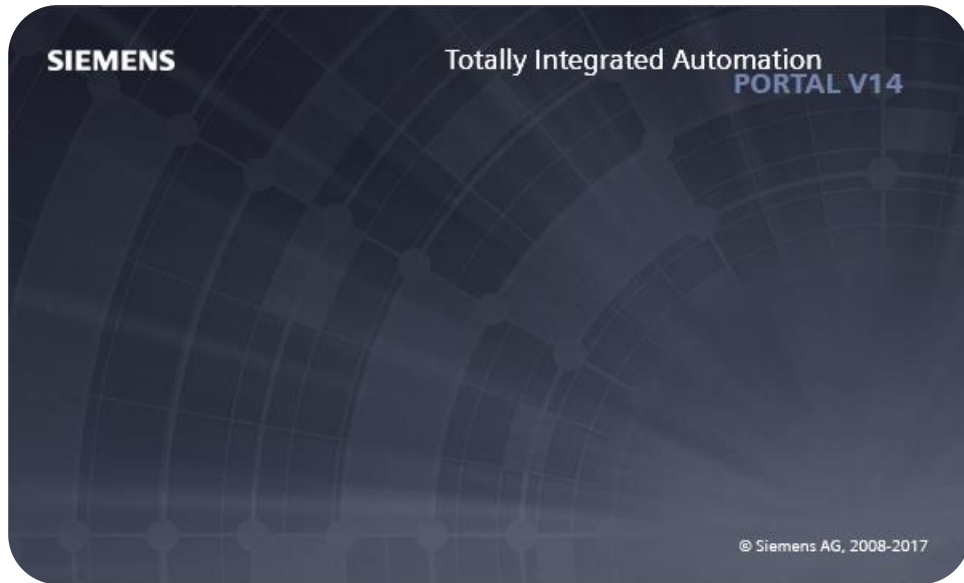


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Appendix

5. PLC Concept & Programming Concept



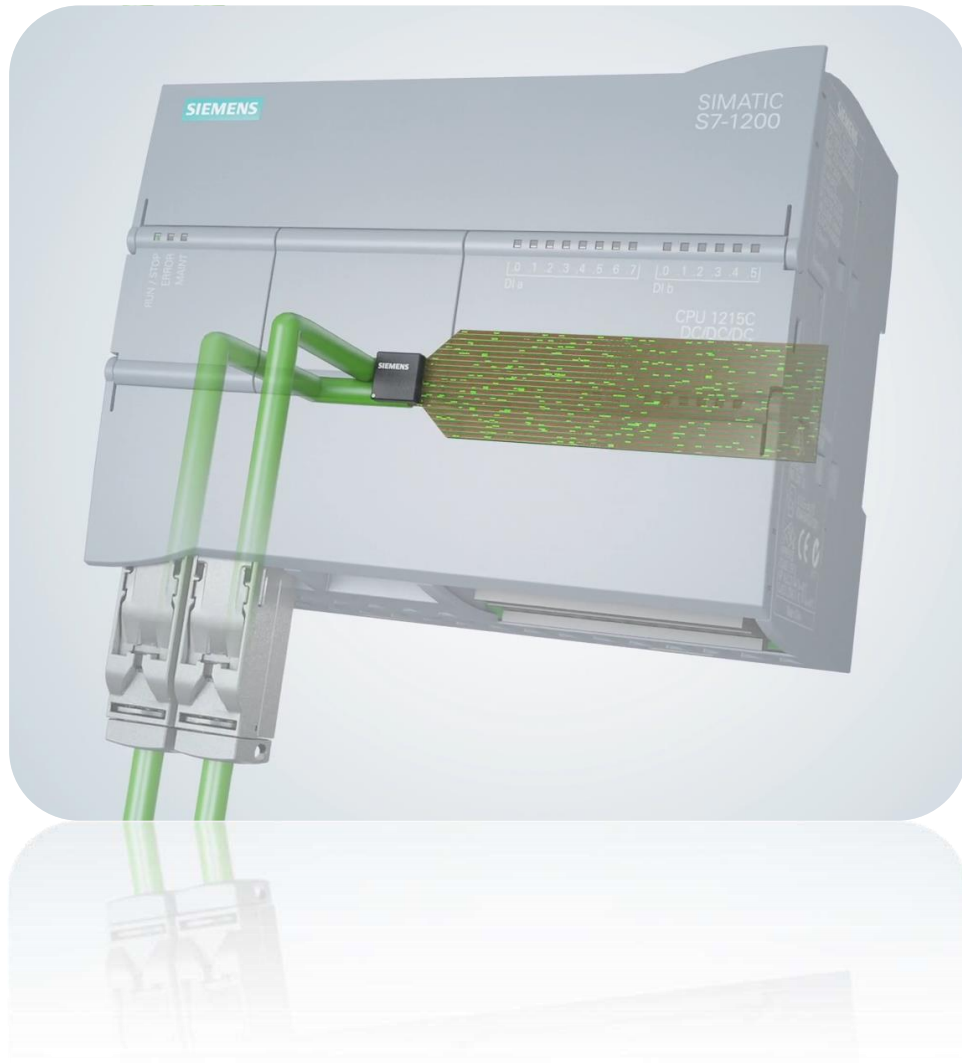
1

PLC concepts

2

Programming concepts

5.1 PLC concepts



Execution of the user program



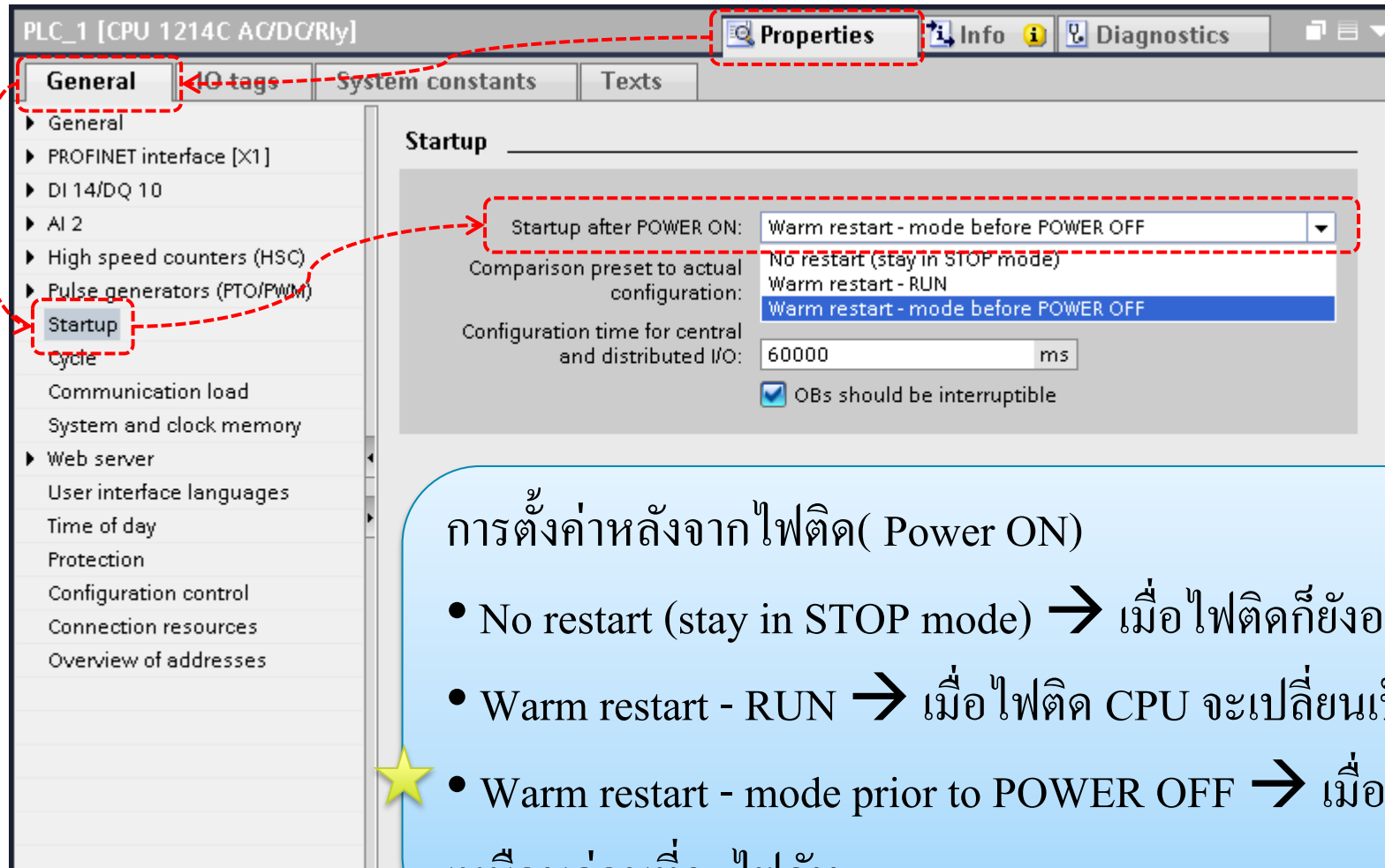
**Data storage, memory areas, I/O and
addressing**

5.1.1 Operating modes of the CPU

CPU ของ Siemens มีโหมดการทำงานอยู่ 3 โหมดด้วยซึ่งมีหลักการทำงานดังนี้

- **STOP mode**
 - CPU จะไม่ทำการประมวลผลโปรแกรม ท่านสามารถดาวน์โหลดโปรเจกต์ลงมายัง CPU ได้ในโหมดนี้
- **STARTUP mode**
 - OBs เริ่มต้น (ถ้ามี) จะประมวลผลไป 1 ครั้ง และจะไม่ประมวลผล interrupt เลขขณะที่อยู่ในโหมดนี้
- **RUN mode**
 - โปรแกรม OBs จะเริ่มทำงานแบบวนซ้ำไปเรื่อยๆ interrupts ก็จะถูกรับรู้และประมวลผลในโหมดนี้ ที่จุดใดก็ได้ของโปรแกรม และสามารถดาวน์โหลดโปรเจกต์ลงไปยัง CPU ได้ (โปรเจกต์ที่เปลี่ยนแปลงเพียงเล็กน้อยหลังจากที่ดาวน์โหลดตอน STOP mode)

5.1.1 Start up

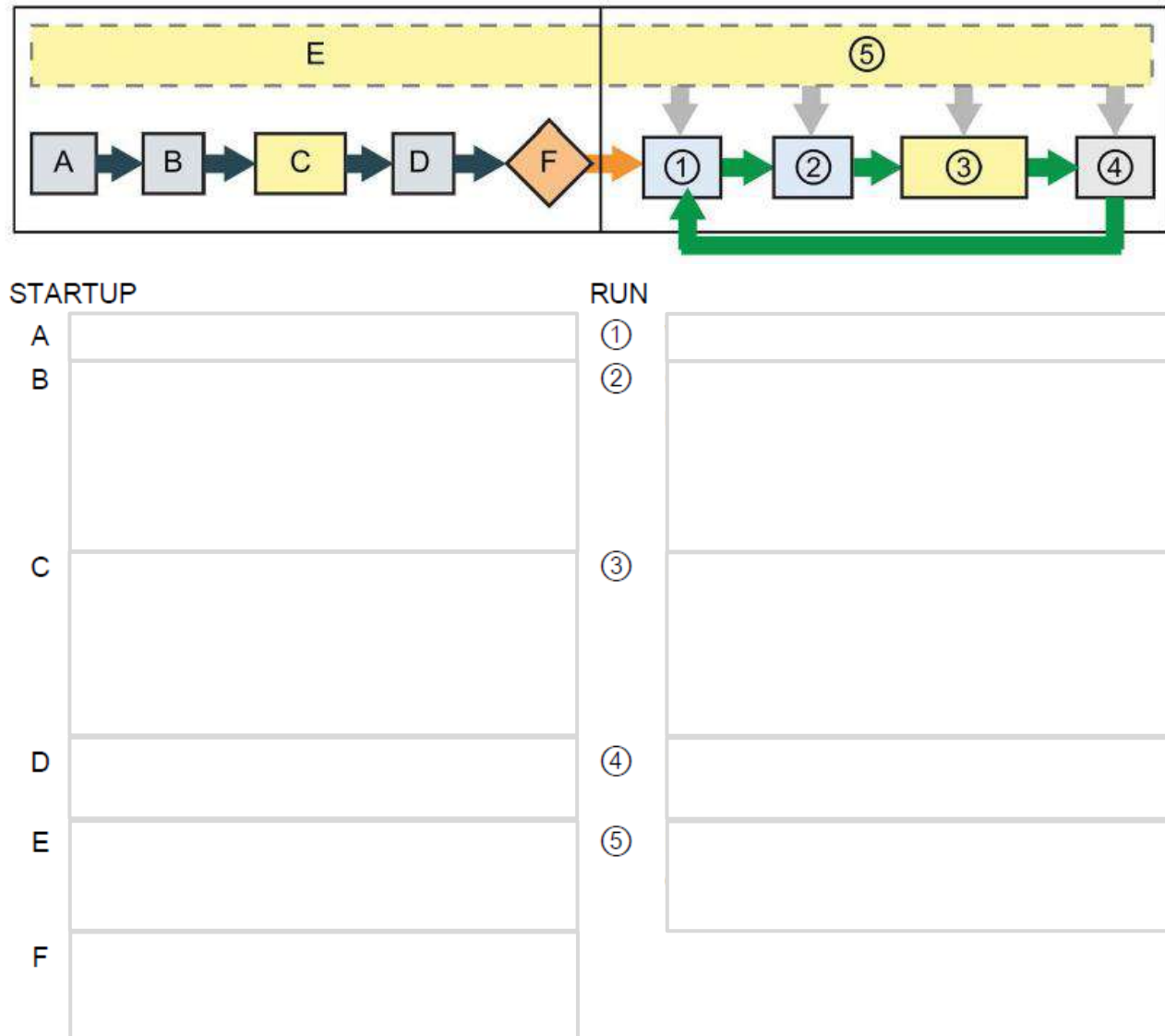


สามารถสั่ง RUN / Stop
ผ่านหน้าจอ HMI ก็ได้

การตั้งค่าหลังจากไฟติด(Power ON)

- No restart (stay in STOP mode) → เมื่อไฟติดก็ยังคงอยู่ในสถานะ STOP mode
- Warm restart - RUN → เมื่อไฟติด CPU จะเปลี่ยนเป็นสถานะ RUN mode
- Warm restart - mode prior to POWER OFF → เมื่อไฟติดขึ้นมา CPU ก็จะอยู่ในสถานะเหมือนก่อนที่จะไฟดับ

5.1.1 Operating modes of the CPU : STARTUP

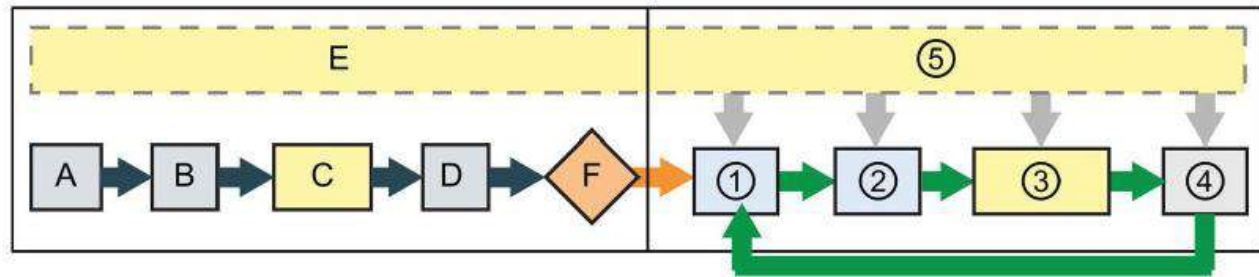


The CPU also performs the following tasks during the startup processing.

- Interrupts are queued but not processed during the startup phase
- No cycle time monitoring is performed during the startup phase
- Configuration changes to HSC (high-speed counter), PWM (pulse-width modulation), and PtP (point-to-point communication) modules can be made in startup
- Actual operation of HSC, PWM and point-to-point communication modules only occurs in RUN

After the execution of the startup OBs finishes, the CPU goes to RUN mode and processes the control tasks in a continuous scan cycle.

5.1.1 Operating modes of the CPU : RUN



STARTUP

- A Clears the I (image) memory area
- B Initializes the Q output (image) memory area with either zero, the last value, or the substitute value, as configured, and zeroes PB, PN, and AS-i outputs
- C Initializes non-retentive M memory and data blocks to their initial value and enables configured cyclic interrupt and time of day events.
Executes the startup OBs.
- D Copies the state of the physical inputs to I memory
- E Stores any interrupt events into the queue to be processed after entering RUN mode
- F Enables the writing of Q memory to the physical outputs

RUN

- ① Writes Q memory to the physical outputs
- ② Copies the state of the physical inputs to I memory
- ③ Executes the program cycle OBs
- ④ Performs self-test diagnostics
- ⑤ Processes interrupts and communications during any part of the scan cycle

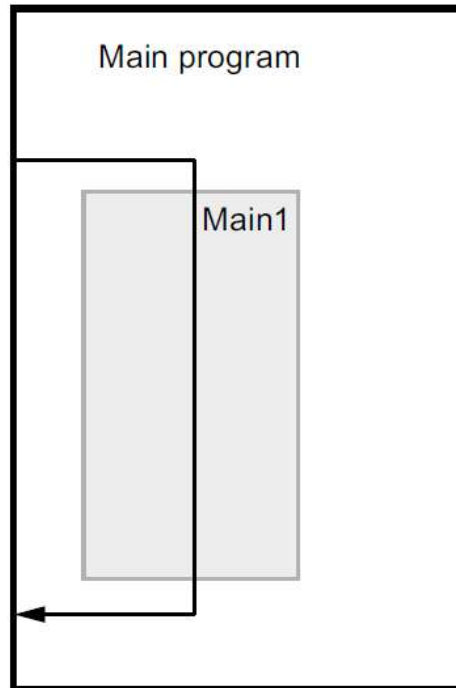
- The system guarantees that the scan cycle will be completed in a time period called the maximum cycle time; otherwise a time error event is generated.
- Communications processing occurs periodically throughout the scan, possibly interrupting user program execution.
- Self-diagnostic checks include periodic checks of the system and the I/O module status checks.
- Interrupts can occur during any part of the scan cycle, and are event-driven. When an event occurs, the CPU interrupts the scan cycle and calls the OB that was configured to process that event. After the OB finishes processing the event, the CPU resumes execution of the user program at the point of interruption.

5.1.2 Programming concept

Linear programming

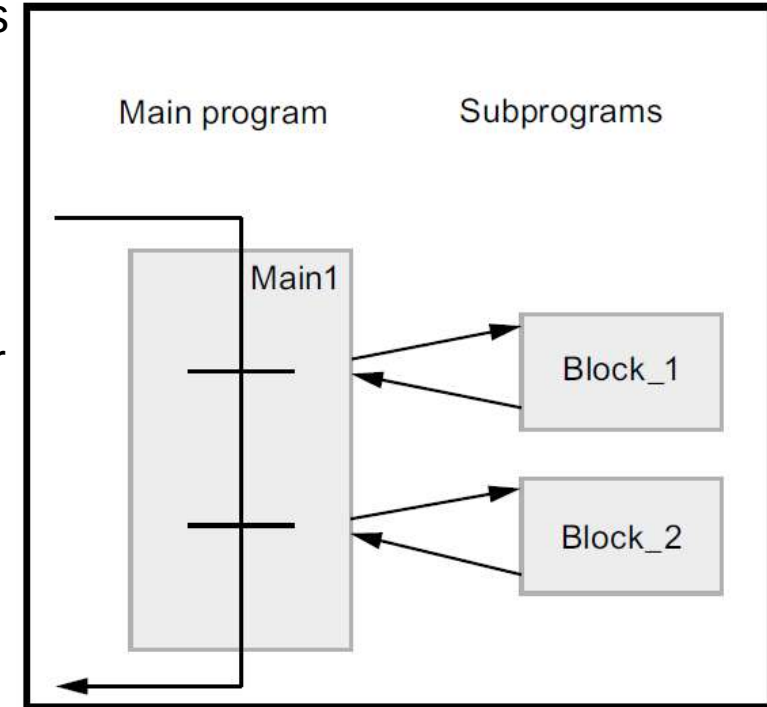
Solutions for small automation tasks can be programmed linearly in a program cycle OB. This is only recommended for simple programs.

The following figure shows a linear program schematically: The "Main1" program cycle OB contains the complete user program.



Structured programming

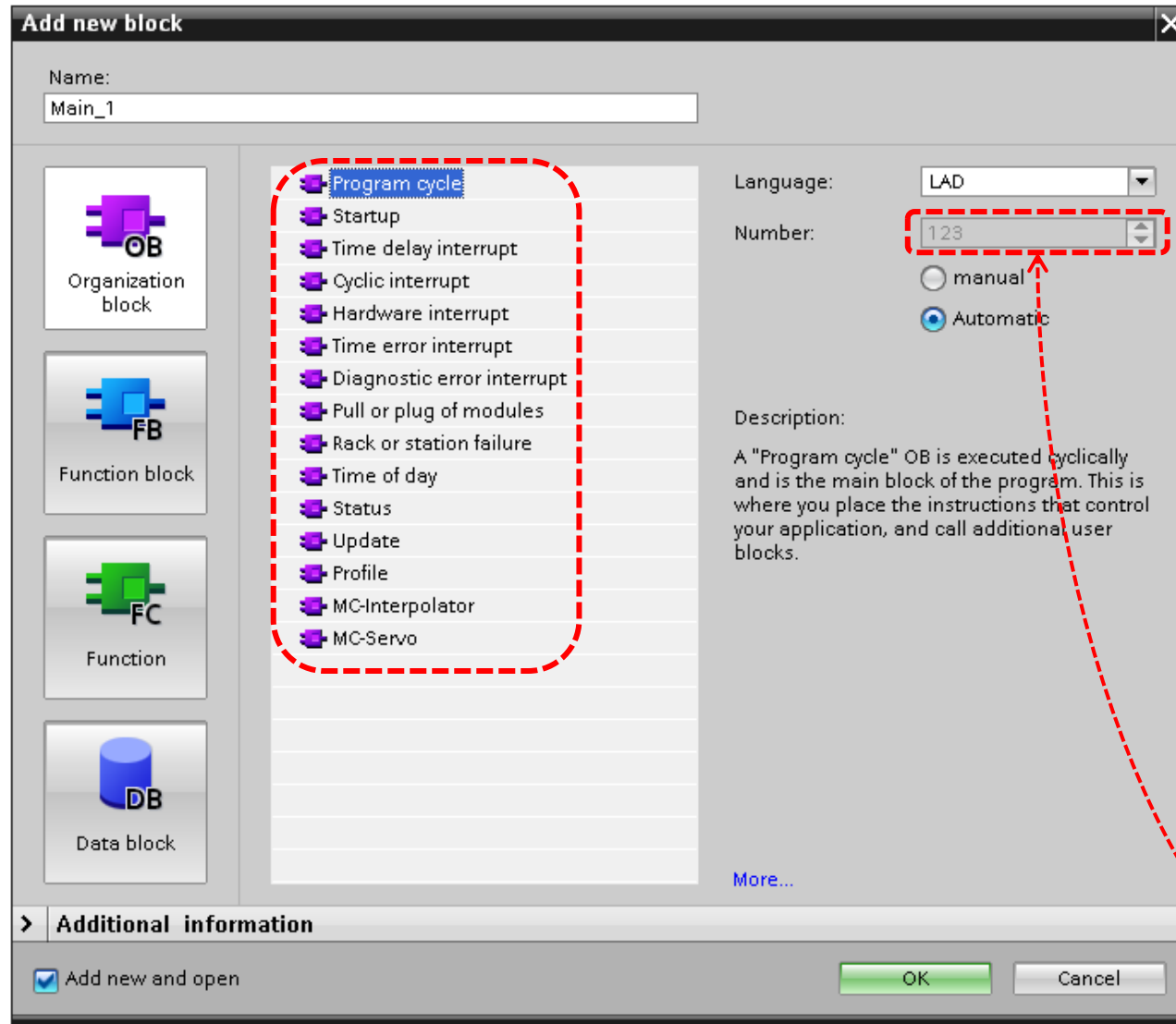
Complex automation tasks can be more easily handled and managed by dividing them into smaller sub-tasks that correspond to the technological functions of the process or that can be reused. These sub-tasks are represented in the user program by blocks. Each block is then an independent section of the user program



5.1.3 Overview of the block types

Block type	Brief description
Organization blocks (OB)	Organization blocks define the structure of the user program.
Functions (FC)	Functions contain program routines for recurring tasks. They <i>have no "memory"</i> .
Function blocks (FB)	Function blocks are code blocks that <i>store their values permanently in instance data blocks</i> , so that they remain available even after the block has been executed.
Instance data blocks (DB)	Instance data blocks are <i>assigned to a function block</i> when it is called for the purpose of storing program data.
Global data blocks (DB)	Global data blocks are data areas for storing data that <i>can be used by any blocks</i> .

5.1.4 Organization blocks



- 1) Program cycle OB ★
- 2) Startup OB ★
- 3) Time delay interrupt OB
- 4) Cyclic interrupt OB ★
- 5) Hardware interrupt OB ★
- 6) Time error interrupt OB
- 7) Diagnostic error interrupt OB
- 8) Pull or plug of modules OB
- 9) Rack or station failure OB
- 10) Time of day OB
- 11) Status OB
- 12) Update OB
- 13) Profile OB
- 14) MC-Interpolator OB
- 15) MC-Servo OB

1,123-32767

100,123-32767

20-23,123-32767

30-38,123-32767

40-47

80

82

83

86

10-17,123-32767

55

56

57

92

91

ตัวเลข OB

5.1.4 Organization blocks : Startup OB

Startup OB

Startup OBs ***execute one time when the operating mode of the CPU changes from STOP to RUN, including powering up in the RUN mode and in commanded STOP-to-RUN transitions.*** After completion, the main "Program cycle" begins executing.

Time delay interrupt OB

Time delay interrupt OBs ***execute after a time delay that you configure.***

Cyclic interrupt OB

Cyclic interrupt OBs ***execute at a specified interval.*** You can configure up to a total of four cyclic interrupt events, with one OB corresponding to each cyclic interrupt event.

5.1.4 Organization blocks

Hardware interrupt OB

Hardware interrupt OBs **execute when the relevant hardware event occurs**. A hardware interrupt OB interrupts normal cyclic program execution in reaction to a signal from a hardware event.

Limits on hardware interrupt events are as follows:

Edges:

- Rising edge events: maximum of 16
- Falling edge events: maximum of 16

HSC events:

- CV=PV: maximum of 6
- Direction changed: maximum of 6
- External reset: maximum of 6

Time error interrupt OB

If configured, the time error interrupt OB (OB 80) **executes when either the scan cycle exceeds the maximum cycle time or a time error event occurs**. If triggered, it executes, interrupting normal cyclic program execution or any other event OB.

The occurrence of either of these events generates a diagnostic buffer entry describing the event. The diagnostic buffer entry is generated regardless of the existence of the time error interrupt OB.

Scan cycle exceeds maximum cycle time

CPU cannot start requested OB

Queue overflow occurred

Exercise – 4 : Startup OB

The image displays the Siemens STEP 7 software interface, illustrating the steps to add a Startup Organization Block (OB) to a PLC program.

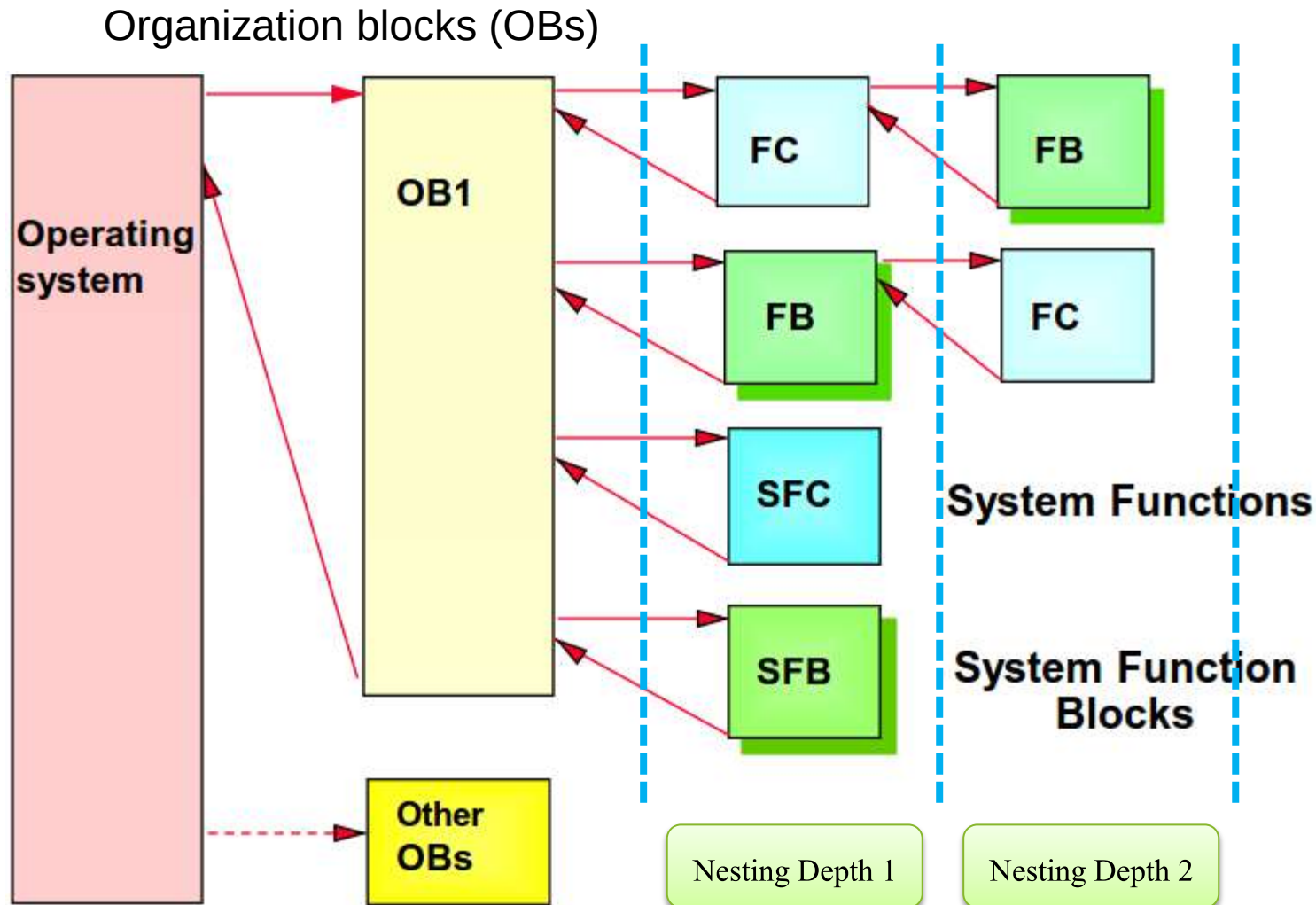
Left Panel (Project tree): Shows the project structure. The "Program blocks" folder is highlighted, and the "Add new block" option is selected.

Center Panel (Add new block dialog): Shows the "Add new block" dialog. The "Name" field is set to "Startup". The "Language" is set to "LAD". The "Number" is set to "100". The "Manual" radio button is selected. The "Description" field contains the text: "A 'Startup' OB will execute one time when the operating mode of the PLC changes from STOP to RUN. After completion, the main 'Program cycle' OB will begin executing."

Right Panel (Block interface): Shows the "Block interface" for the "Startup [OB100]" block. The "Block title" is "Complete Restart". The "Network 1" is defined by the ladder logic: $\%Q0.0$ (Set coil) and "Run_M1" (Normally open contact).

Annotations: Red dashed lines and boxes highlight the "Add new block" option in the Project tree, the "Startup" block in the "Add new block" dialog, the "Startup [OB100]" block in the Project tree, and the "OK" button in the "Add new block" dialog.

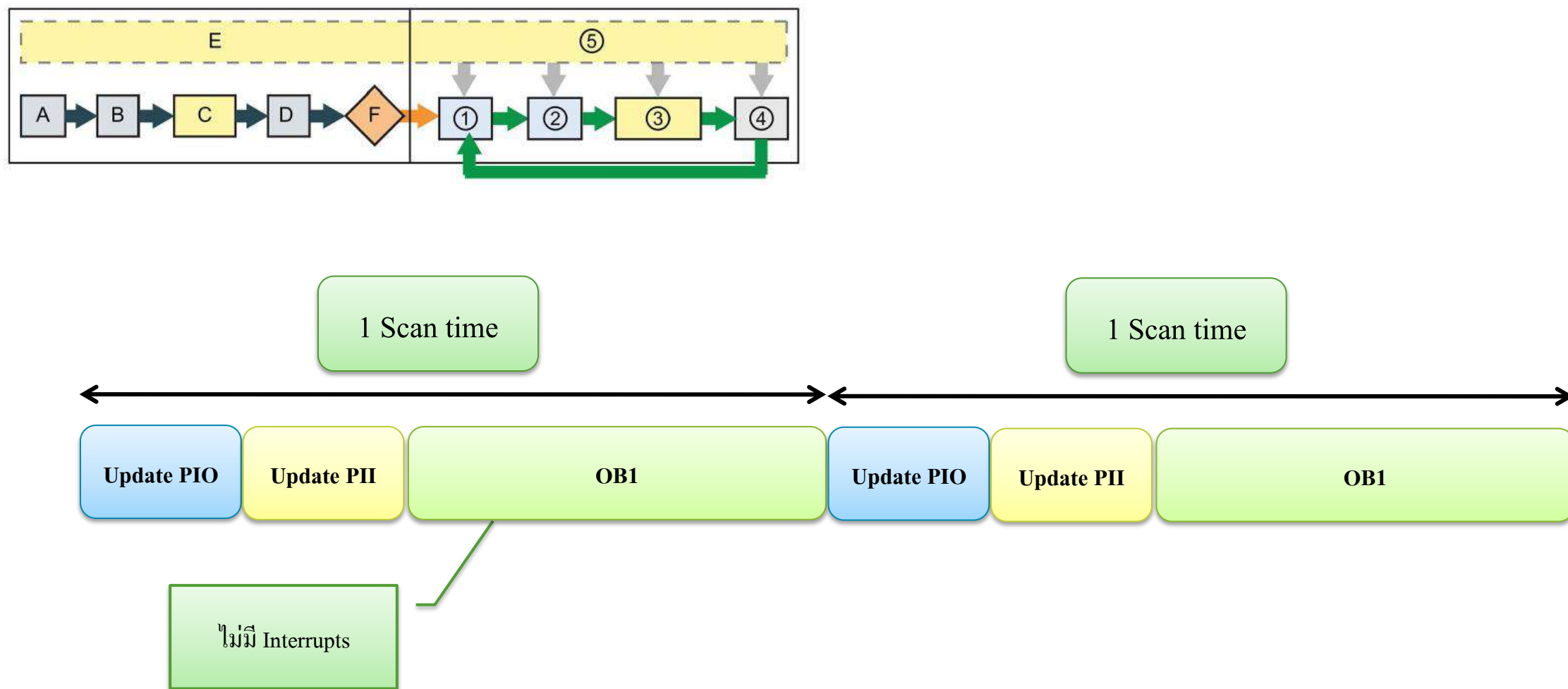
5.1.5 Operating system



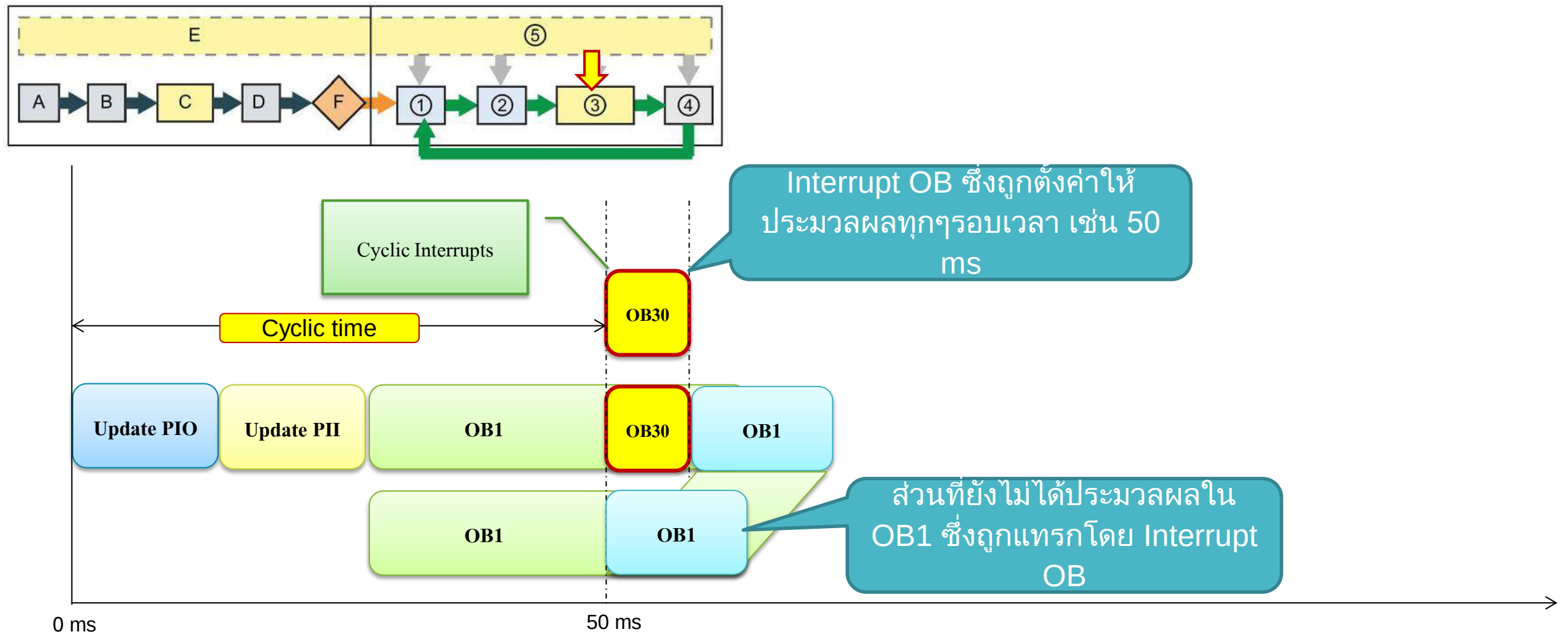
OB1 คือ OB ที่ทำหน้าที่หลักในการประมวลผล
OBs ไม่สามารถที่จะเรียก OBs ด้วยกันได้ และไม่สามารถเรียกจาก FC หรือ FB ได้เช่นกัน
OBs จะทำงานตามเงื่อนไขและคุณสมบัติของแต่ละ OBs

จากตัวอย่าง FB ,FC,SFC,SFB เป็น
Instructions ย่อยใน OB1 ซึ่งมีการ รับ-ส่ง
ข้อมูลกันภายใน (ลูกศร) จากในภาพจะมี
Nesting Depth = 2 หมายถึง มีความลึกใน
การจัดการข้อมูลสองชั้น

5.1.6 OB Execution : Without interrupts



5.1.6 OB Execution : With interrupts



Exercise – 5 : Cyclic Interrupt OB

The screenshot illustrates the configuration of a Cyclic Interrupt Organization Block (OB) in Siemens STEP 7. The 'Add new block' dialog is open, showing the 'Cyclic interrupt' block selected from the list. The 'Language' is set to LAD, and the 'Number' is 30. The 'Cyclic time (ms)' is set to 750. A yellow tooltip indicates that the value must be between 1 and 60000. The 'Description' field contains text about cyclic interrupts. The 'OK' button is highlighted with a red dashed box. The 'Project tree' on the right shows the 'Cyclic interrupt [OB30]' block added to the 'Program blocks' of the 'PLC_Training_B01_01' project. The 'Block interface' on the right shows the ladder logic diagram for the cyclic interrupt, with inputs '%M2.5' (labeled 'Enable_Flash') and '%Q0.3' (labeled 'Run_M4') connected to the output '%Q0.3' (labeled 'Run_M4').

Add new block dialog:

- Name: Cyclic interrupt
- Language: LAD
- Number: 30
- Cyclic time (ms): 750
- Description: A "Cyclic interrupt" C programs at periodic intervals independently of cyclic program execution. The intervals can be entered in this dialog or in the properties of the OB.
- Buttons: OK, Cancel, more...

Project tree:

- Basic program_FW42
 - Devices & networks
 - PLC_Training_B01 [CPU 1212C AC/DC/Rly]
 - PLC_Training_B01_01 [CPU 1212C AC/DC/Rly]
 - Program blocks
 - Add new block
 - Cyclic interrupt [OB30]

Block interface:

- Block title: Cyclic interrupt [OB30]
- Network 1:
 - Inputs: %M2.5 (Enable_Flash), %Q0.3 (Run_M4)
 - Output: %Q0.3 (Run_M4)

5.1.6 OBs priority

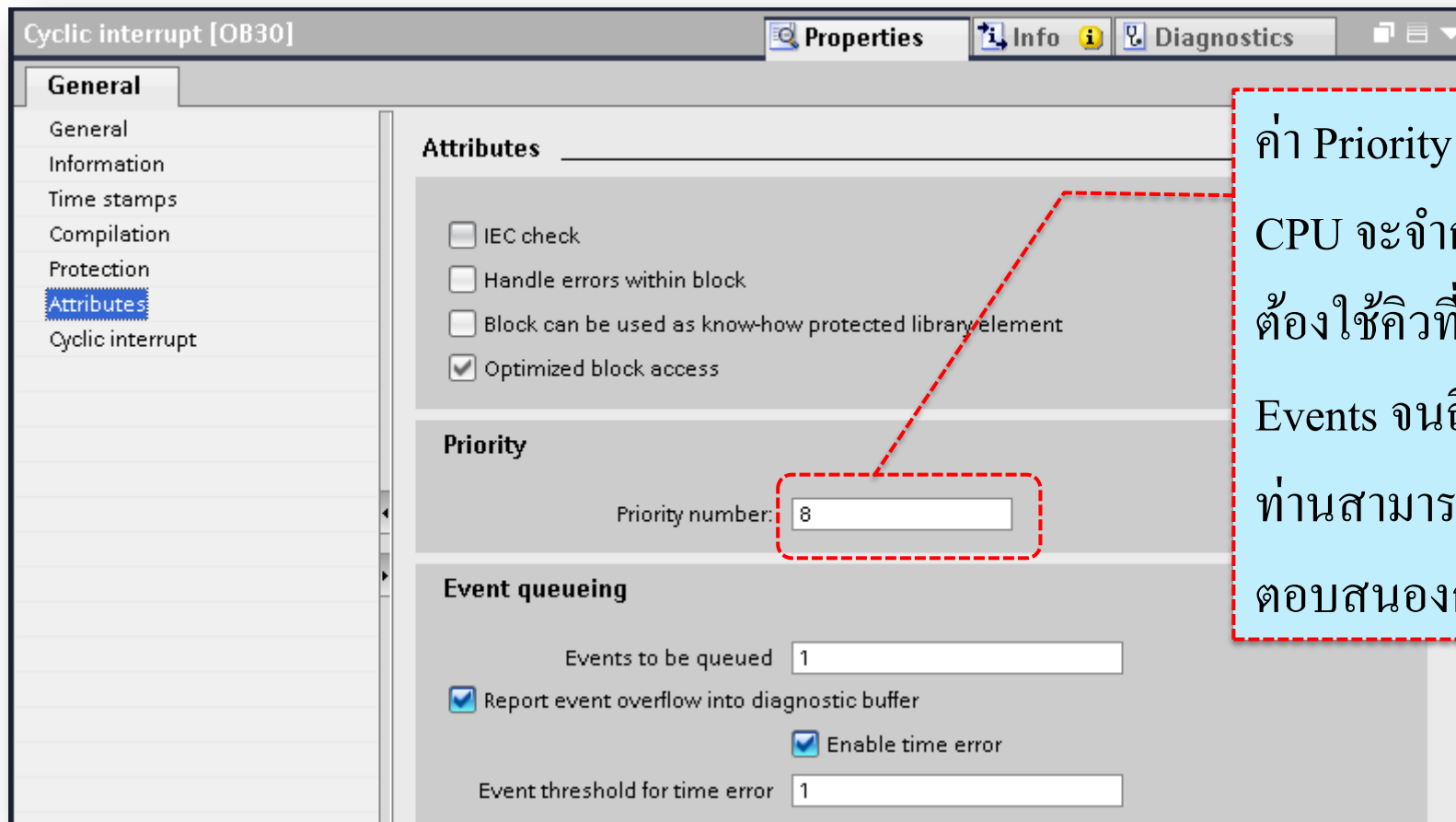
Event	Quantity allowed	Default OB priority
Program cycle	1 program cycle event Multiple OBs allowed	1 ⁴
Startup	1 startup event ¹ Multiple OBs allowed	1 ⁴
Time delay	Up to 4 time events 1 OB per event	3
Cyclic interrupt	Up to 4 events 1 OB per event	8
Hardware interrupt	Up to 50 hardware interrupt events ² 1 OB per event, but you can use the same OB for multiple events	18
		18
Time error	1 event (only if configured) ³	22 or 26 ⁴
Diagnostic error	1 event (only if configured)	5
Pull or plug of modules	1 event	6
Rack or station failure	1 event	6
Time of day	Up to 2 events	2
Status	1 event	4
Update	1 event	4
Profile	1 event	4
MC-Servo	1 event	25
MC-Interpolator	1 event	24

ตัวเลขมาก Priority สูงกว่า
 ตัวเลขน้อย Priority น้อยกว่า
 เช่น Hardware interrupt มี Priority สูงกว่า
 Cyclic interrupt

- 1) Program cycle กับ Startup จะไม่ทำงานพร้อมกัน Startup จะทำงานจนเสร็จก่อนจากนั้น Program cycle จึงจะทำงาน
- 2) ท่านสามารถใช้ hardware interrupt OBs มากกว่า 50 ตัว ถ้าท่านใช้คำสั่ง DETACH หรือ ATTACH
- 3) ท่านสามารถตั้งค่าให้ CPU ยังคงอยู่ในสถานะ RUN ได้ ถ้า scan cycle เกินเวลาที่ตั้งไว้ หรือท่านสามารถใช้ คำสั่ง RE_TRIGR เพื่อรีเซ็ตเวลาของ cycle อย่างไรก็ตามถ้าเกิด Time Error ขึ้นเป็นครั้งที่ 2 CPU จะ STOP
- 4) ถ้าเป็น V4.0 , V4.1 จะเป็น 22 ถ้าเปลี่ยนจาก V3.0 เป็น V4.0 ,V4.1 จะเป็น 26 แต่สุดท้ายก็สามารถเปลี่ยนค่าได้ แต่ต้องอยู่ระหว่าง 22-26

The **lowest priority class is 1** (for the main program cycle), and the **highest priority class is 26**.

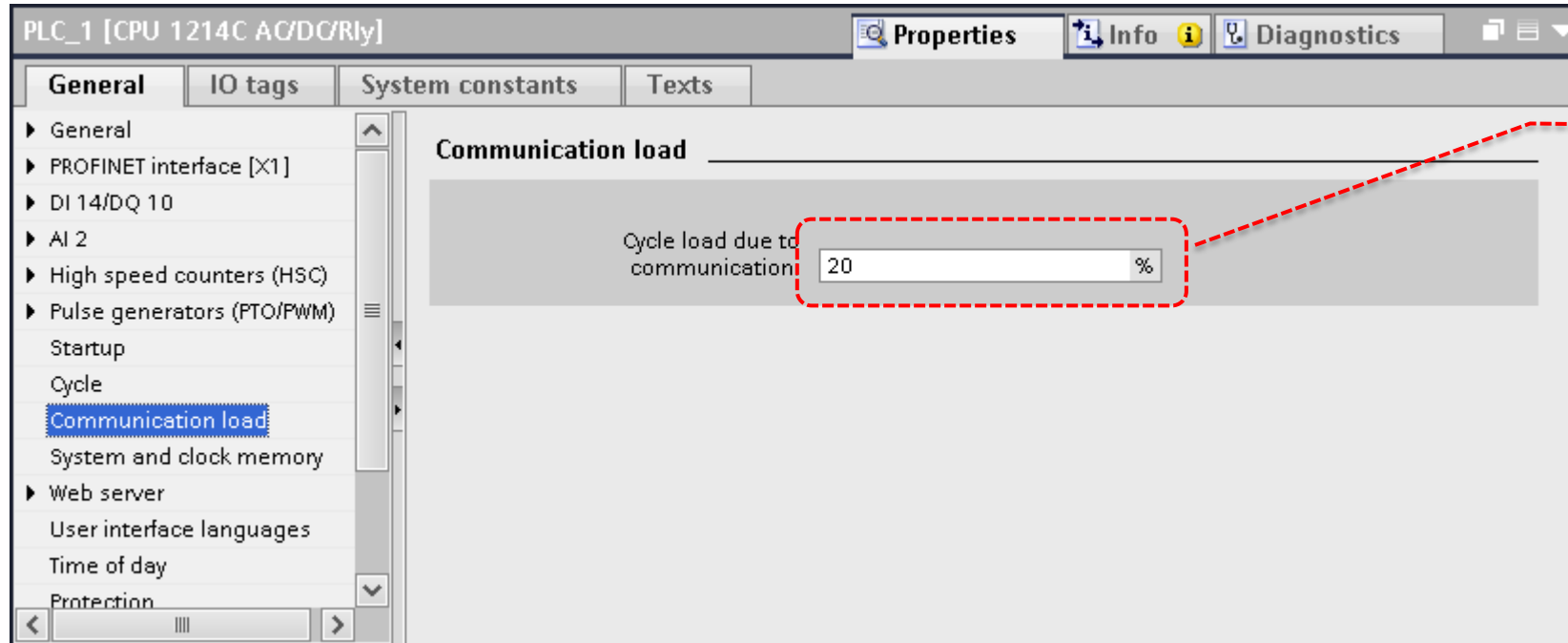
5.1.6 Setting OB's priority



ค่า Priority (ระดับความสำคัญ)

CPU จะจำกัดจำนวนคิวของ Events จากแหล่งเดียว
ต้องใช้คิวที่ไม่เหมือนกันในแต่ละ Event และถ้าใช้
Events จนถึงจุดที่จำกัดไว้แล้ว Event ต่อไปก็จะมี
ท่านสามารถใช้ time error interrupt OB เพื่อให้
ตอบสนองกับคิวที่เกินมา

5.1.7 Communication load



ค่า % ของเวลาที่ใช้ในการสื่อสาร
สามารถใส่ค่าได้ 15-50 %
(ค่าปกติ 20%)

5.1.8 Accessing the data of the S7-1200

STEP 7 ท่านสามารถสร้าง tags สำหรับ address ของข้อมูลซึ่งเกี่ยวข้องกับ address ของ memory โดยจะแบ่งข้อมูลออกเป็นดังนี้

1. Global memory → สำหรับเก็บ memory ของ Inputs ,Outputs และ memory Blocks ทุกBlocks สามารถเข้าถึงได้นำไปใช้ได้
2. PLC tag table → ท่านสามารถใส่ symbolic names ใน tag table นี้กำหนดที่อยู่ให้กับ memory และก็จะเป็น global tag ในโปรแกรมซึ่งท่านสามารถใช้ symbolic names ในการเรียกเพื่อใช้งาน tag ได้เลย
3. Data block (DB) → ท่านสามารถเพิ่ม DBs ในโปรแกรมของท่านได้เพื่อที่จะใส่ข้อมูลสำหรับ block ที่มีการคำนวณ ซึ่งข้อมูลนั้นก็ยังคงอยู่ใน DB แม้ว่า block ที่ได้ใช้ข้อมูลจาก DB นี้ได้ทำงานไปแล้ว ซึ่ง DB จะมีอยู่ 2 ประเภทคือ
 - I. Global DB ซึ่ง block ทุก block สามารถเข้ามาใช้งานได้
 - II. Instance DB เป็น DB เฉพาะของแต่ละ FB ซึ่งมีไว้ใส่ parameter ของ FB นั้นเอง
4. Temp memory → มีไว้เพื่อเก็บ block ที่เรียกมา execute ถ้าทำเสร็จแล้วก็จะนำไปเก็บไว้ที่อื่น แล้วนำ block ต่อไปมา execute ต่อ

5.1.9 CPU memory

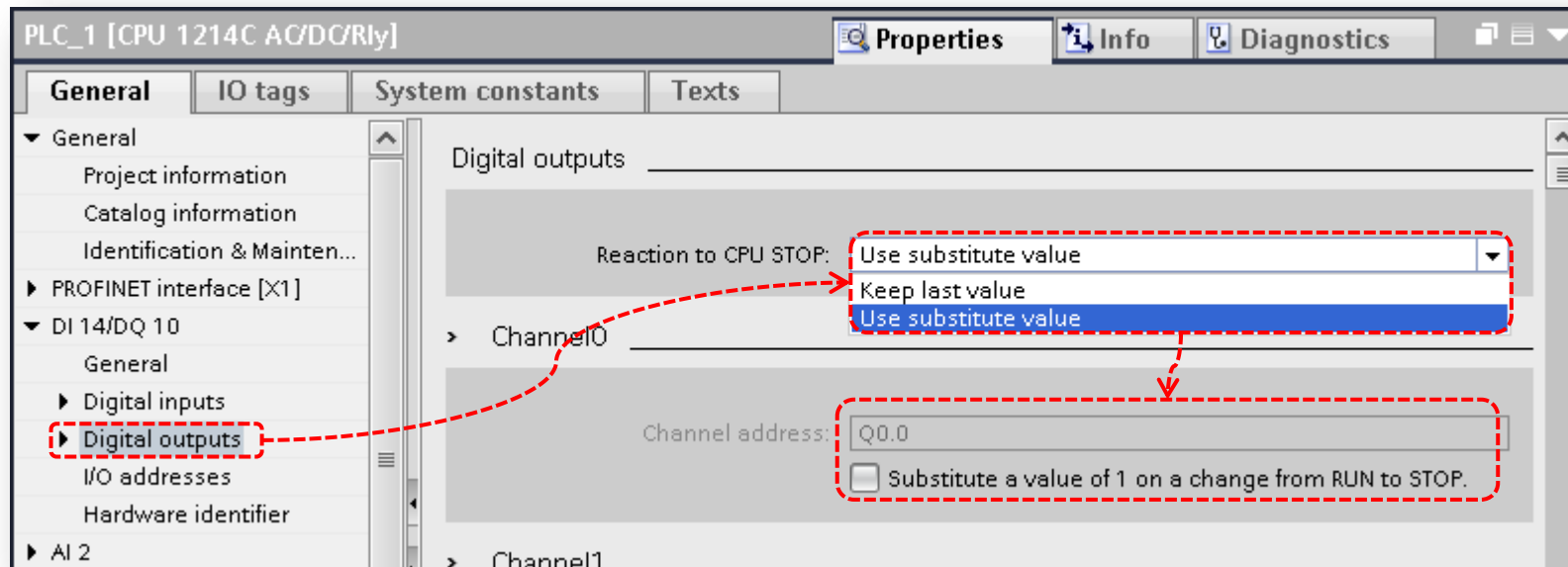
- **Load memory** is non-volatile storage for the user program, data and configuration. When you download a project to the CPU, the CPU first stores the program in the Load memory area. This area is located either in a memory card (if present) or in the CPU. The CPU maintains this non-volatile memory area through a power loss. The memory card supports a larger storage space than that built-in to the CPU.
- **Work memory** is volatile storage for some elements of the user project while executing the user program. The CPU copies some elements of the project from load memory into work memory. This volatile area is lost when power is removed, and is restored by the CPU when power is restored.
- **Retentive memory** is non-volatile storage for a limited quantity of work memory values. The CPU uses the retentive memory area to store the values of selected user memory locations during power loss. When a power down or power loss occurs, the CPU restores these retentive values upon power up.

5.1.10 Diagnostics buffer

Diagnostics buffer สามารถดูได้ขณะที่ online เท่านั้น

CPU จะมี buffer สำหรับ diagnostics เพื่อเก็บ event ที่เกิดขึ้น โดยจะแบ่งเป็น เกิด event อะไรขึ้น เวลาไหนวันไหน เป็น event แบบไหนและจะมีคำอธิบายสำหรับแต่ละ event ว่าเกิดอะไรขึ้น โดยแต่ละ event จะเรียงลำดับก่อนหลังโดยจะเก็บได้มากที่สุด 50 อันดับถ้าเกินกว่านี้ อันดับเก่าๆจะหายไปอันดับใหม่จะมาแทนที่ เช่น CPU errors , Module errors ,START , STOP and RUN

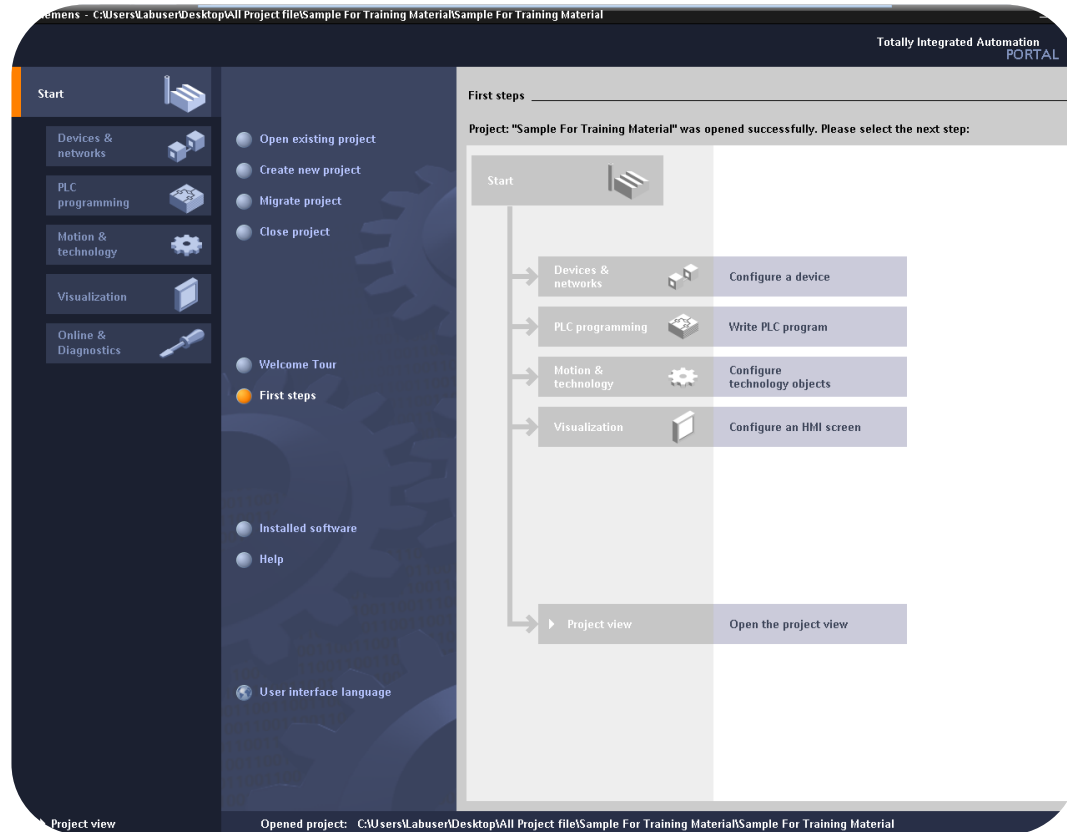
5.1.11 Configuring the outputs on a RUN-to-STOP transition



เป็นการตั้งค่าเมื่อ CPU จะเปลี่ยนสถานะจาก RUN ไปยัง STOP ว่าจะให้ CPU คงค่าอะไรไว้

1. Keep last value ค้างค่าล่าสุดที่ได้ประมวลผลไว้
2. Use substitute value ให้ใช้ค่าที่กำหนดไว้

5.2 Programming concepts



1

Structuring your user program

2

Using blocks to structure your program

3

Programming language

5.2.1 Structuring your user program

การสร้างโปรแกรม

- **Organization Block (OB)**

เป็นการทำงานในรูปของ event ที่เจาะจงและสามารถ interrupt user program ได้ ซึ่งจะเรียกได้ว่า OB คือการทำงานหลักของ CPU ส่วน FB FC คือฟังก์ชันย่อยเหมือน block ให้ OB เรียกมาใช้

- **Function Block (FB)**

คือการเขียนฟังก์ชันการทำงานย่อย หรือ Subroutine เป็นเหมือนการกำหนด Block ขึ้นมาเอง เพื่อให้ OB ได้เรียก Block ที่เราสร้างขึ้นนี้ไปใช้งานซึ่งจะมี instance DB ไว้เพื่อเก็บค่าพารามิเตอร์ส่วนตัวด้วย

- **Function (FC)**

คล้ายๆกันกับ FB แต่จะไม่มี instance DB เป็นของตัวเอง จึงใช้ global DB ในการเก็บข้อมูล

5.2.2 FB and Instance DB

Siemens - D:\Training project files\Basic_01\Basic program\Basic

Project Edit View Insert Online Options Tools Window

Save project

Project tree

Devices

Name

Basic program

Add new device

Devices & networks

PLC_Training_B01 [CPU 1214C AC/DC]

Device configuration

Online & diagnostics

Program blocks

Add new block

Main [OB1]

Add new block

Name: Block_1

Language: LAD

Number: 1

Manual

Automatic

Description: Function blocks are code blocks that store their values permanently in instance data blocks, so that they remain available after the block has been executed.

OB

Function block

FC

Function

DB

Data block

Additional information

Title:

Comment:

Version: 0.1

Family:

Author:

User-defined ID:

OK

Cancel

Add new and open

Basic program > i device > PLC_Training_B01 [CPU 1214C AC/DC/Rly] > Program blocks > FB_001_Start motor [FB1]

FB_001_Start motor

Name	Data type	Default value	Retain	Accessible f...	Visible in ...	Setpoint	Comment
Input							
2 Start	Bool	false	Non-ret...	☒	☒		
3 Stop	Bool	false	Non-retain	☒	☒		
Output							
5 Running	Bool	false	Retain	☒	☒		
6 InOut							
7 <Add new>							
8 Static							
9 <Add new>							
10 Temp							
11 <Add new>							
12 Constant							
13 <Add new>							

ประกาศตัวแปรใน FB

Network 1:

Comment

#Start

#Stop

#Running

#Running

โปรแกรมใน FB

5.2.2 FB and Instance DB

Approved
Partner

Factory
Automation

SIEMENS

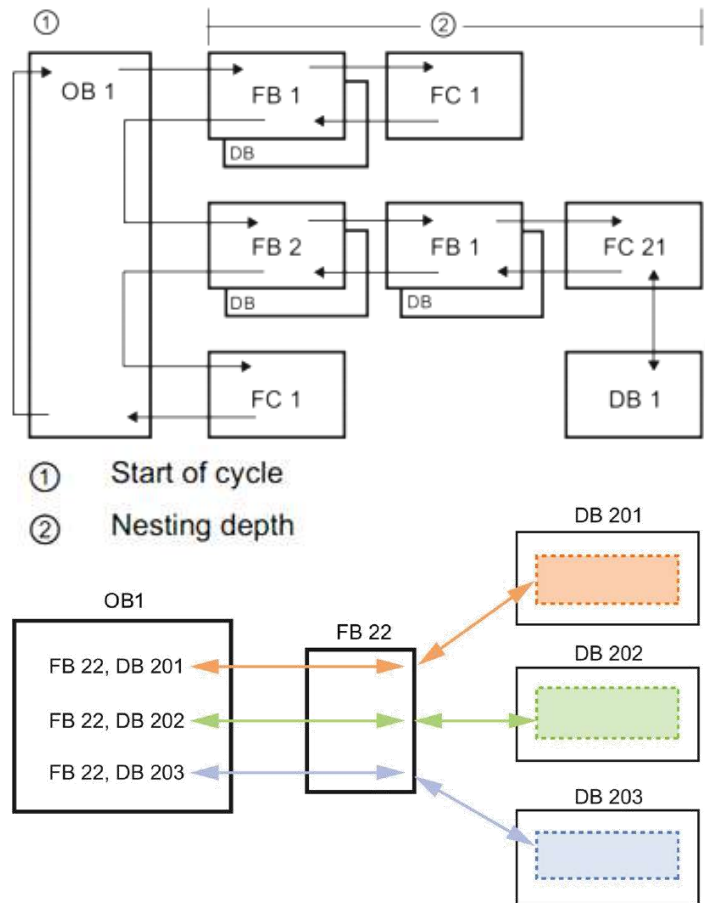
The screenshots illustrate the following steps:

- Project Tree:** The 'FB_001_Start motor' function block is selected in the 'Program blocks' folder.
- Network Editor:** The function block is inserted into Network 6. The inputs are 'start_M2' (Start) and 'stop_M2' (Stop). The output is 'M2_Running'.
- Call Options:** The 'Call options' dialog is open. Under 'Data block', the 'Name' is 'FB_001_Start motor_DB_M2' and the 'Number' is '3'. The 'Automatic' option is selected.
- Define Tag:** The 'Define tag' dialog is open. The tags 'start_M2', 'stop_M2', and 'M2_Running' are defined with addresses '%M2.3', '%M2.4', and '%M2.5' respectively, all of type 'Bool'.

The final result shows the function block 'FB_001_Start motor' with its data block 'FB_001_Start motor_DB_M2' and the tags 'start_M2', 'stop_M2', and 'M2_Running' defined in the project.

5.2.2 Using blocks to structure your program

You can nest the block calls for a more modular structure. In the following example, the nesting depth is 3: the program cycle OB plus 3 layers of calls to code blocks.



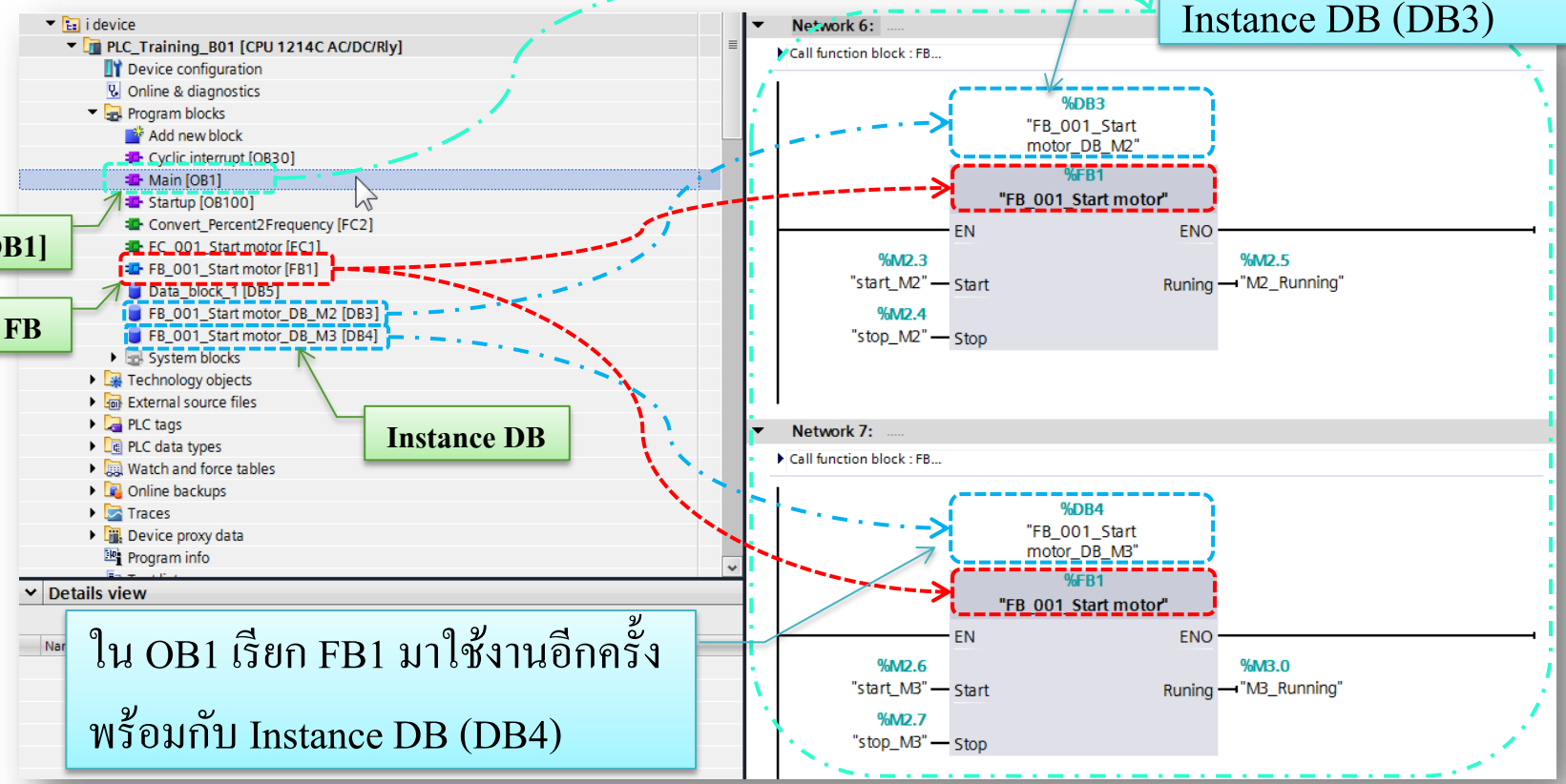
Main [OB1]

FB

Instance DB

ใน OB1 เรียก FB1 มาใช้งานอีกครั้ง
พร้อมกับ Instance DB (DB4)

ใน OB1 เรียก FB1 มา
ใช้งานพร้อมกับ
Instance DB (DB3)



Exercise – 6 : FB (Function block)

Siemens - D:\Training project files\Basic_01\Basic program_FW42\Basic program_FW42

Project Edit View Insert Online Options Tools Window Help

Save project

Go online Go offline

Search in project

Project tree

Devices

Basic program_FW42

Add new device

Devices & networks

PLC_Training_B01 [CPU 1212C AC/DC/Rly]

PLC_Training_B01_01 [CPU 1212C AC/DC/Rly]

Device configuration

Online & diagnostics

Program blocks

Add new block

Main [OB1]

Start motor [FB1]

System blocks

Technology objects

External source files

Watch and force tables

Add new watch table

Force table

Online backups

Traces

Device properties

Details view

Name Address

Basic program_FW42 > PLC_Training_B01_01 [CPU 1212C AC/DC/Rly] > Program blocks > Start motor [FB1]

Start motor

Name	Data type	default value	Retain	Accessible f...	Writa...	Visible in ...	Setpoint
1	Input						
2	CMD_Start	Bool	false	Non-retain			
3	CMD_Stop	Bool					
4	CMD_HMI_Start	Bool					
5	Status_Fault	Bool					
6	Output						
7	Run	Bool					
8	InOut						
9	<Add new>						
10	Static						

ประกาศตัวแปรต่างๆ พร้อมกำหนด Data type

สร้าง Function block พร้อมตั้งชื่อ

Block title: ...

Comment

Network 1: ...

Start / stop motor

#CMD_Start

#CMD_Stop

#Status_Fault

#Run

#Run

#CMD_HMI_Start

เขียนเงื่อนไข หรือ โปรแกรม

Drag & Drop

Start motor_DB_M2 [DB4]

Network 4: ...

Control M2 using FB

%QB4

"Start motor_DB_M2"

%FB1

"Start motor"

EN

ENO

%Q0.1

Run → "Run_M2"

"CMD_Start_M2" → CMD_Start

%Q0.5

"CMD_Stop_M2" → CMD_Stop

%M2.3

"CMD_HMI_Start_M2" → CMD_HMI_Start

%M2.4

"Status_Fault_M2" → Status_Fault

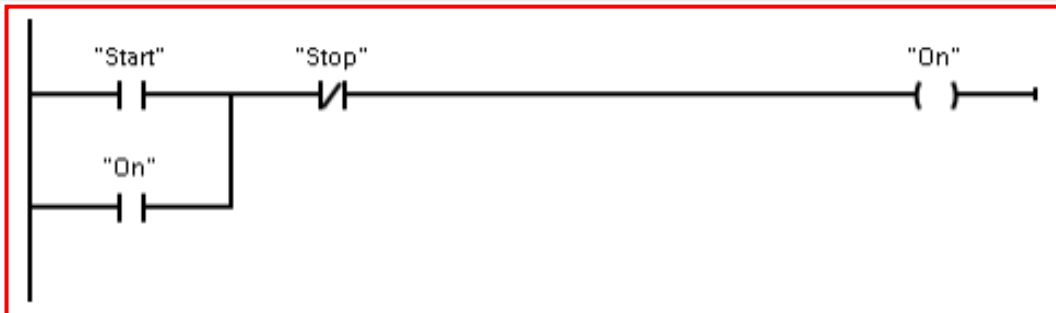
เรียก Function block มาใช้งาน

5.2.3 Programming language

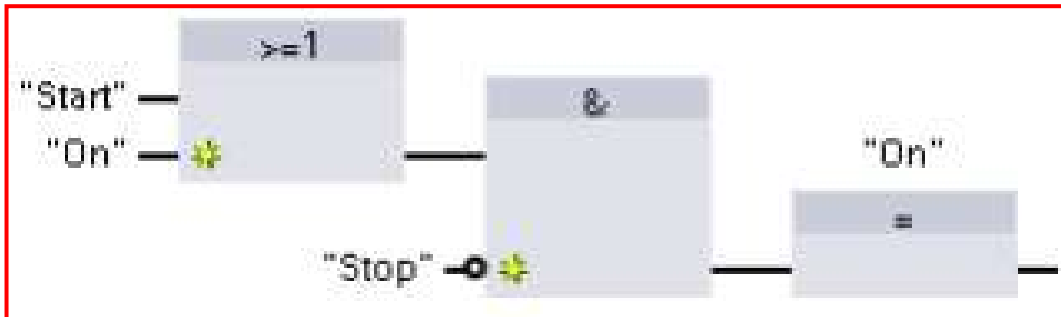
ภาษาที่ใช้ในการเขียนโปรแกรมจะมีด้วยกัน 3 ภาษา คือ

- 1.LAD (ladder logic)
- 2.FBD (Function Block Diagram)
- 3.SCL (structured control language) เป็นภาษาระดับสูงสำหรับผู้ถนัดเขียนแบบโปรแกรม

1



2



3

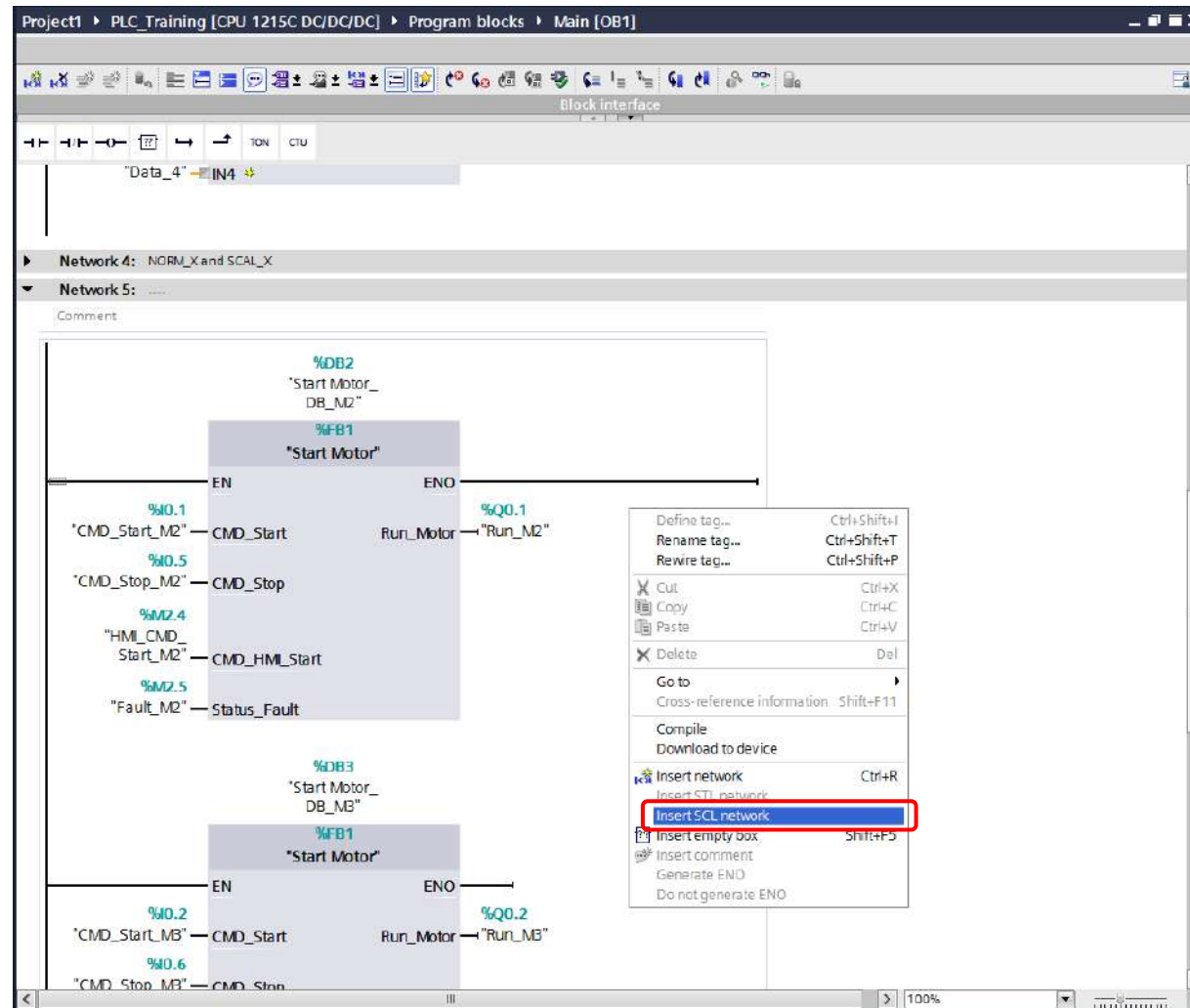
The screenshot shows the Siemens STEP 7 software interface for editing a function block. The top part is a table with columns for Name, Data type, and Comment. The bottom part is a code editor showing the structured control language (SCL) code for the function block.

	Name	Data type	Comment
1	Input		
2	StartStopSwitch	Bool	
3	Output		
4	RunYesNo	Bool	
5	InOut		
6	<Add new>		
7	Temp		
8	<Add new>		
9	Constant		
10	<Add new>		
11	Return		
12	Function_1	Void	

```
1 IF condition THEN
2   // Statement section IF
3   ;
4 END_IF;
```

Exercise – 7 : SCL

- การเพิ่ม SCL นั้น เราสามารถคลิกขวาที่ Network ล่าสุดแล้วเลือก Insert SCL network ได้ทันที



Define tag...	Ctrl+Shift+I
Rename tag...	Ctrl+Shift+T
Rewire tag...	Ctrl+Shift+P
Cut	Ctrl+X
Copy	Ctrl+C
Paste	Ctrl+V
Delete	Del
Go to	
Cross-reference information	Shift+F11
Compile	
Download to device	
Insert network	Ctrl+R
Insert STL network	
Insert SCL network	
Insert empty box	Shift+F5
Insert comment	
Generate ENO	
Do not generate ENO	

Exercise – 7.1 : SCL

- เริ่มต้นทดลองเขียนโปรแกรมด้วย SCL แบบง่ายๆ โดยเขียนโปรแกรมบวกค่าตั้งแต่ 1 ไปจนถึง 10 ซึ่งผลบวกรวมจะต้องได้เท่ากับ 55 โดยเขียนโปรแกรมดังนี้ ($1+2+3+4+5+6+7+8+9+10 = 55$)

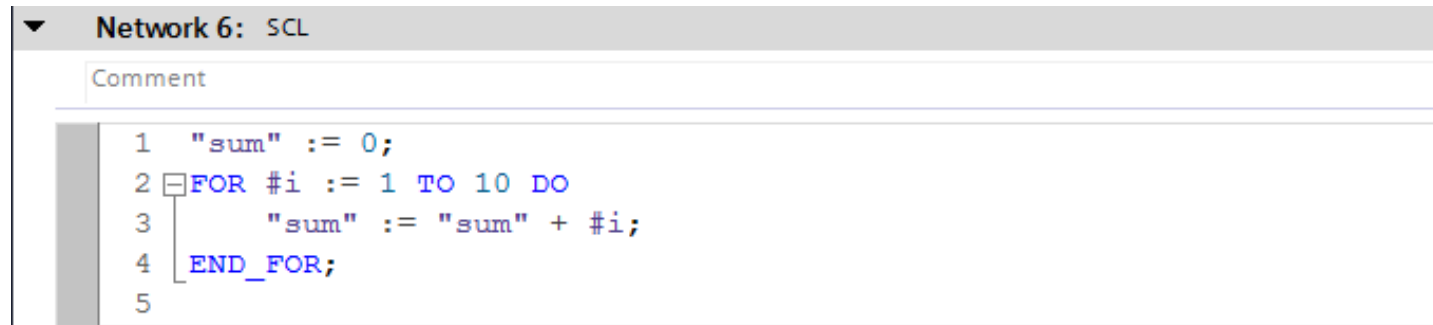
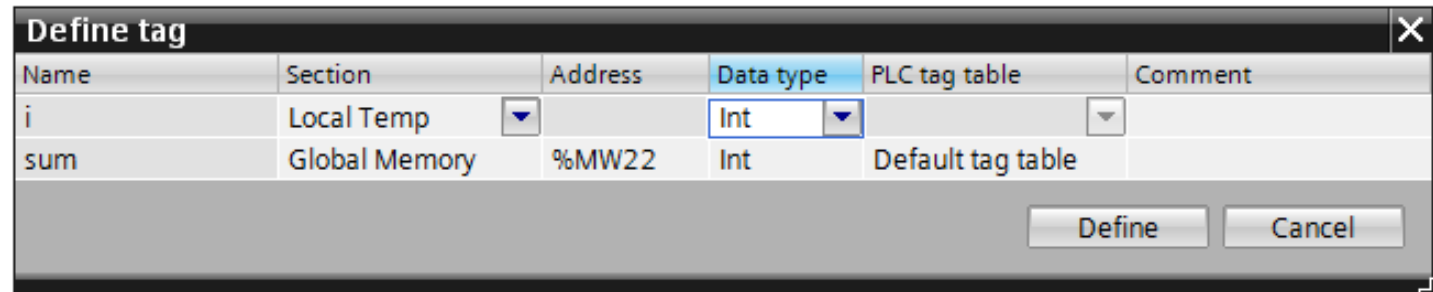
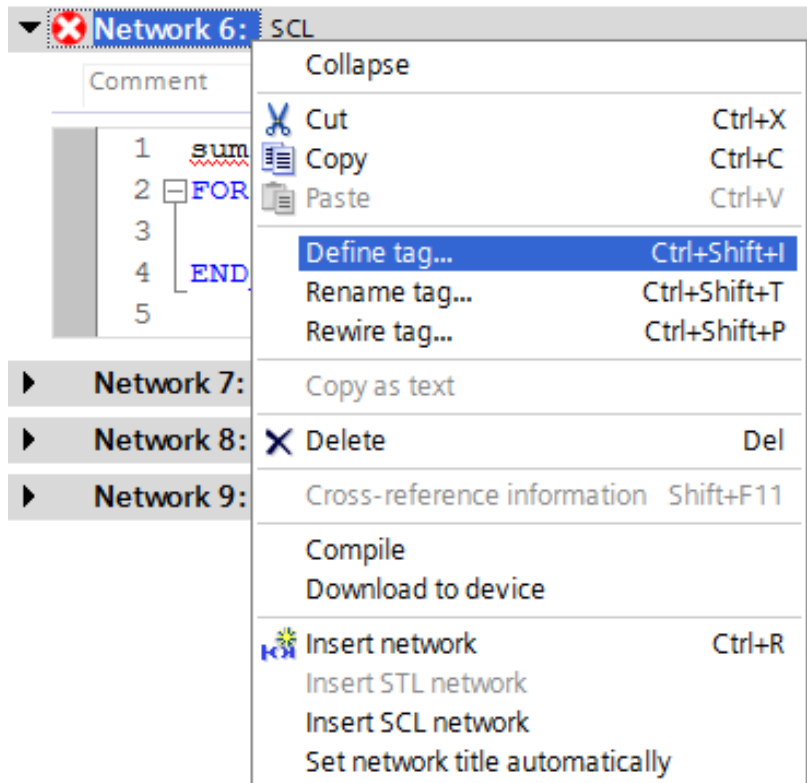


```
Network 6: SCL
Comment
1  sum := 0;
2  FOR i := 1 TO 10 DO
3      sum := sum + i;
4  END_FOR;
5
```

- ซึ่งจะเห็นได้ว่า ตัวแปร sum และ i นั้น เราพิมพ์ขึ้นมาลอยๆเอง แต่ยังไม่ได้เชื่อมกับ tag ใดๆของ PLC จึงทำให้โปรแกรมยังมี error อยู่ ดังนั้นในขั้นตอนต่อไปเราจะทำการ Define tag ให้กับตัวแปร sum และ i ซึ่งหลักการก็เหมือนขั้นตอนการ Define tag ตอนทำ ladder ทุกประการ

Exercise – 7.1 : SCL

- เนื่องจากตัวแปร i เป็นตัวแปรที่เอาไว้วนลูปเท่านั้น เราสามารถใช้เป็น Local Temp ได้เพราะแม้ว่าจะมีการเคลียร์ค่าทุกๆ scan cycle ก็ไม่ได้กระทบอะไรกับการทำงาน



Exercise – 7.1 : SCL

- เมื่อ download project ลง PLC และทำการ monitor ตัวแปร จะเห็นว่าผลลัพธ์ที่ได้ต้องเท่ากับ 55

Project1 ▶ PLC_Training [CPU 1215C DC/DC/DC] ▶ PLC tags ▶ Default tag table [82]

Tags User constants

Default tag table

	Name	Data type	Address	Retain	Acces...	Writa...	Visibl...	Monitor value
42	Data_5	Real	%MD18	☐	☒	☒	☒	0.0
43	Enable flash	Bool	%M2.3	☐	☒	☒	☒	☐ FALSE
44	HMI_CMD_Start_M2	Bool	%M2.4	☐	☒	☒	☒	☐ FALSE
45	Fault_M2	Bool	%M2.5	☐	☒	☒	☒	☐ FALSE
46	HMI_CMD_Start_M3	Bool	%M2.6	☐	☒	☒	☒	☐ FALSE
47	Fault_M3	Bool	%M2.7	☐	☒	☒	☒	☐ FALSE
48	sum	Int	%MW22	☐	☒	☒	☒	55
49	<Add new>			☐	☒	☒	☒	

Exercise – 7.1 : SCL

- TIPS : เราไม่จำเป็นต้องจำคำสั่งต่างๆของ SCL เช่น FOR..TO..DO หรือ WHILE..DO หรือ IF..THEN..ELSE เป็นต้น เพราะสามารถลากจาก Program control operations มายัง Network ได้เลย

The screenshot displays the Siemens STEP 7 interface. The main window shows the 'Main [OB1]' program block. Network 6 contains SCL code for a loop that calculates the sum of numbers from 1 to 10. Network 7 is currently selected and shows a template for a 'FOR' loop. A red arrow points from the 'FOR ... TO ... DO' instruction in the 'Instructions' list on the right to the 'DO' keyword in the Network 7 code.

Network 6: SCL

```
1 "sum" := 0;  
2 FOR #i := 1 TO 10 DO  
3   "sum" := "sum" + #i;  
4 END_FOR;  
5
```

Network 7:

```
1 FOR counter := _start_count TO _end_count DO  
2   // Statement section FOR  
3   ;  
4 END_FOR;  
5
```

Instructions List:

Name	Description	Version
Bit logic operations		V1.0
Timer operations		V1.0
Counter operations		V1.0
Comparator operations		
Math functions		V1.0
Move operations		V2.1
Conversion operations		
Program control operations		V1.1
scl IF ... THEN ...	Run conditionally	
scl IF ... THEN ... ELSE ...	Branch conditionally	
scl IF ... THEN ... ELSIF ...	Branch conditionally m...	
scl CASE ... OF ...	Create multiway branch	
scl FOR ... TO ... DO ...	Run in counting loop	
scl FOR ... TO ... BY ... DO ...	Run in counting loop w...	
scl WHILE ... DO ...	Run if condition is met	
scl REPEAT ... UNTIL ...	Run if condition is not ...	
scl CONTINUE	Recheck loop condition	
scl EXIT	Exit loop immediately	
scl RETURN	Exit block	
scl (* *)	Insert a comment secti...	
scl REGION	Organize source code	
Runtime control		

Exercise – 7.2 : SCL

- ต่อมาเราจะลองทำโปรแกรมคล้ายๆเดิม โดยประยุกต์ใช้กับ Array กันบ้าง เพื่อให้เห็นข้อดีของการใช้ SCL เพื่อสังวนรูปในตัว Array ซึ่งเป็นรูปแบบที่มีประสิทธิภาพมากกว่าการเขียนในภาษาอื่นๆ
- ให้ทำการสร้าง Data block ตัวใหม่เพื่อสมมติให้เป็นชุดข้อมูลของ array ที่มีการอ่านค่าจาก sensor มาแล้ว

Project tree

Project1 ► PLC_Training [CPU 1215C DC/DC/DC] ► Program blocks

Devices

Name

- PLC_Training [CPU 1215C DC/DC/DC]
- Device configuration
- Online & diagnostics
- Program blocks
 - Add new block
 - Flash_750 [OB30]
 - Main [OB1]
 - Startup [OB100]
 - Start Motor [FB1]
 - Data [DB4]**
 - Start Motor_DB_M2 [DB2]
 - Start Motor_DB_M3 [DB3]
- System blocks

Data

Name	Data type	Start value
Static		
▼ Sensor	Array[0..4] of Int	
Sensor[0]	Int	2
Sensor[1]	Int	3
Sensor[2]	Int	8
Sensor[3]	Int	10
Sensor[4]	Int	9
Sum2	Int	0
Average2	Real	0.0
<Add new>		

สมมติว่าให้เป็นค่าที่อ่านมา
ได้จาก sensor โดยอ่านเป็น
Array จำนวน 5 ค่า

Exercise – 7.2 : SCL

- สังเกตว่าหน้าต่างของการโปรแกรม SCL ตัวแรกกับตัวที่สองนั้นคล้ายกันมาก แต่ตัวที่สองเป็นการวนลูปให้กับ Array ซึ่งเหมาะสมกับการใช้งานจริงมากกว่าแบบแรกที่เป็นค่าคงที่

```
▼ Network 6: SCL1
Comment
1  "sum" := 0;
2  FOR #i := 1 TO 10 DO
3    "sum" := "sum" + #i;
4  END_FOR;
5

▼ Network 7: SCL2
Comment
1  "Data".Sum2 := 0;
2  FOR #i := 0 TO 4 DO
3    "Data".Sum2 := "Data".Sum2 + "Data".Sensor[#i];
4  END_FOR;
5  "Data".Average2 := "Data".Sum2 / 5.0;
```

- ความหมายของ SCL2 ก็คือให้ทำการบวกค่าใน Data.Sensor[0] จนถึง Data.Sensor[4] แล้วนำค่าไปใส่ไว้ใน Data.Sum2 จากนั้นนำมาหาค่าเฉลี่ยโดยหาร 5 ไปเก็บไว้ใน Data.Average2 นั่นเอง

Exercise – 7.2 : SCL

- ผลที่ได้จากการคำนวณ สามารถตรวจสอบได้จากการ monitor ตัว Data block ได้โดยตรง

Project1 ▶ PLC_Training [CPU 1215C DC/DC/DC] ▶ Program blocks ▶ Data [DB4]						
 Keep actual values  Snapshot  Copy snapshots to start values  Load						
Data						
		Name	Data type	Start value	Snapshot	Monitor value
1		Static				
2		Sensor	Array[0..4] of Int			
3		Sensor[0]	Int	2		2
4		Sensor[1]	Int	3		3
5		Sensor[2]	Int	8		8
6		Sensor[3]	Int	10		10
7		Sensor[4]	Int	9		9
8		Sum2	Int	0		32
9		Average2	Real	0.0		6.4
10		<Add new>				

Exercise – 7 : SCL

- สรุปการใช้งาน SCL

- การใช้งาน SCL นั้นเป็นรูปแบบการใช้งานแบบ high-level programming ซึ่งเหมาะกับผู้ที่มีการใช้งาน programming พื้นฐานจากภาษาอื่นๆมาบ้างแล้ว
- เราสามารถแทรก SCL เข้าไปที่ Network ถัดไปได้เลย ทำให้การใช้งานสะดวกมากขึ้น ส่วนไหนที่ใช้ SCL แล้วทำงานง่ายกว่าก็สามารถแทรกเข้าไปได้เลย
- ผู้ใช้งานไม่จำเป็นต้องจำคำสั่งของ SCL เพราะสามารถลากมาใช้งานจาก Basic instructions -> Program control operations ได้
- ไม่ได้มีข้อจำกัดตายตัวว่าการแก้ไขปัญหาแบบไหนต้องเขียนโปรแกรมแบบไหน ขึ้นอยู่กับทักษะและความถนัดของผู้ใช้มากกว่า

Exercise – 8 : Char & String

Project tree: Basic program_FW42 > PLC_Training_B01_01 [CPU 1212C AC/DC/Rly] > Program blocks > Data_block_1_Myname [DB2]

1. Add new block

2. Data_block_1_Myname

Name	Data type	Start value	Retain
Static			
MyChar	Array[0..6] of Char		
MyChar[0]	Char	'C'	
MyChar[1]	Char	'h'	
MyChar[2]	Char	'a'	
MyChar[3]	Char	'n'	
MyChar[4]	Char	't'	
MyChar[5]	Char	'i'	
MyChar[6]	Char	'd'	
MyName	String		

3. Chars_TO_Strg String

4. Chars_TO_Strg String

Instructions: Chars_TO_Strg Convert Array of CHAR to character string V1.2

Chars_TO_Strg means Characters to String

The instruction converts an Array of CHAR/BYTE into an ASCII string starting at a given array element.

S7-1200, S7-1500

Chars_TO_Strg: Convert Array of CHAR to character string

DELETE Delete characters in a character string V1.3

Contents

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1

Introduction to SIMATIC Step 7

2

Programming Procedure

3

Basic programming 1

4

Basic programming 2

5

PLC Concept & Programming Concept

6

Technical note

7

Appendix

6.1 Program information

The screenshot displays the Siemens TIA Portal interface for a PLC program. The left sidebar shows the project tree with the following structure:

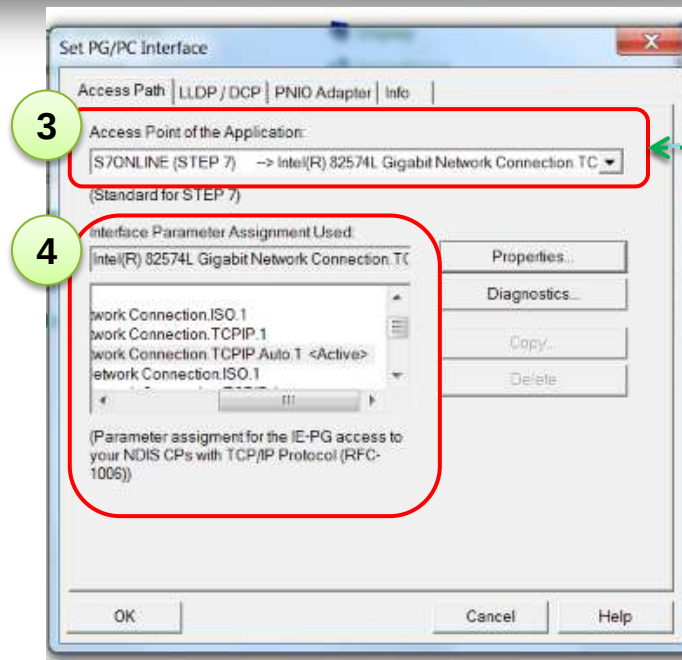
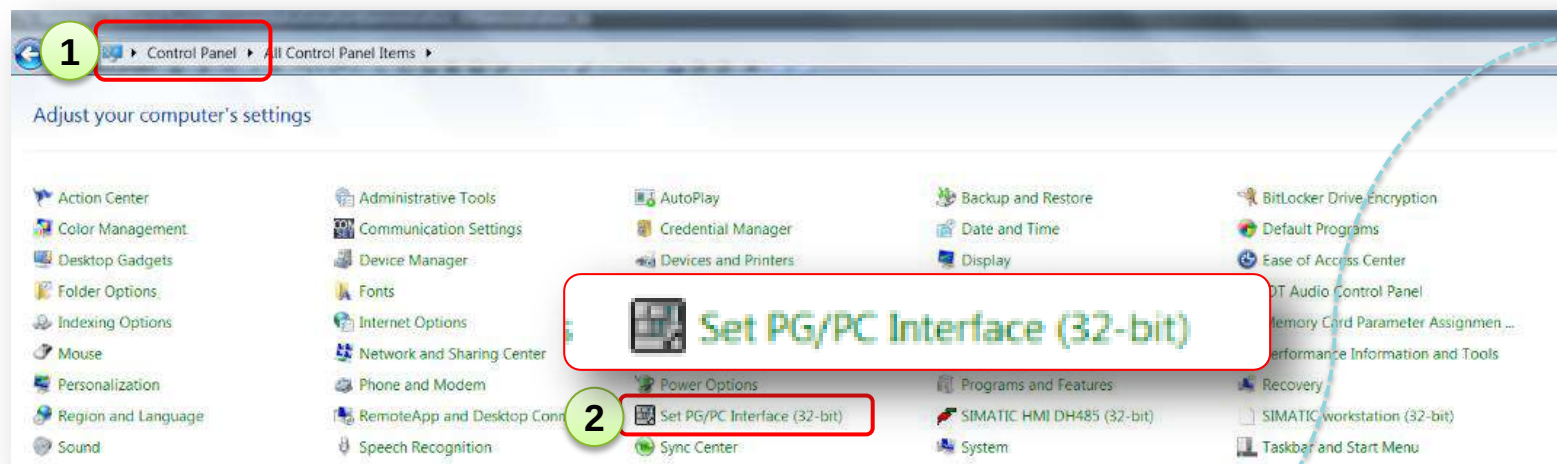
- Basic program
 - Add new device
 - Devices & networks
 - PLC_Training_B01 [CPU 1212C AC/DC/Rly]
 - PLC_Training_B01_01 [CPU 1212C AC/DC/Rly]
 - Device configuration
 - Online & diagnostics
 - Program blocks
 - Technology objects
 - External source files
 - PLC tags
 - PLC data types
 - Watch and force tables
 - Online backups
 - Traces
 - Device proxy data
 - Program info** (highlighted with a red dashed box)
 - PLC alarm text lists
 - Local modules

The main window shows the 'Program info' tab for 'PLC_Training_B01_01 [CPU 1212C AC/DC/Rly]'. The 'Resources' tab is selected, showing the following data:

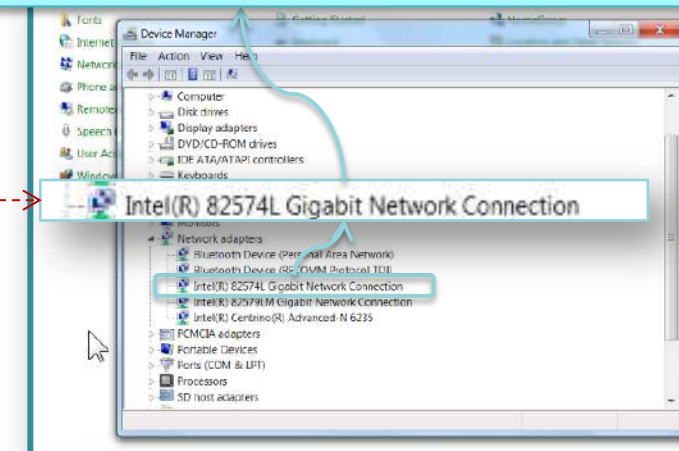
Objects	Load memory	Work memory	Retain memory	I/O	DI	DO	AI	AQ	
1	1 %	0 %	0 %		8 %	4 %	0 %	0 %	
2									
3	Total:	1 MB	76800 bytes		Configured:	24	24	6	2
4	Used:	9020 bytes	147 bytes		Used:	2	1	0	0
5	Details								
6	OB	5687 bytes	133 bytes						
7	FC	-	-						
8	FB	-	-						
9	DB	747 bytes	14 bytes						
10	Objects for Motion Technology	-	-						
11	Data types	-	-						
12	PLC tags	2586 bytes							

The 'Assignment list' tab is also shown, displaying the 'Assignment list of PLC_Training_B01_01'. The list is divided into 'Input, Output' and 'Bit memory' sections. The 'Input, Output' section shows addresses from IB0 to IB1011, and the 'Bit memory' section shows addresses from MB2 to MB21. The list is filtered by '<No filter>'.

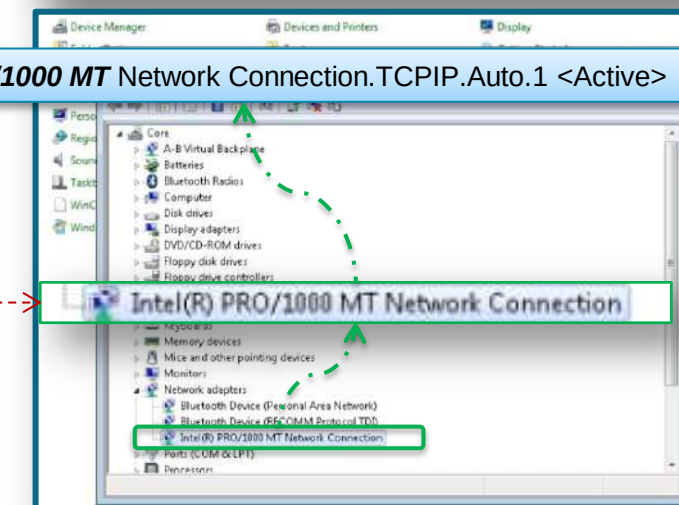
6.2 Set PG/PC Interface



Intel(R) 82574L Gigabit Network Connection.TCPIP.Auto.1 <Active>

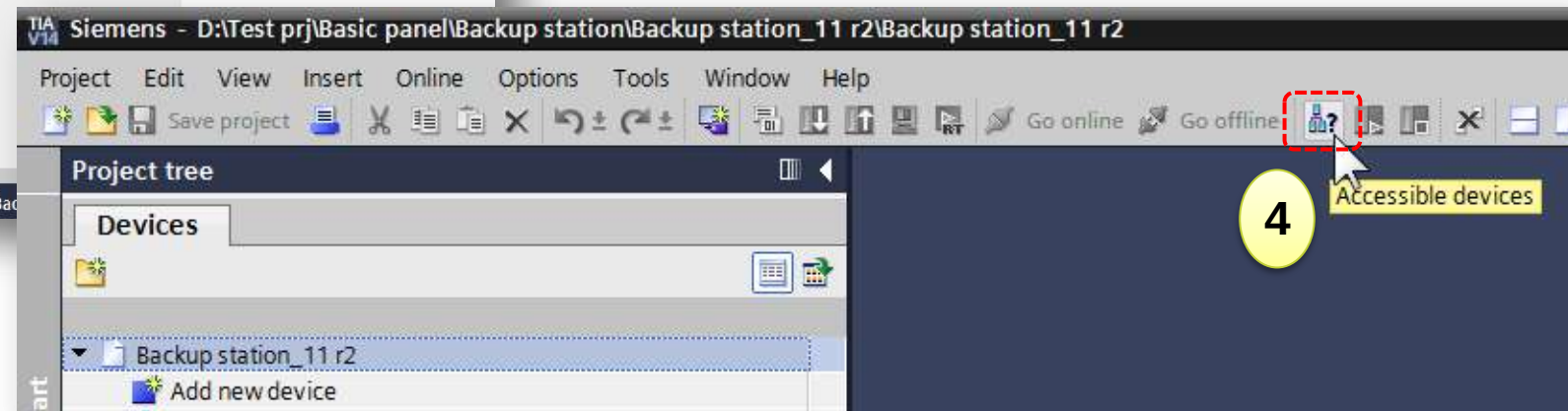
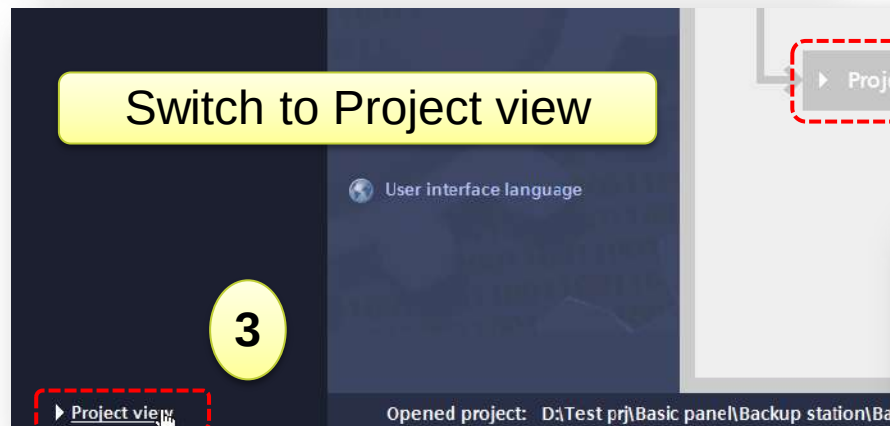
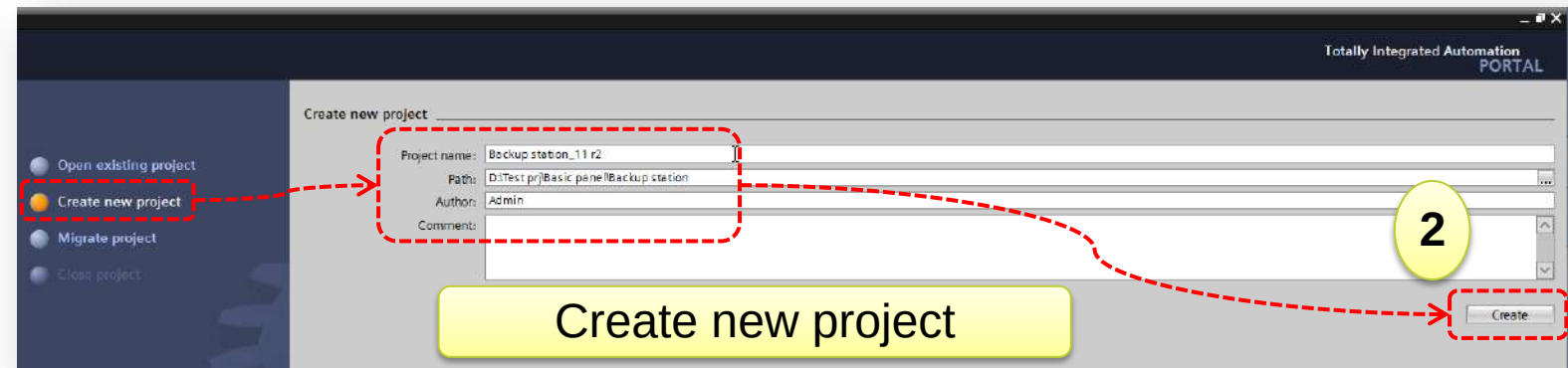


Intel(R) PRO/1000 MT Network Connection.TCPIP.Auto.1 <Active>



Depended on your network adapters

6.3 Backup PLC Station

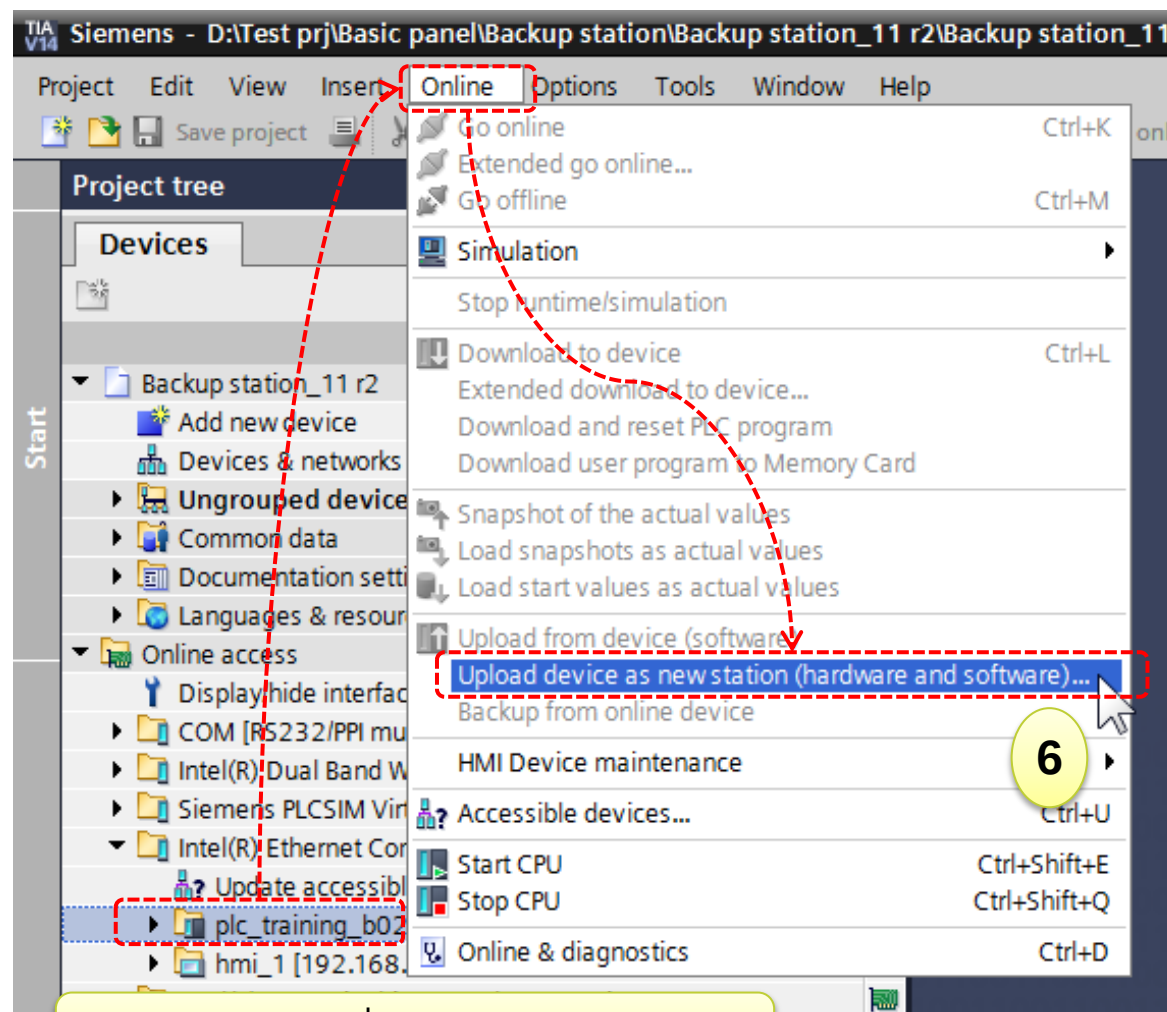
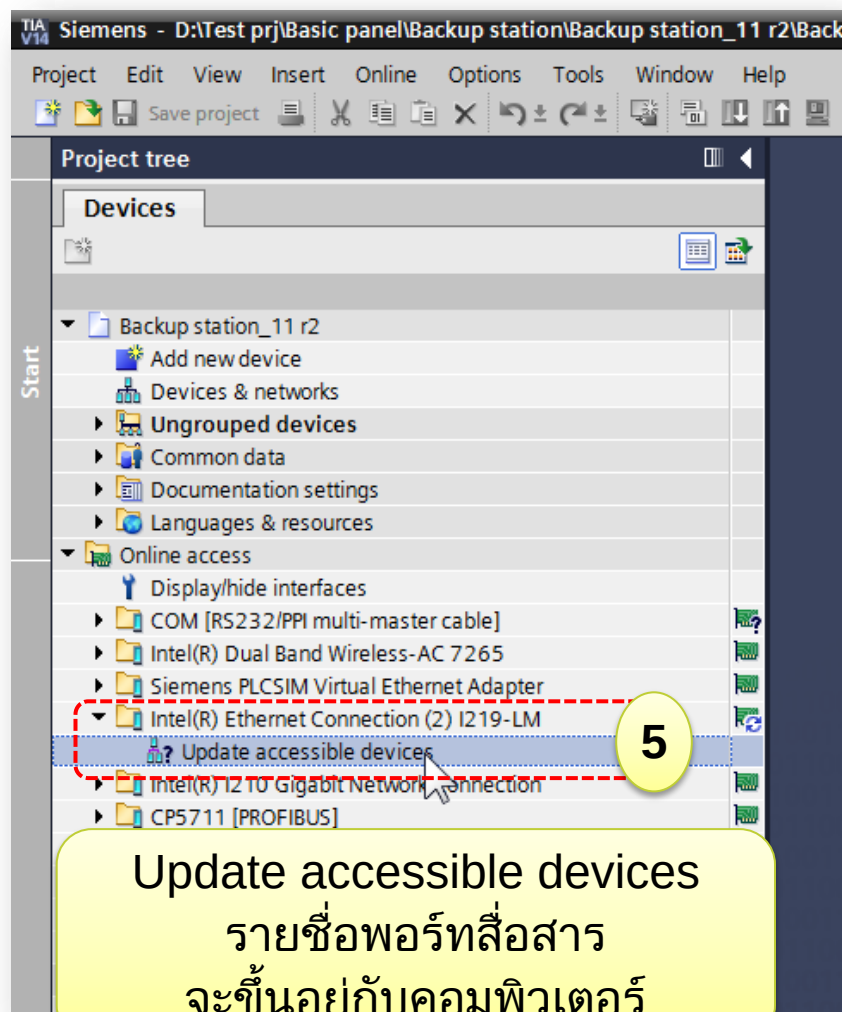


Backup PLC Station

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Backup PLC Station

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The screenshot displays the Siemens TIA Portal interface for a project named 'Backup station_11 r2'. The main window shows the 'PLC Training_B02_AIO [CPU 1215C DC/DC/DC]' device in 'Rack_0'. The device is configured with a CPU 1215C and various modules, including a CM 1241 (RS485) and an AI 4x138IT/AQ 2x145IT. The 'Device overview' table on the right lists the modules and their addresses.

Module	Slot	I address
CM 1241 (RS422/485)_1	103	
	102	
	101	
PLC Training_B02_AIO	1	
DI 14/DQ 10_1	1.1	0...1
AI 2/AQ 2_1	1.2	64...67
	1.3	
HSC_1	1.16	1000...10...
HSC_2	1.17	1004...10...
HSC_3	1.18	1008...10...
HSC_4	1.19	1012...10...
HSC_5	1.20	1016...10...
HSC_6	1.21	1020...10...
Pulse_1	1.32	
Pulse_2	1.33	
Pulse_3	1.34	
Pulse_4	1.35	
PROFINET interface_1	1.X1	
AI 4x138IT/AQ 2x145IT_1	2	96...103
	3	
	4	
	5	

The bottom of the screen shows the 'General' tab of the 'Properties' window, displaying a list of messages related to the backup process. The messages indicate that the backup was successful and that the device was scanned and configured correctly.

Message	Go to ?	Date	Time
Project Backup station_11 r2 created.		4/24/2017	6:32:41 PM
Scanning for devices on interface Intel(R) Ethernet Connection (2) I219-LM was started.		4/24/2017	6:33:22 PM
Scanning for devices completed for interface Intel(R) Ethernet Connection (2) I219-LM. Fou		4/24/2017	6:33:29 PM
Start upload from device		4/24/2017	6:33:50 PM
Hardware configuration uploaded successfully.		4/24/2017	6:33:59 PM
Main was loaded successfully.		4/24/2017	6:34:07 PM
Upload from device finished (errors: 0; warnings: 0).		4/24/2017	6:34:07 PM
Scanning for devices on interface Intel(R) Ethernet Connection (2) I219-LM was started.		4/24/2017	6:34:41 PM
Scanning for devices completed for interface Intel(R) Ethernet Connection (2) I219-LM. Fou		4/24/2017	6:34:48 PM
Start upload from device		4/24/2017	6:35:14 PM
Hardware configuration uploaded successfully.		4/24/2017	6:35:18 PM
Main was loaded successfully.		4/24/2017	6:35:19 PM
Upload from device finished (errors: 0; warnings: 0).		4/24/2017	6:35:19 PM

6.4 Factory reset

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The screenshot illustrates the process of performing a factory reset on a PLC module within the Siemens TIA Portal. The interface is divided into several panes: Project tree, Online access, Diagnostics, and Online tools.

Step 1: In the Project tree, the 'Online & diagnostics' folder is selected under the 'plc_training_b01_01' device.

Step 2: In the 'Online & diagnostics' pane, the 'Functions' sub-pane is selected.

Step 3: The 'Reset to factory settings' button is highlighted in the 'Functions' pane.

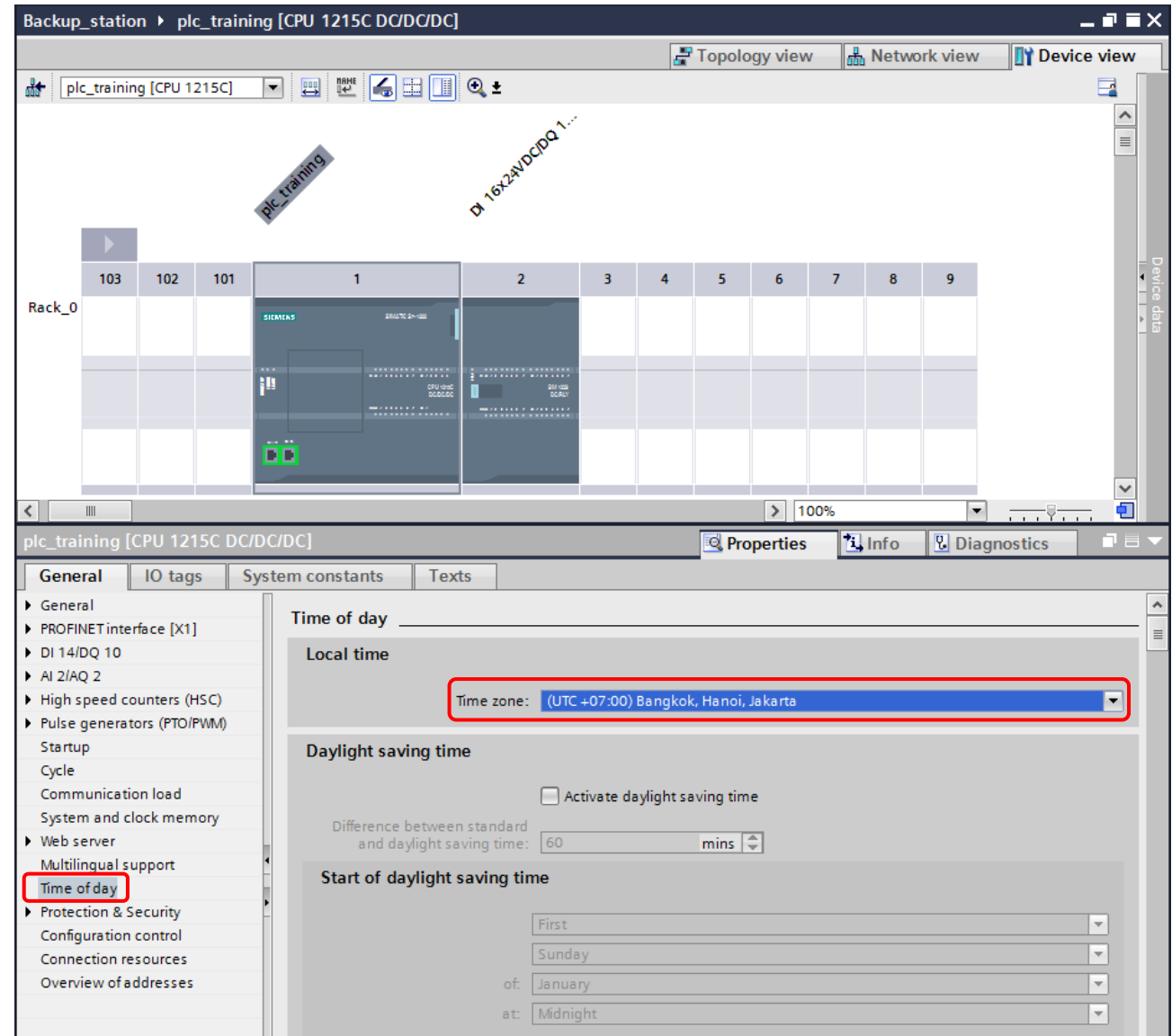
Step 4: The 'Reset' button is highlighted in the 'Reset to factory settings' dialog box.

Step 5: A confirmation dialog box titled 'Online & diagnostics (0241:000056)' is displayed, asking 'Do you really want to reset the module?'. The 'Yes' button is highlighted.

The 'Online access' pane shows the connection to the PLC module 'plc_training_b01_01' [192.168.0.71]. The 'Diagnostics' pane shows the 'General' tab with the 'Reset to factory settings' button. The 'Online tools' pane shows the 'CPU operator panel' with 'RUN / STOP', 'ERROR', and 'MAINT' buttons.

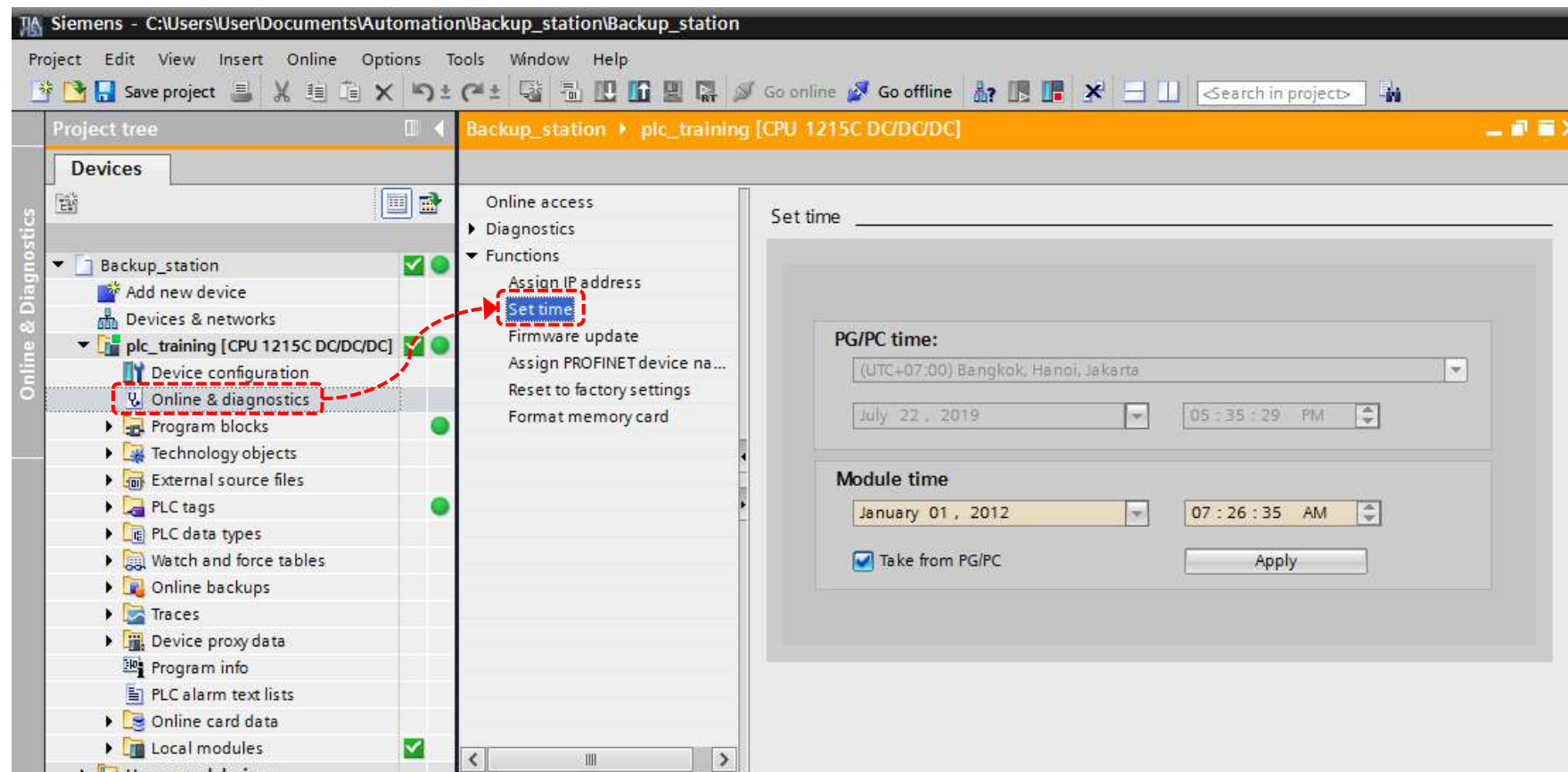
6.5 Set time to PLC

- การตั้งเวลาให้กับ PLC นั้น แนะนำว่าควรจะทำให้การตั้งค่า Time zone ให้ PLC ได้รับทราบก่อนว่า PLC ที่เราใช้งานนั้นอยู่ใน Time zone ใด เพราะการอ่านเขียนเวลานั้นมีทั้งการอ่านเขียนเวลา Universal Time และเวลา Local time
- การตั้ง Time zone ที่ถูกต้องจะทำให้การอ่านเขียนเวลาถูกต้องทั้ง Universal Time และ Local time
- จากนั้นทำการ download hardware configuration ลงไปใน PLC



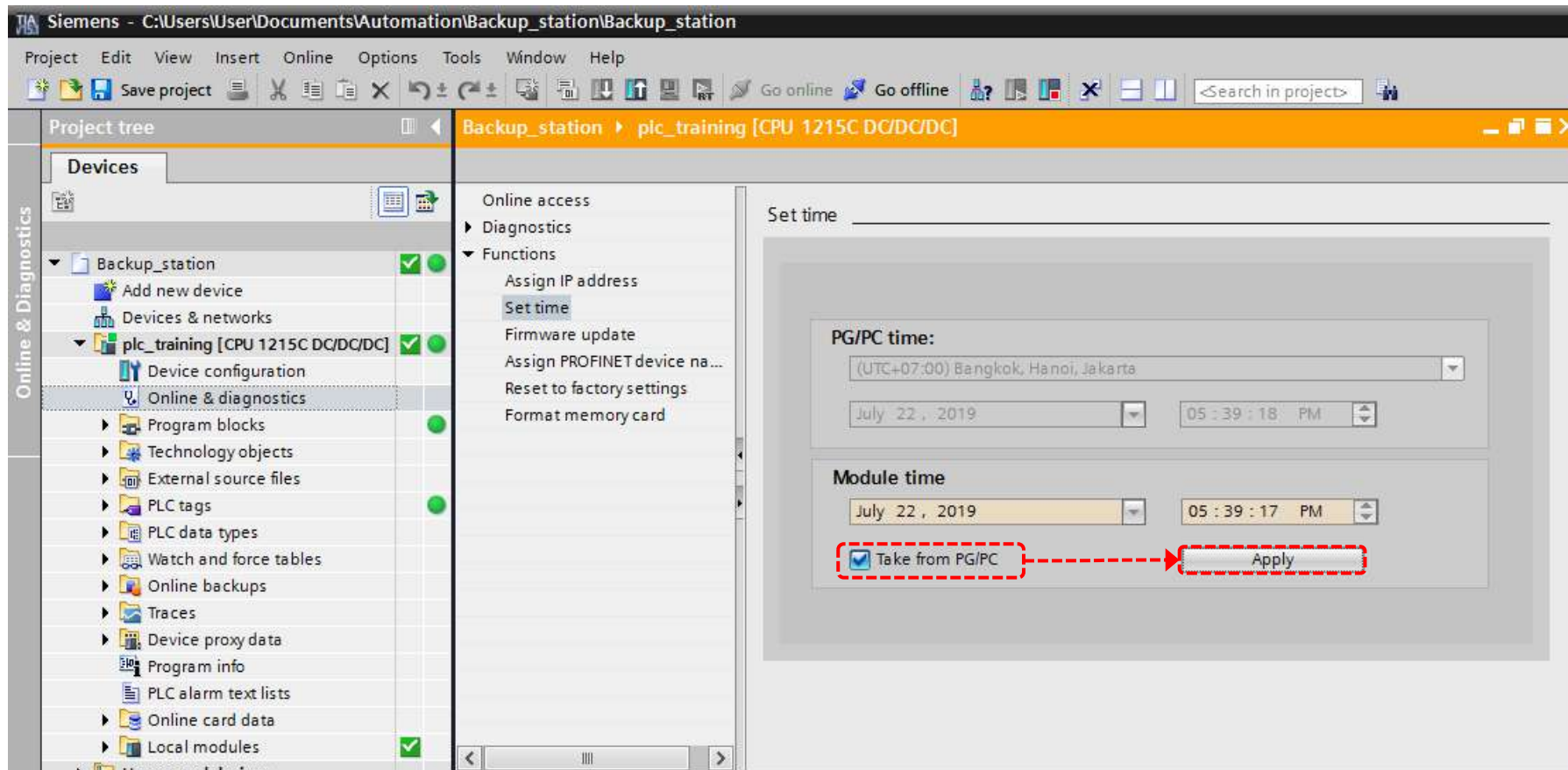
Set time to PLC

- Online PLC แล้วไปที่ Online & diagnostics -> Set time
 - เวลาที่ PG/PC time คือเวลาในคอมพิวเตอร์ที่เราใช้งานอยู่ ส่วนเวลาที่ Module time คือเวลาของ PLC ในขณะนั้น



Set time to PLC

- เลือกที่ Take from PG/PC แล้วกดปุ่ม Apply ก็จะนำเวลาจากคอมพิวเตอร์มาเขียนลงให้ PLC ทันที



6.7 Update firmware

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The screenshot displays the Siemens TIA Portal interface with the following components and annotations:

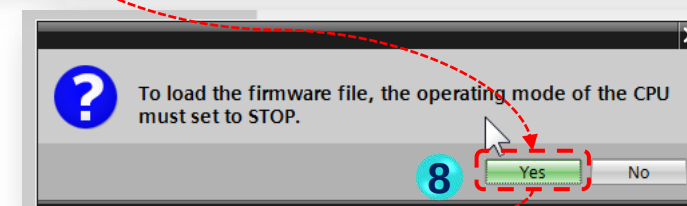
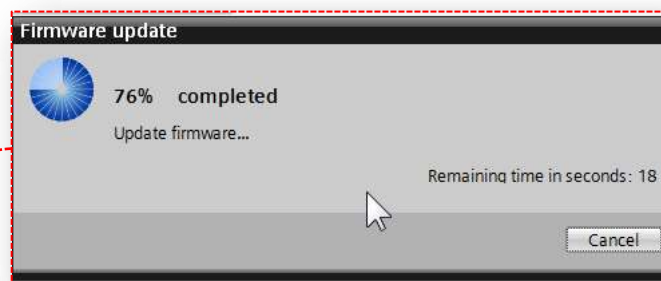
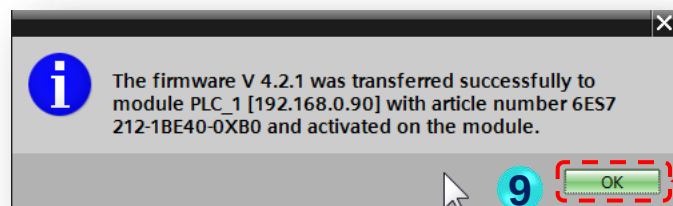
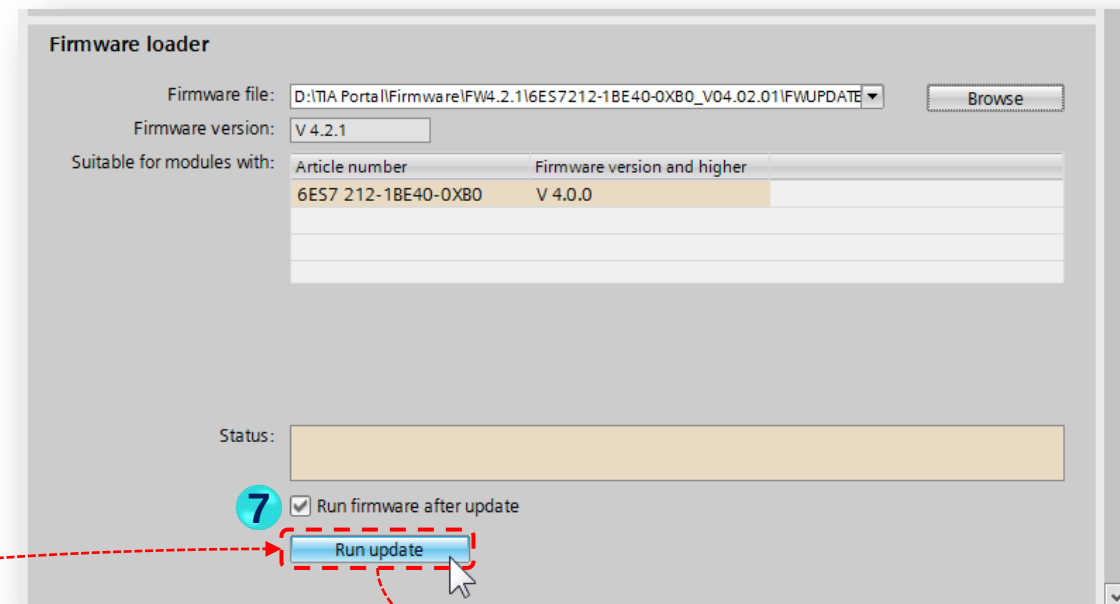
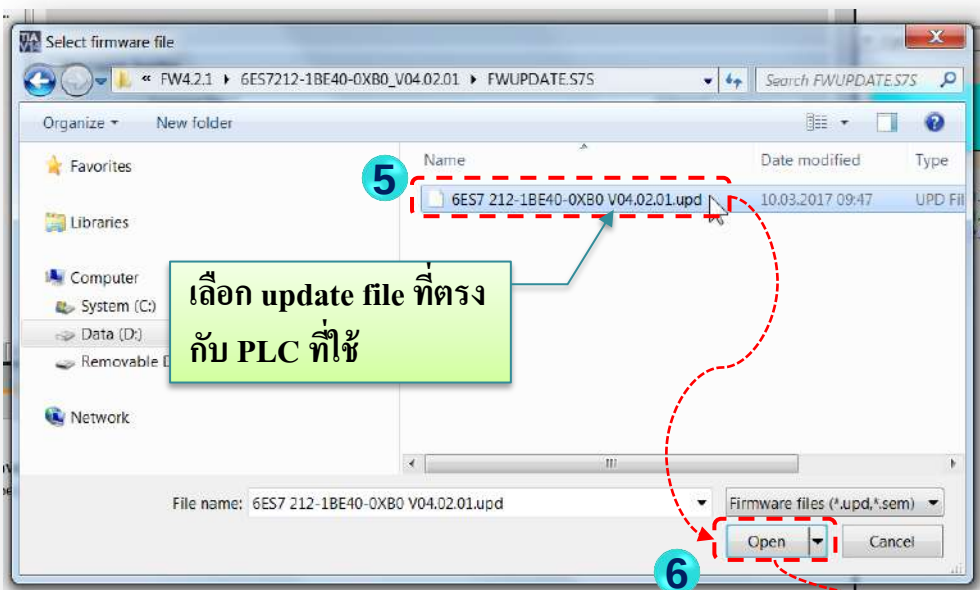
- Project Tree (Left):** Shows the project hierarchy. A red dashed box labeled '1' highlights the 'Online access' folder. A red dashed box labeled '2' highlights the 'Online & diagnostics' folder under 'plc_1 [192.168.0.90]'.
- Online access pane (Middle):** A red dashed box labeled '3' highlights the 'Firmware update' option under the 'Functions' menu.
- Online data pane (Right):** Displays the 'Firmware update' configuration for 'PLC_1 [192.168.0.90]'. Fields include:
 - Article number: 6ES7 212-1BE40-0XB0
 - Firmware: V 4.1.3
 - Name: PLC_1 [192.168.0.90]
 - Rack: 0
 - Slot: 1
- Firmware loader (Bottom Right):** Contains a 'Firmware file' field and a 'Browse' button. A red dashed box labeled '4' highlights the 'Browse' button. A green callout box with an arrow pointing to the button contains the text: 'Click เพื่อเลือก update file'.

Update firmware

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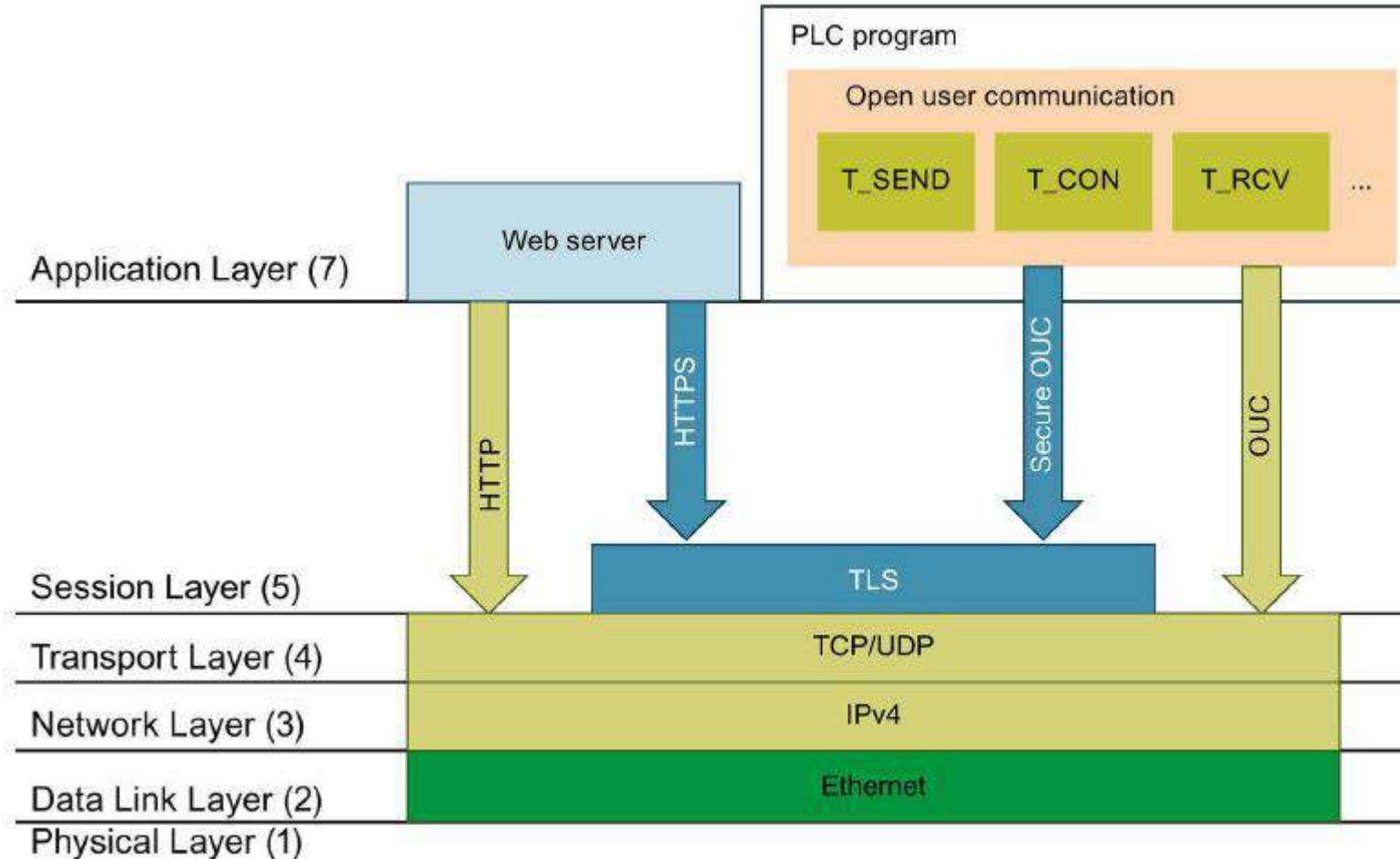


6.7 Web Server

The screenshot shows the Siemens STEP 7 software interface. The Project tree on the left (labeled 1) shows the project structure. The main workspace displays a rack configuration with modules: PLC Training_B01_01 [CPU 1212C AC/DC/Rly], DI 16x24VDC..., and AI 4x13BIT/AQ... (labeled 2). The Properties window at the bottom right (labeled 3) shows the 'Web server' section. The 'General' tab is selected, and the 'Web server' checkbox is checked (labeled 4). The 'Automatic update' section is also visible.

4. เปิดใช้งาน Web Server

Web Server: Secure communication



Web Server: User management

User management

Name	Access level	Password
Everyone	Minimum	
Add new user		

5. เพิ่มชื่อผู้ใช้งาน

6. กำหนดสิทธิ์สำหรับแต่ละผู้ใช้งาน

7. ตั้งรหัสผ่านสำหรับแต่ละผู้ใช้งาน

Enter password: *****

Confirm password: *****

Add user

Setting authorized user

Setting password for user

Web Server: User management

1. Add user

User management

Name	Access level	Password
Everybody	Minimum	
<Add new user>		

User management

Name	Access level	Password
Everybody	Minimum	
siemens	Minimum	*****
<Add new user>		

เพิ่มผู้ใช้งาน

5

3. Setting Password

User management

Name	Access level	Password
Everybody	Minimum	
siemens	Administrative	*****
<Add new user>		

ตั้งค่ารหัสผ่านสำหรับผู้ใช้งาน

7

Enter password: *****

Confirm password: *****

2. Setting authorized

The user is authorized to...

- ☒ query diagnostics
- ☒ read tags
- ☒ write tags
- ☒ read tag status
- ☒ write tag status
- ☐ acknowledge alarms
- ☒ open user-defined web pages
- ☒ write in user-defined web pages
- ☒ read files
- ☒ write/delete files
- ☒ change operating mode
- ☒ flash LEDs
- ☒ perform a firmware update
- ☐ change system parameters
- ☐ change application parameters
- ☒ create a backup of the PLC
- ☒ restore the PLC by a backup file
- ☐ perform changes as F-Admin

New FW4.2

6

กำหนดสิทธิ์ในการใช้งาน

Web Server: Default web pages

The screenshot shows the Siemens S7-1200 station web interface. The browser address bar indicates the URL <https://192.168.0.71/Portal/Portal.mw>. The page title is "S7-1200 station_2 / PLC_Training_B01_01". The user is logged in as "User: siemens" with a "Logout" link. The page displays the following information:

- General:**
 - TIA Portal: V14.0
 - Station name: S7-1200 station_2
 - Module name: PLC_Training_B01_01
 - Module type: CPU 1212C AC/DC/RLY
- Status:**
 - Operating Mode: RUN
 - Status: OK
- CPU operator panel:**
 - RUN
 - STOP
 - LED flashes

The left sidebar contains the following navigation links:

- Start Page
- Diagnostics
- Diagnostic Buffer
- Module Information
- Communication
- Tag status
- Watch tables
- Online backup
- User-defined pages
- File Browser
- Introduction

The right sidebar, titled "The user is authorized to...", lists the following permissions:

- ☒ query diagnostics
- ☒ read tags
- ☒ write tags
- ☒ read tag status
- ☒ write tag status
- ☐ acknowledge alarms
- ☒ open user-defined web pages
- ☒ write in user-defined web pages
- ☒ read files
- ☒ write/delete files
- ☒ change operating mode
- ☒ flash LEDs
- ☒ perform a firmware update
- ☐ change system parameters
- ☐ change application parameters
- ☒ create a backup of the PLC
- ☒ restore the PLC by a backup file
- ☐ perform changes as F-Admin

A yellow box at the bottom of the page contains the text: "Update firmware อยู่ใน Module information".

Web Server: Diagnostics

SIEMENS S7-1200 station_2 / PLC_Training_B01_01

User: siemens Logout

09:57:27 am 5/3/2017 UTC English

Diagnostic Buffer

Diagnostic buffer entries 1-25

Number	UTC Time	UTC Date	Event
1	02:52:57.929 am	1/1/2012	Wire break -
2	12:57:14.984 am	1/1/2012	IO device failure -
3	12:51:51.364 am	1/1/2012	Follow-on operating mode change - CPU changes from STARTUP to RUN mode
4	12:51:51.351 am	1/1/2012	IO device failure - IO device not found -
5	12:50:51.412 am	1/1/2012	Wire break -
6	12:50:51.262 am	1/1/2012	Wire break -
7	12:50:51.060 am	1/1/2012	Follow-on operating mode change - CPU changes from STOP to STARTUP mode
8	12:50:50.952 am	1/1/2012	New startup information - Current CPU operating mode: STOP
9	12:50:50.951 am	1/1/2012	Follow-on operating mode change - CPU changes from STOP (initialization) to STOP mode
10	12:50:39.572 am	1/1/2012	Power on - CPU changes from NO POWER to STOP (initialization) mode
11	12:48:56.054 am	1/1/2012	Power off - CPU changes from STARTUP to NO POWER mode
12	12:48:56.050 am	1/1/2012	Supply voltage missing -
13	12:48:45.704 am	1/1/2012	Wire break -

Details: 1

Error: Wire break
HW_ID= 00280, Input channel number 00001

Incoming event

Event ID: 16# 06:018

status ของ fault
Incoming event / Out going

Click ดูรายการของ fault

ข้อมูลและรายละเอียดของ fault

Diagnostic Buffer

6.8 Watch table

1. Double click เพื่อเปิด Force table ซึ่งจะแสดงใน Work area

2. Click เพื่อเปิด Tag table ซึ่งจะแสดงใน Details view

3. เลือก Tag ที่ต้องการใน Details view

4. Drag & Drop Tags ที่เลือกไว้ใน Details view แล้ว Drop ลงใน Force/Watch table ใน Work area

Name	Address	Display format	Monitor value	Force value	Comment
"CMD_Start_M1":P	%I0.0:P	Bool			
"CMD_Start_M2":P	%I0.1:P	Bool			
"CMD_Start_M3":P	%I0.2:P	Bool			
"CMD_Start_M4":P	%I0.3:P	Bool			
"CMD_Stop_M1":P	%I0.4:P	Bool			
"CMD_Stop_M2":P	%I0.5:P	Bool			
"CMD_Stop_M3":P	%I0.6:P	Bool			
"CMD_Stop_M4":P	%I0.7:P	Bool			
"Run_M1":P	%Q0.0:P	Bool			
"Run_M2":P	%Q0.1:P	Bool			
"Run_M3":P	%Q0.2:P	Bool			
"Run_M4":P	%Q0.3:P	Bool			
"Count_M1"	%MW4	DEC #			
"Data_01"	%MD6	Floating-point nu...			
"Data_02"	%MD10	DEC #			
"Data_03"	%MD12	Floating-point nu...			
"Data_04"	%MW16	DEC #			
"Data_05"	%MD18	Floating-point nu...			

Name	Data type	Details	Com...
CMD_Stop_M4	Bool	%I0.7	
Count_M1	Real	%MW4	
Data_01	Real	%MD6	
Data_02	Int	%MW10	
Data_03	Real	%MD12	
Data_04	Int	%MW16	
Data_05	Real	%MD18	
FirstScan	Bool	%M1.0	
HMI_CMD_Start_M1	Bool	%M2.1	
Local	HW_SubMo...	49	
Local-AL_2_1	HW_SubMo...	258	
Local-AL_9x13BIT_AQ...	HW_SubMo...	271	
Local-Common	HW_SubMo...	50	
Local-Configuration	HW_SubMo...	33	
Local-Device	HW_Device...	32	

Watch Table: using in Web Server

Basic program_FW42 ▸ PLC_Training_B01_01 [CPU 1212C AC/DC/Rly]

Topology view | Network view | Device view

PLC_Training_B01_01 [CPU]

Rack_0

103 102 101 1 2 3

SIEMENS 2012C AC/DC/Rly

DI 16x24VDCI... AI 4x13BIT/AQ...

100%

Properties | Info | Diagnostics

3. เลือก Force/Watch table ที่ต้องการ
เปิดใช้งานบน Web server

4. กำหนดสิทธิ์ในการใช้งาน

1. Web server

2. Watch tables

Watch tables

Name	Access
Force table	Read/Write
<Add new watch table>	

Name	Access
Force table	Read/Write
<Add new watch table>	
	Read
	Read/Write

Watch Table: using in Web Server

SIEMENS S7-1200 station_2 / PLC_Training_B01_01

User: siemens Logout

12:44:10 am 1/1/2012 UTC English

Watch tables

Force table

Name	Address	Display Format	Monitor Value	Modify Value	Comment
CMD_Start_M1.P	%I0.0.P	BOOL	☒ true		Go
CMD_Start_M2.P	%I0.1.P	BOOL	☒ true		Go
CMD_Start_M3.P	%I0.2.P	BOOL	☒ true		Go
CMD_Start_M4.P	%I0.3.P	BOOL	☒ true		Go
CMD_Stop_M1.P	%I0.4.P	BOOL	☐ false		Go
CMD_Stop_M2.P	%I0.5.P	BOOL	☐ false		Go
CMD_Stop_M3.P	%I0.6.P	BOOL	☐ false		Go
CMD_Stop_M4.P	%I0.7.P	BOOL	☐ false		Go
Run_M1.P	%Q0.0.P	BOOL	☒ true		Go
Run_M2.P	%Q0.1.P	BOOL	☐ false		Go
Run_M3.P	%Q0.2.P	BOOL	☐ false		Go
Run_M4.P	%Q0.3.P	BOOL	☐ false		Go
Count_M1	%MW4	DEC+/-	2		Go
Data_01	%MD6	Floating_Point	100.0		Go
Data_02	%MW10	DEC+/-	0		Go
Data_03	%MD12	Floating_Point	15.0		Go
Data_04	%MW16	DEC+/-	0		Go
Data_05	%MD18	Floating_Point	1500.0		Go

Refresh

พิมพ์ค่าที่ต้องการแก้ไข
แล้วกด GO หรือ Apply
(ในกรณีที่ต้องการแก้ไข
หลายข้อมูลพร้อมกัน)

DEC+/-	2	Go
Floating_Point	100.0	Go
DEC+/-	0	Go
Floating_Point	15.0	Go
DEC+/-	0	Go
Floating_Point	1500.0	Go

10

5

Watch Table: using in Web Server

The screenshot displays the Siemens TIA Portal interface, specifically the Watch Tables and Ladder Logic editor. The Watch Tables window on the left shows a list of variables and their current values. The Ladder Logic editor on the right shows a network with a CALCULATE block. Red dashed boxes highlight the connection between the Watch Tables and the Ladder Logic network.

Watch Tables

Force table

Name	Address	Display Format	Monitor Value
"CMD_Start_M1":P	%I0.0:P	BOOL	true
"CMD_Start_M2":P	%I0.1:P	BOOL	true
"CMD_Start_M3":P	%I0.2:P	BOOL	true
"CMD_Start_M4":P	%I0.3:P	BOOL	true
"CMD_Stop_M1":P	%I0.4:P	BOOL	false
"CMD_Stop_M2":P	%I0.5:P	BOOL	false
"CMD_Stop_M3":P	%I0.6:P	BOOL	false
"CMD_Stop_M4":P	%I0.7:P	BOOL	false
"Run_M1":P	%Q0.0:P	BOOL	true
"Run_M2":P	%Q0.1:P	BOOL	false
"Run_M3":P	%Q0.2:P	BOOL	false
"Run_M4":P	%Q0.3:P	BOOL	false
"Count_M1"	%MW4	DEC+/-	2
"Data_01"	%MD6	Floating_Point	100.0
"Data_02"	%MW10	DEC+/-	10
"Data_03"	%MD12	Floating_Point	15.0
"Data_04"	%MW16	DEC+/-	5
"Data_05"	%MD18	Floating_Point	1100.0

Ladder Logic Network 3:

Calculate Real

OUT := (IN1+IN2)*(IN3-IN4)

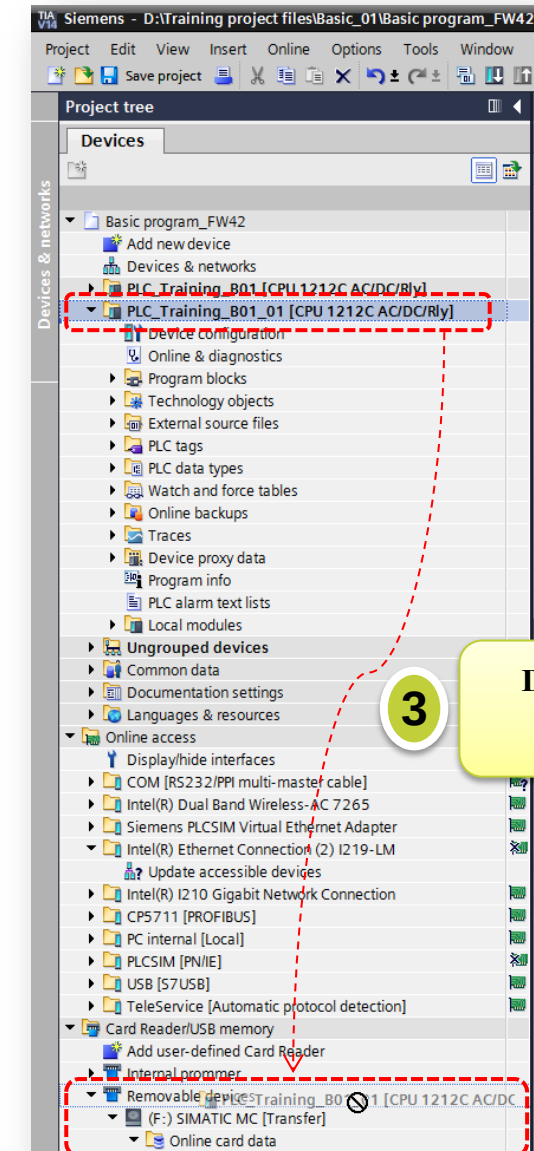
IN1: 100.0 (%MD6) "Data_01"

IN2: 10 (%MW10) "Data_02"

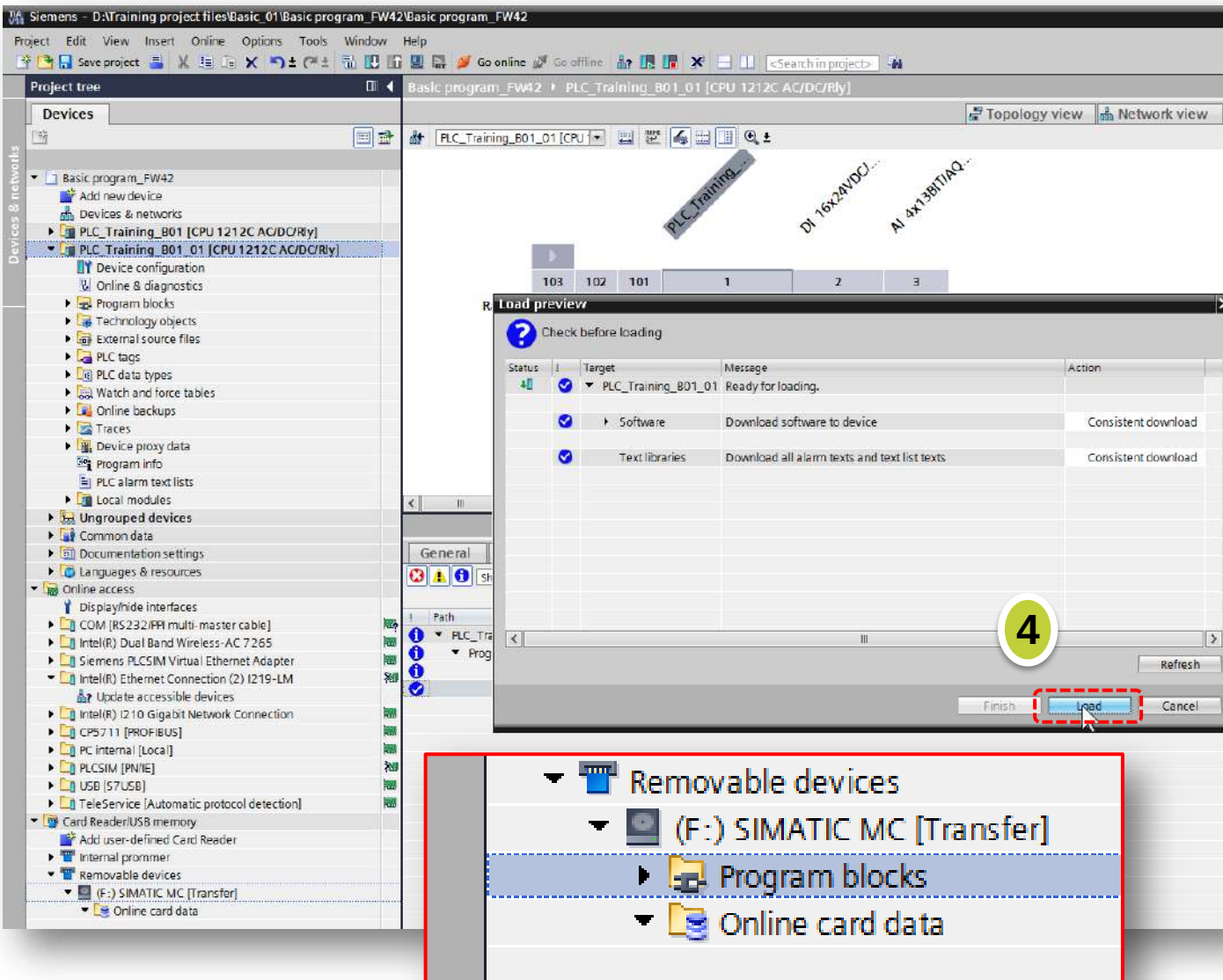
IN3: 15.0 (%MD12) "Data_03"

IN4: 5 (%MW16) "Data_04"

OUT: 1100.0 (%MD18) "Data_05"



Maintenance concept using Memory card



5

ปิดไฟที่จ่ายให้ PLC

6

ใส่ Card ในช่อง MC



7

เปิดไฟที่จ่ายให้ PLC

8

รอนจนกระทั่ง ไฟ RUN/STOP สีส้มค้าง
และไฟ MAINT สีส้มกระพริบ



9

ปิดไฟที่จ่ายให้ PLC แล้วถอด Card ออก
เปิดไฟให้ PLC แล้วสั่ง RUN PLC

Information & Resources

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<https://support.industry.siemens.com>



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Edition V4.4 11/2019

SIMATIC

S7

S7-1200 Programmable controller

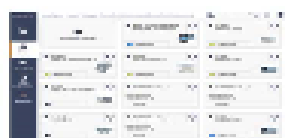
System Manual

Begin >

TIA Selection Tool cloud

> TIA Selection Tool cloud

Information and start - options



The “TIA Selection Tool cloud” allows you to select, configure and order devices for Totally Integrated Automation. The “TIA Selection Tool cloud” comes with an assistant to help you choose the right devices, configure module and accessory selections and check that your selection will function correctly as well as providing an interface for you to export your order to the Industry Mall shopping cart.

You can find additional information on: www.siemens.com/tia-selection-tool . You can download a Windows version from <http://www.siemens.com/tia-selection-tool-standalone>.

[Start](#)

TIA Portal - An Overview of the Most Important Documents and Links - Controller

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Entry ID: 65601780,

Entry date: 06/19/2020



Controller Software in TIA Portal	Controller Hardware SIMATIC S7-1200/ S7-1500
Downloads / Updates for TIA Portal	Downloads / HSP for S7-1200/ S7-1500
Videos for TIA Portal	Videos for S7-1200/ S7-1500
Product Information/ Licenses for TIA Portal	Product Information/ Licenses for S7-1200/ S7-1500
Manuals for TIA Portal	Manuals for S7-1200/ S7-1500
FAQs for TIA Portal	FAQs for S7-1200/ S7-1500
Forum for TIA Portal	Forum for S7-1200/ S7-1500
Training / Courses	Training / Courses



> Visualisation



> Motion



> Communication



> Safety



> Controls



> Migration

TIA Portal Tutorial Center : Entry ID: 106656707, Entry date: 10/15/2020



Browser address bar: [https://support.industry.siemens.com/cs/document/106656707/the-tia-portal-tutorial-center-\(videos\)?dti=0&lc=en-WW](https://support.industry.siemens.com/cs/document/106656707/the-tia-portal-tutorial-center-(videos)?dti=0&lc=en-WW)

Page title: The TIA Portal Tutorial Cen...

Navigation: File Edit View Favorites Tools Help

Search: Search...

Language: EN - Fokusthemen

Industry Online Support Product Support

Navigation: Home > Product Support

Entry type: Application example Entry ID: 106656707, Entry date: 10/15/2020

Rating: ★★★★★ (81) > Rate

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The TIA Portal Tutorial Center (videos)

Entry Associated product(s)

The TIA Portal Tutorial Center is a media system with various videos which gives you an overview of the general functions and tools of the TIA Portal.

Introduction

TIA Portal is the Siemens engineering framework for all automation tasks in manufacturing automation. It seamlessly integrates controllers, distributed I/O, HMI, drives, motion control and motor management in one engineering environment. TIA Portal and its uniform operating concept offers users the ideal tool for any task and any project phase to ensure an optimized engineering workflow.

The TIA Portal Tutorial Center is a media system with various videos which gives you an overview of the general functions and tools of the TIA Portal.

Contents

The Videos

- provide you with an overview of functions and tools in the TIA Portal
- provide you with an overview of the smooth interaction between configuration and programming tools
- show how to work effectively with TIA Portal
- show how to realize various automation tasks in the TIA Portal

This media system is intended for system and project planners as well as for maintenance and service staff. The included videos mainly focus on technical content and offer support for all phases of system service life.

Structure

mySupport Co

- > Add to mySupport f
- > Add to mySupport c
- > Favorites
- > My requests
- > CAX downloads
- > My Products / Clipb

Product inform

- + Presales info
- + Catalog and orderin
- + Technical info
- + Support
- + Service offer
- + Training
- + Contact & partners

SIEMENS

TIA Portal - Tutorial Center

Overview TIA Portal

SIMATIC S7- Architecture

Library concept

Visualization

Communication

Diagnostics

Programming Languages

Industrial Security

Migration

Motion Control

Energy Management

Tools & Apps

SIMATIC STEP 7 Basic/Professional V16 and SIMATIC WinCC V16

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SIMATIC

STEP 7 and
WinCC Engineering V16

System Manual

Printout of the online help

11/2019

System overview of STEP 7 and WinCC	1
What's new in TIA Portal	2
Readme	3
Installation	4
Migrating projects and programs	5
Introduction to the TIA Portal	6
Editing projects	7
Editing project data	8
Using user administration	9
Using libraries	10
Editing devices and networks	11
Programming the PLC	12
Visualizing processes	13
Using technology functions	14
Using online and diagnostics functions	15
Processing energy data and optimizing energy consumption	16
Using Team Engineering	17
Working with the TIA project server	18

Hardware / software	Requirement
Processor	Intel® Core™ i5-6440EQ (up to 3.4 GHz)
RAM	16 GB (min. 8 GB, 32 GB for large projects)
Hard disk	SSD with 50 GB free storage space
Network	1 Gbit (for multi-user)
Monitor	15.6" full HD display (1920 x 1080 or more)
Operating system	Windows 7 (64-bit)** - Windows 7 Home Premium SP1 * - Windows 7 Professional SP1 - Windows 7 Enterprise SP1 - Windows 7 Ultimate SP1 Windows 10 (64-bit) - Windows 10 Home Version 1809, 1903 * - Windows 10 Professional Version 1809, 1903 - Windows 10 Enterprise Version 1809, 1903 - Windows 10 IoT Enterprise 2015 LTSB - Windows 10 IoT Enterprise 2016 LTSB - Windows 10 IoT Enterprise 2019 LTSB Windows Server (64 bit) - Windows Server 2012 R2 StdE (full installation) - Windows Server 2016 Standard (full installation) - Windows Server 2019 Standard (full installation) *only for Basic Edition **not for "STEP 7 Basic/Professional and WinCC Professional", only for "STEP 7 Basic/Professional incl. Safety and WinCC Basic/Comfort/Advanced and WinCC Unified"

Entry type: **Manual Entry**
ID: **109773506**, Entry
date: **11/20/2019**



Manual

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S7
S7-1200 Programmable controller

System Manual

Preface	
Product overview	1
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STEP 7 programming software	3
Installation	4
PLC concepts	5
Device configuration	6
Programming concepts	7
Basic instructions	8
Extended instructions	9
Technology instructions	10
Communication	11
Web server	12
Communication processor and Modbus TCP	13
TeleService communication (SMTP email)	14
Online and diagnostic tools	15
Technical specifications	A
Calculating a power budget	B
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Device exchange and spare parts compatibility	D



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S7-1200
Easy Book

Manual



Preface	
Introducing the powerful and flexible S7-1200	1
STEP 7 makes the work easy	2
Getting started	3
PLC concepts made easy	4
Easy to create the device configuration	5
Programming made easy	6
Easy to communicate between devices	7
PID is easy	8
Web server for easy Internet connectivity	9
Motion control is easy	10
Easy to use the online tools	11
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Exchanging a V3.0 CPU for a V4.1 CPU	B

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Programming Guideline for S7-1200/1500

TIA Portal

<https://support.industry.siemens.com/cs/ww/en/view/81318674>

Siemens Industry Online Support

Getting started with S7-1200

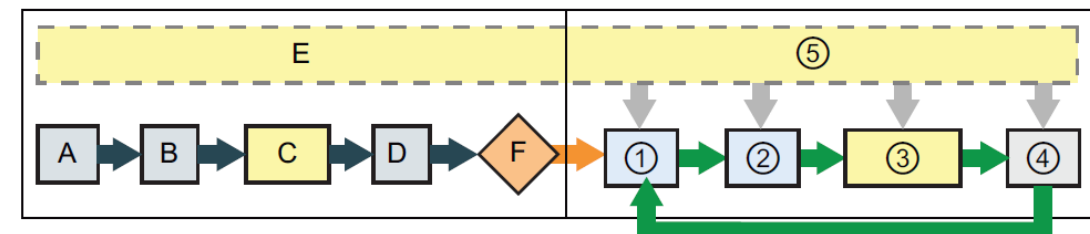
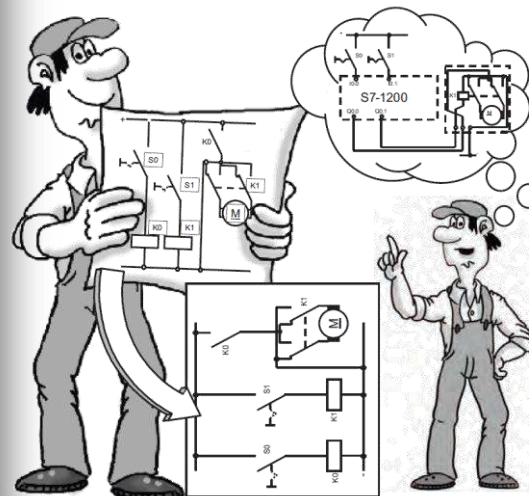
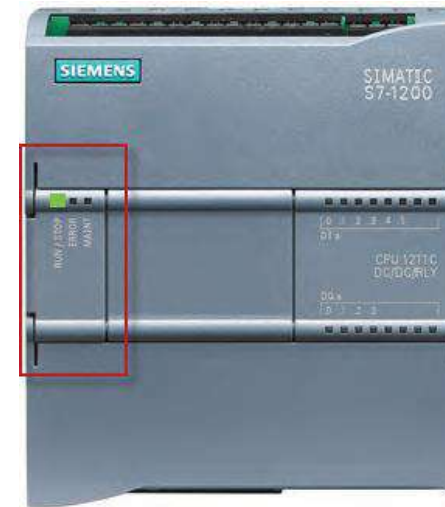
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S7-1200
Getting started with S7-1200

Getting Started

Preface	
Quick review	1
Installation	2
Create a simple latch circuit	3
Complete the user program	4
Use a watch table for monitoring	5



STARTUP
mode

- A The input area of the process image (I memory) is cleared.
- B The outputs are initialized with the last value.
- C Any start-up logic (contained in special code blocks) is executed.
- D The state of the physical inputs is copied to I memory.
- E Any interrupt events are queued for processing during RUN mode.
- F The writing of the output area of the process image (Q memory) to the physical outputs is enabled.

SIMATIC Automation Tool

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SIMATIC Automation Tool

The easy support for commissioning, adjusting and service

You can now access all connected controllers simultaneously. Operating and maintaining automation networks could be done without engineering framework. With the scan function you can see at a glance which controllers have an IP address, by clicking on the DOWNLOAD icon, you can assign IP addresses to the remaining machines. You can also set the time for all machines. If you want to stop all the controllers and then restart them, that can also be done easily at the press of a button. If an incident requiring service occurs while the plant is in operation, order to download a diagnostics buffer from the controller. Naturally you can trace all the changes you've made at any time.



Functionality in combination with S7-1200 and S7-1500 Controllers

All named features can operate in parallel to various CPUs:

- Scan an Ethernet network and identify all S7-1200 and S7-1500 CPUs connected to the Network
- Assign addresses (IP, subnet, gateway) and station name (PROFINET device) to a CPU
- Set the time in a CPU to the current time in your PG/PC
- Download a program to a CPU
- Put a CPU in RUN or STOP mode
- Flash the lights on a CPU to help physically locate the CPU
- Upload fault information from a CPU
- Read the diagnostic log from a CPU



Entry type: Download, Entry ID: 98161300, Entry date: 10/01/2015

SIMATIC Automation Tool - the commissioning and service operation tool for S7-1200, S7-1500 including ET 200SP CPU and decentralized IO modules

Entry Associated product(s)

Hereby it takes place the deployment of the SIMATIC Automation Tool.

After a control program is created and verified with the Siemens TIA Portal software, the SIMATIC Automation Tool can be used in the field for operating and maintaining automation networks.

Automation Tool operations:

- Scan an Ethernet network and identify all devices connected to the network
- Assign addresses (IP, subnet, gateway) and station name (PROFINET device)
- Set the time in a CPU to the current time in your PG/PC
- Download a program to a CPU (no failsafe programs)
- Put a CPU in RUN or STOP mode
- Backup/Restore CPU data to/from a backup file (no failsafe)
- Flash the lights on a CPU to help physically locate the CPU
- Upload fault information from a CPU
- Read the diagnostic log from a CPU
- Reset a CPU to factory default settings
- Update firmware in a CPU and attached modules
- Document and save your network information
 - in a standard text .csv file or
 - an encrypted & password protected .sat file



Download
SIMATIC Automation Tool

PRONETA

Commissioning and diagnostics tool for PROFINET networks



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PRONETA simplifies commissioning and configuration of your PROFINET network. The topology of your network is read automatically. You can manually adapt the address parameters of every PROFINET device or simply apply the parameters from a template, which can also be created with PRONETA.

You can use PRONETA to configure, control and monitor I/O modules of the SIMATIC ET 200SP, ET200M, ET 200MP, ET 200AL, ET 200eco PN and ET 200S distributed I/O systems. The test results are provided in an easy to view log.

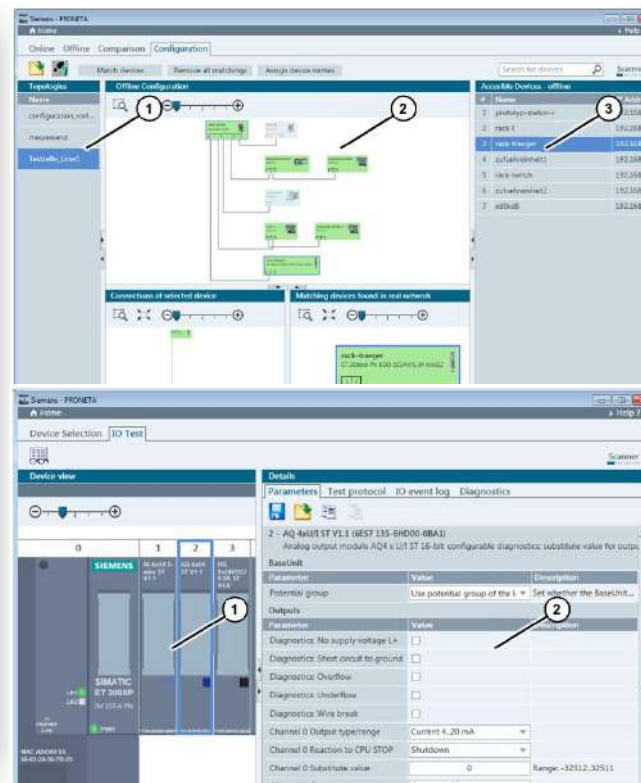
With PRONETA, the configuration and testing of your control cabinet can begin during installation. This means there is nothing more in the way of fast and successful commissioning!



IO test of an ET 200SP with PRONETA



Comparison of two PROFINET topologies with PRONETA



Entry type: Application example, Entry ID: 67460624, Entry date: 02/03/2016

PRONETA 2.3.0.26 Commissioning and Diagnostics Tool for PROFINET

Entry Associated product(s)

PRONETA is a free tool for the analysis and configuration of PROFINET networks with special support for ET 200 distributed IO. The IO Test provides quick and comfortable examination and documentation of a plant's wiring.



Technical information

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The screenshot displays the Siemens Industry Online Support website. The browser's address bar shows the URL: <https://support.industry.siemens.com/cs/sc/3058/important-issues-at-a-glance?lc=en-US>. The page title is "Important issues at a glance". The main content area lists various support topics under different categories: TIA Portal, SIMATIC HMI, PC-based Automation, SIMOTION, Industrial Identification, Micro Automation, and SIMATIC PCS 7. A sidebar on the left contains a search bar and a list of navigation links. A right sidebar titled "mySupport Cockpit" shows user-specific information and links. The footer contains copyright information: "© Siemens AG 2009-2016".

Content Search

All

- > Service Cooperation
- > Plant Data Services
- > Training Services
- > Support & Consulting Services
- > Online Support
 - > Finding solutions
 - > Finding application examples
 - > Easy testing of compatibility
 - > Discovering services
 - > Support for my product
 - > Discussing in the Forum
 - > Important issues at a glance
- > Working with the portal
- > Our info services for you
- > Technical Support
- > Engineering Support
- > SIMATIC Migration Services
- > SIMATIC System Audit
- > SIMATIC Inventory Baseline Services
- > Lifecycle Information Services
- > Spare Part Services
- > Repair Services
- > Field & Maintenance Services
- > Retrofit & Modernization Services
- > Service Programs & Agreements

Important issues at a glance

We have compiled overview pages for your frequently asked questions and comprehensive issues from our range of information. The overview pages show you all information on an issue at a glance – available at just one click.

TIA Portal

- > TIA Portal - The new Automation Framework
- > Automation Tasks in 10 minutes or less - TIA Portal
- > Migrating Production Plants with SIMATIC (TIA Portal) - Overall systems, Control, Visualization, Communication

SIMATIC HMI

- > SIMATIC HMI in Online Support

PC-based Automation

- > PC-based Automation: An Overview of the Most Important Documents and Links

SIMOTION

- > SIMOTION at a glance

Industrial Identification

- > Industrial Identification mit SIMATIC Ident

Micro Automation

- > Applications and Getting Started with LOGO! and S7-1200

SIMATIC PCS 7

- > SIMATIC PCS 7 in Industry Online Support

mySupport Cockpit

Chantid Boun-Aht

- > Add to mySupport favorites
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- > Personal messages
- > My requests
- > CAX downloads
- > Entries last viewed
- > My tags
- > User online (14)

Product information

- > Presales info
- > Catalog and ordering system online
- > Technical info
- > Support
- > Service offer
- > Training
- > Contact & partners

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<https://support.industry.siemens.com/cs/sc/3058/important-issues-at-a-glance?lc=en-US>

<https://support.industry.siemens.com>

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1. Access Industry Online Support web site
<https://support.industry.siemens.com/cs/products>

2. Type product

3. Click to search

4. Click "CAX data"

Filter criteria for entries

Product tree: All

Enter search term: 6ES7215-1AG40-0XB0

Product: 6ES7215-1AG40-0XB0

Entry type: All

Date: From To

Search product

Product Support

6ES7215-1AG40-0XB0

CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO

SIMATIC S7-1200, CPU 1215C, COMPACT CPU, DC/DC/DC, 2 PROFINET PORT, ONBOARD I/O: 14 DI 24V DC; 10 DO 24V DC 0.5A 2 AI 0-10V DC, 2 AO 0-20MA DC, POWER SUPPLY: DC 20.4 - 28.8 V DC, PROGRAM/DATA MEMORY: 125 KB

Product details Technical data CAX data

5. Click your expected information

6ES7215-1AG40-0XB0

CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO

Product details Technical data CAX data

CAX data

Here you collect your CAX data for free. Select the Siemens CAX data types you need for your CAD or CAE system.

Graphical CAX data for this product

Here you can view the existing graphical data for this product and you can download it directly.

Product Picture
Dimensioned Drawing
3D Model
EPLAN Macro

Compile all available CAX data for this product

The CAX

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Product Picture
Dimensioned Drawing
3D Model
EPLAN Macro

Data protection Terms of use Digital ID

Cx Data

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➤ Product Picture

➤ Dimensioned Drawing

➤ 3D Model

➤ EPLAN Macro

➤ Product Picture

➤ Dimensioned Drawing

➤ 3D Model

➤ EPLAN Macro

Compile all available Cx data for this product

The Cx Download Manager is at your disposal with this direct link:

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Do you want to open or save **G_ST70_XX_99516V.edz** (524 KB) from **automation.siemens.com**?

Product information

+ Presales info

+ Catalog and ordering system online

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Save

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➤ EPLAN Macro

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6. Save file

Sales status and Technical data

6ES7215-1AG40-0XB0

CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO

Product details

Technical data

CAX data



SIMATIC S7-1200, CPU 1215C, COMPACT CPU, DC/DC/DC, 2 PROFINET PORT, ONBOARD I/O: 14 DI 24V DC; 10 DO 24V DC 0.5A 2 AI 0-10V DC, 2 AO 0-20MA DC, POWER SUPPLY: DC 20.4 - 28.8 V DC, PROGRAM/DATA MEMORY: 125 KB

Product life



6ES7215-1AG40-0XB0

CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO

Product details

Technical data

CAX data

Technical data



SIMATIC S7-1200, CPU 1215C, COMPACT CPU, DC/DC/DC, 2 PROFINET PORT, ONBOARD I/O: 14 DI 24V DC; 10 DO 24V DC 0.5A 2 AI 0-10V DC, 2 AO 0-20MA DC, POWER SUPPLY: DC 20.4 - 28.8 V DC, PROGRAM/DATA MEMORY: 125 KB

General information

Product type designation	CPU 1215C DC/DC/DC
Firmware version	V4.1
Engineering with	
• Programming package	STEP 7 V13 SP1 or higher

Display

with display	No
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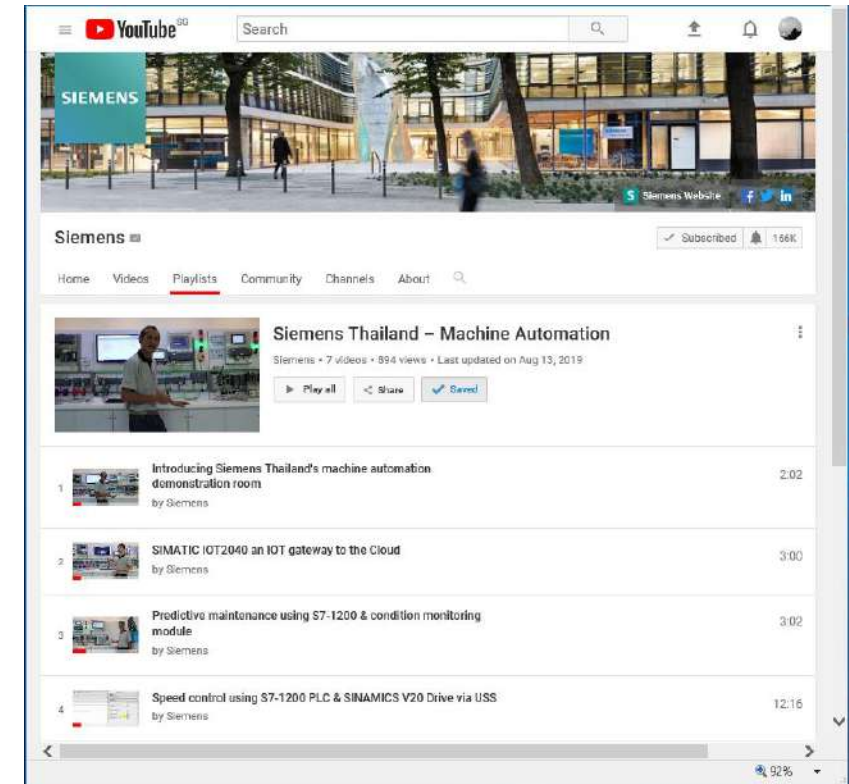
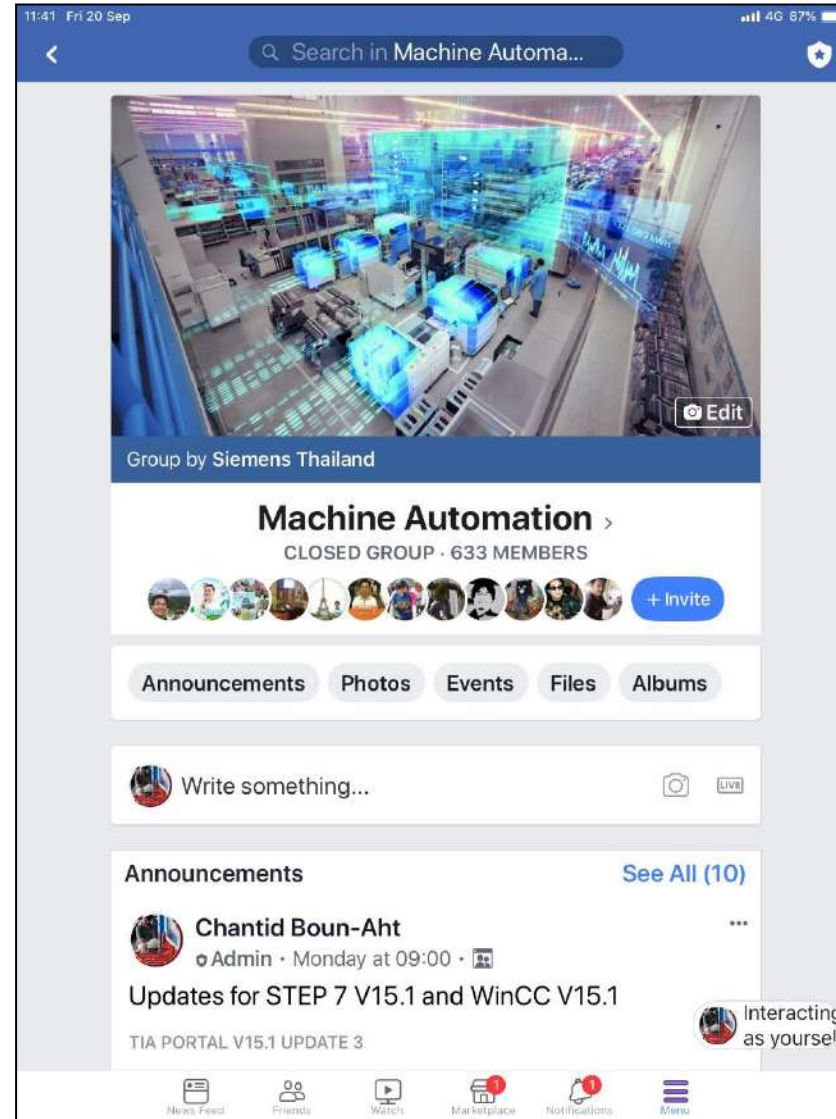
Supply voltage

Rated value (DC)	
• 24 V DC	Yes
permissible range, lower limit (DC)	20.4 V
permissible range, upper limit (DC)	28.8 V
Reverse polarity protection	Yes

Social education : Facebook group “Machine Automation”



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Thank you for your attention

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<https://support.industry.siemens.com>



Chantid Boun-Aht
Product Manager



Arkom Pithakwirakun
Technical consultant



Nattapong Thiangdee
Technical consultant



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